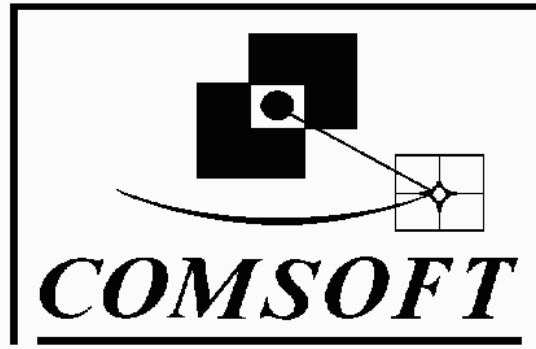




PC-TCS™ Version 6.x
User's Manual

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Chapter 1

PC-TCS Overview

1.1 How to Avoid Reading this Manual

Almost everyone would rather learn about a new piece of software at their computer terminal than by reading a manual. PC-TCS is a menu-driven program designed for those users with a background in astronomy and little or no experience with computers. With only a brief introduction about how to move around in the PC-TCS menus, the user can be commanding the telescope to move to and track their favorite astronomical objects.

As a cursory introduction to PC-TCS operations, it is suggested you read all of chapter 2 – **Getting Started**. With the information presented in this section you should be able to test drive PC-TCS with no further eyestrain. As you explore, you will probably find that PC-TCS will try and prevent you from doing silly (and perhaps dangerous) things like point your telescope at the floor, or use your instrument as a Roman catapult. In normal operating modes, PC-TCS is designed to inhibit the user from damaging himself and his telescope within reasonable limits. The program will tell you if it doesn't like what you are asking it to do by “beeping” and/or displaying an appropriate error message. However, it is possible to make selections in PC-TCS that might allow the telescope to assume positions or speeds that may be hazardous. Be careful, and if you are unsure of what a given option or command does in PC-TCS, please...

⇒ **STOP and READ the Manual!** ⇐

Eventually you will want to learn more about some of the sophisticated capabilities of PC-TCS. This manual therefore, is organized in a manner which provides a quick reference type access to information, as well as step by step directions for some of the more common operations. The appendices in the back of this manual cover information as to error messages and system trouble shooting.

1.2 What is PC-TCS?

PC-TCS is a *Personal Computer based Telescope Control System* designed for use with moderate to large professional astronomical telescopes. The program will run on all IBM-PC class microcomputers and compatibles using a 80286 or better processor with a math coprocessor.

Its primary duty is to control the telescope drive system and its related hardware (such as the paddle interface and limits detection circuitry). Its true utility is, however, the sophisticated user interface that allows the observer to perform functions previously unavailable in any observatory class control system.

Version 6.X/DCS of PC-TCS requires one Custom Dual Axis DC Servo Control Card available from COMSOFT; version 6.X/CT requires a third party counter / timer control card available from a number of vendors. Optionally, PC-TCS can use a single 4-Port Serial Card (Personal Computing Tools, Inc.) to accommodate up to 8 channels of remote serial access to the control system.

When commanded to move the telescope, PC-TCS's proprietary servo algorithm will dynamically increase or decrease the control signals to the motor amplifiers to ensure a smooth departure from the initial coordinates and subsequent approach to the commanded position.

1.3 What Does PC-TCS do?

PC-TCS provides the observer with a highly flexible battery of motion control functions handsomely matching the demands of astronomical observing. As much as possible, system complexity is minimized by the use of a "Menu Driven" (See section 2.3) user interface that, in most cases requires only single keystrokes to make the telescope perform complex tasks.

For example PC-TCS is capable of controlling automated slews from one object to another – even if the object is below the observer's astronomical pole. The activation of sidereal, lunar, solar, cometary and asteroidal, as well as user definable tracking rates is as simple as just one keystroke. Also among PC-TCS's repertoire of programmed motions are raster and spiral search patterns useful in finding that odd comet or planetoid with uncertain coordinates. And, at the end of a long night's observing, PC-TCS can perform all the necessary shutdown procedures with just a simple keystroke.

In addition, PC-TCS has the Messier, FK5, Yale Bright Star, SAQ, PPM, GCVS, Oke-Stone spectroscopic standard star and Revised New General object (RNGC) catalogs accessible from disk – and a utility software to generate your own customized catalogs. The current positions of the Sun, Moon, Comets, Asteroids, and all eight visible major planets can also be computed in a fraction of a second. PC-TCS accurately processes all motion requests compensating the input coordinates for the object's proper motion as well as the effects of precession, nutation and annual aberration, atmospheric refraction and parallax. PC-TCS even has the ability to correct for the telescope's inherent

mechanical flexure. Given an accurate flexure map, PC-TCS can display the true “astronomic” coordinates the telescope is currently pointing.

Manual control of the telescope is available from either a hardware “paddle” which contains four direction buttons, a two position Guide/Drift switch, a Slew/Slew Permit buttons, and a PANIC switch or a standard PC game-type Joy-stick providing smooth motion control of telescope available only with such an analog type device. Using these inputs, the observer can move the telescope manually at two user definable rates (Drift and Guide) and slew speed. The PANIC buttons is provided for user security and allows emergency drive shutdown while away from the PC-TCS keyboard.

For the more sophisticated user, PC-TCS has the ability to be remotely controlled via up to 8 serial communications channels. (If you have a modem, this means you can even control your telescope over the phone lines!) In addition, user created disk files containing powerful PC-TCS commands can be used to completely automate a night’s observing – from first star acquisition to predawn – all without touching a single button.

Some features of PC-TCS version 6.X . . .

- **Completely Menu and window driven.** “Hot-key” and “direct - key” access to menuing options along with full mouse support.
- **Displays over 50 telescope and telescope subsystem related items** including Current Telescope RA and DEC, Hour angle, Altitude and Azimuth, Parallax Angle, airmass (secant Z), Local Siderial Time, Universal Time, Julian Date, Calendar Date; Telescope focus position, Dome position, Instrument rotation subsystem position catalog object Equatorial position, name and magnitude; Time will rise/set of Current; Next and Reference telescope positions. User definable “Next” and “Reference” equatorial positions; user definable paddle guide and drift rates; User definable “wobble” and “Offset” vectors. Time till rise/set of Current, Next and Reference Position.
- **Automated slewing and tracking** to any planet, the sun, moon or other solar system object given object orbital elements. Automatically computes DEC and RA rates for such objects for “hands-off” tracking of non-stellar objects.
- **Precise telescope pointing** with corrections to computed accounting for Proper Motion, Precession, Nutation, Annual Aberration, Atmospheric Refraction and repeatable telescope mechanical flexure and mount misalignments.
- **Tracking corrections** for differential Refraction, Flexure, RA gear Periodic Error and declination gear backlash.

- **True “Astronomic” Coordinate Display** of telescope position for any epoch the user desires – deconvolutions for precession, aberration, refraction and telescope flexure.
- **Supports all types of equatorial mounts** – including English yoke and german designs with user definable manual and automatic german “flips”.
- **User definable limits** in hour angle, declination, horizon and airmass (secant Z).
- **On-line catalogs of objects** including the Yale Bright Star catalog, Revised New General Catalog (RNGC), Messier catalog, Smithsonian Astrophysical Observatory catalog (SAO), FK5 catalogs of bright stars, Positions and Proper Motions catalog (PPM). Users can make there own catalogs with an included catalog generation utility program.
- **Supports Remote serial access** on up to 8 serial ports for control of the telescope via modem from a remote site or by other computers and peripheral “smart” instruments.
- **Powerful file oriented “script language”** for self-contained telescope automation.
- **Semi-automated pointing data collection and analysis** utilities for generating telescope flexure maps.
- **Automated Raster and Spiral search scan modes** for wide field photometry and other surveys.
- **Automated drift scan capabilities** for extended object spectroscopy.
- **Support for joy-stick and paddle manual control** of telescope motion.
- **Earth Orbiting Satellite tracking** capability.
- **User definable security system** for restricting level of program operation and access at sites that have multiple users and operators.

1.4 What to do if you have Problems

As with any complex system – problems may arise, and here at COMSOFT we are anxious to avoid any user hostility. If you find a genuine “creepy – crawly” (or “undocu-

mented feature” as programmers like to refer to them) please drop us a line explaining the anomaly. We will fix it and send you an update at no charge. If you have suggestions for future PC-TCS improvements or would like technical assistance with this product . . .

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Vail, Arizona

Chapter 2

Getting Started

2.1 PC-TCS Conventions

We begin by starting a few conventions of this manual. Table 2.1 describes the characters used to represent certain keystrokes on your computer keyboard. In addition, certain “Function” keys (labeled F1, F2, F3, etc.) will have the effect of directly invoking PC-TCS functions without having to highlight the menu item and press ←. These and other special keys are documented in the reference section of this manual under the individual functions they invoke.

Throughout this manual when sample user input is being displayed, the type style keys denoting user keystrokes will be *slanted and emphasized*. All computers generated output will be in **Bold Face** type style.

2.2 PC-TCS System Requirements

PC-TCS is designed to operate under the MS-DOS or PC-DOS operating system (versions 3.3 and up). Note that PC-TCS can be run under Microsoft’s Windows 98se provided that the program is run in “exclusive” mode. PC-TCS should not be run as a background job or simultaneously with any other Windows 98 or DOS program. Although Windows 98 is capable of “running” more than one program simultaneously, because of the way Windows handles task scheduling and interrupts, PC-TCS operation will be severely compromised if run as a non-exclusive task.

PC-TCS Keystroke Conventions	
←	The ENTER or CR key. Used to activate a menu selection or complete data entry. Same as the left mouse button.
↑	The Up Arrow key. Used to move the highlighted option indicator up a vertical menu or move back a full field in a data entry window. Same as up mouse motion.
↓	The Down Arrow key. Used to move the highlighted option indicator down in a vertical menu or forward a full field in a data entry window. Same as down mouse indicator.
←	The Left Arrow key. Used to move the highlighted option indicator left in a horizontal menu or backward one character in a data entry field. Same as the left mouse motion.
→	The Right Arrow key. Used to move the highlighted option indicator right in a horizontal menu or forward one character in a data entry field. Same as the right mouse option.
ESC	The Esc or ESCAPE key. Used to abort any data entry in progress and to return to the previous level in all menu operations. Same as right mouse button.
Alt	The Alt or ALT key. Used in conjunction with one other key to directly invoke a PC-TCS function from the keyboard and bypass standard menu selection.
Ctrl	The Ctrl or CONTROL key. Used in conjunction with one other key to directly invoke a PC-TCS function from the keyboard and bypass standard menu selection.
Ctrl ←	The Ctrl and ENTER keys pressed simultaneously. Used to force an update of multi-query windows after one or more fields have been updated. Allows the user to complete the data entry process in multi-query windows having to process all fields.
BS	The Backspace key. Used to delete characters before the cursor during data entry.

Table 2.1: PC-TCS Keystroke Conventions

- **Processor:** A IBM PC class machine with either an 80486 micro processor installed. PC-TCS will function nominally on machines that have clock rates as low as 100 Megahertz, a speed of 400 Megahertz or greater is HIGHLY recommended.
- **Co-Processor:** A math coprocessor is REQUIRED.
- **Memory:** At least 610 kilobytes of Transient Program memory must be available for PC-TCS to be loaded. Command files will consume more at run time and it is suggested that a minimum of 612 Kb be available for PC-TCS use with command files and to perform satellite tracking.
- **Storage:** At least one (1) 360K floppy disk drive and a hard disk drive of at least 10 Gigabytes capacity is necessary. PC-TCS can be installed and run from a floppy – however, all catalog files will not fit on even the highest density floppies and the user will have minimal catalog support.
- **Video Display / Adapter:** A monochrome display and driver card capable of at least 25 lines by 80 columns is required. PC-TCS was originally designed to use a color display and adapter (EGA/ VGA class) and users with this option will find PC-TCS more visually pleasing.
- **Serial Ports:** (Optional) None are required, however, users wishing to take advantage of PC-TCS remote access capabilities must have at least one (1) RS-232 port available. (Up to eight (8) serial communications ports are supported with optional third party hardware.)
- **Drive Interface Board:** Version 6.X/DCS requires a single Dual Axis DC servo Control Card to drive DC Servo amplifiers. This board is available only through COMSOFT. The Stepper Motor implementation of PC-TCS (Version 6.X/CT) requires a single third party interface board available from a number of vendors. Contact COMSOFT for further information.
- **Analog Interface Board:** (Optional) Version 6.X requires a single multichannel ADC board if the user wishes to take advantage of the COMSOFT custom Joy-stick interface, focus and dome control/ read-out and other analog input capabilities. This board is available through several third party vendors – contact COMSOFT for more information.
- **Mouse:** (Optional) Version 6.X can be used with a mouse to move around the PC-TCS window and menu environment. A mouse is NOT required but will make data

entry option selection more efficient. If a serial mouse is used – one of the serial communications channel may not be available to PC-TCS for remote access.

- **Joy-stick:** (Optional) Version 6.2 can be used with a standard IBM-PC Game type joy-stick or custom joy-stick supplied by COMSOFT. The joy-stick can be used in place of, or supplement the paddle to move the telescope manually. If the Game type joy-stick is used, the target computer must have the appropriate game port installed.

- **UTC Clock:** (Optional) Version 4.5X is designed to maintain accurate Universal Time via an external clock is accomplished through a single RS-232 port at 9600 baud – allowing PC-TCS to maintain time to approximately 1 millisecond. This feature is optional.

2.3 PC-TCS Menus and Windows

Given the limited space available on the PC computer CRT screen and the large variety of operations that can be accomplished with PC-TCS, feature selection and information display in PC-TCS is handled via a number of “menus” and “windows”.

The menus are designed to give minimum keystroke access to PC-TCS’s many functions and utilities. Choices are displayed in linear or columnar format and selections are made using the appropriate arrow keys (see Table 2.1) or mouse motion to move the highlighted menu cursor. Some menu selection invoke still other menus. Each succeeding level of menu is referred to as a “submenu” of the one before. The Runtime Operations menu has a submenu for each menu entry item and most of these submenus have submenus, etc. In this manual, the level of menu nesting for any documented function can be gleaned from the number of “subs” stated before its title.(e.g. **Paddle I/O** subsubsubmenu is a menu three from the root or Runtime Operations Menu.) As the observer becomes familiar with a given menu option’s function, a single “Hot Key” – (shown to the immediate right of the menu option and documented in the reference section) may be used to invoke the function directly.

When single keystroke are not sufficient to convey all the necessary information, PC-TCS will prompt the user for more input or display additional data using “windows” which nondestructively obscure part of the Runtime Display. Both the menus and data windows remain visible only for as long as the observer needs them and not interfere, in anyway, with the operation of PC-TCS’s primary function – that of telescope will continue to smoothly track, the paddle inputs will continue to function, and telescope position status will continue to be displayed.

In choosing the menu and windowing environment for data entry and display PC-TCS has provided the observer with simple productivity. Tedious user data entry and unnecessary computer inquiries have been eliminated. Once familiar with the functionality of PC-TCS’s menus, the observer may quickly execute most PC-TCS features with the single touch of a key.

2.3.1 PC-TCS File Window

Some of the menu items described in Chapter 3 of this manual will bring up a window displaying “files”. The user select the desired file using the arrow keys to highlight the appropriate selection then hitting the ← key.

The *Del* key has a special functionality during the processing of this window type. This key will have the effect of “Deleting” the file currently highlighted.

2.4 The PC-TCS Runtime Display

The screenshot displays the PC-TCS Runtime Display window with the following information:

UT DATE: 02/02/2011 UT: 05:09:05.2 LST: 06:33:18 LHA: +11:59:35.50
 JD: 2455594.71
 ACCESS: NOVICE RA << LIMIT >> DEC Tube WEST

ARF Epoch: J2000.0 18:34:16.40 +69:29:30.5 El: 11.5
 Az: -0.0
 Secz: 5.01
 PA: +0.1

P Commanded: 18:32:08.62 SLEW PERMIT +69:49:20.9 Mag: 16.30

Next: # 35 09:32:15.03 H0928+50 U +49:50:53.4
 Reference: 06:28:05.15 +32:10:35.7
 Offset Vector: +00:00:00.00 +00:00:00.0
 δ Wobble Vector: +00:00:11.68 +00:00:00.0 MC: Closed
 δ Difference: +921.5 +282584.2 sFoc: 0
 Dome: -30.0

Bias: +0.0000 +0.0000
 δ Guide: +5.0000 +5.0000
 δ Drift: +50.0000 +50.0000 9600

Coordinate Mode: File Command File: SUN.TCS
 Coordinate File: CBA.CAT Current Command: WAITSUNE \$1
 Coordinate Epoch: TOD MPNARF*p Command Count: 7 09:14:18

Runtime Operations Menu

Motion deClare mOve Position Rates Source Time pARameters shutDown

The PC-TCS Runtime Display contains the information needed by the observer to continually monitor all aspects of telescope motion, position, the current date, UT and

Sidereal times, catalog and user defined coordinates and vectors, data and command file status and information, remote access activity, external paddle status, and user menus.

Figure 2.1 shows the general lay-out of the PC-TCS Runtime Display Screen. The following section attempts to describe the meaning of each field. For those observers with color monitors, field labels, when present, are shown in green or white-on-black (half intensity on monochrome screens) while data and information fields are displayed in bright yellow, red, magenta and cyan (full intensity on monochrome screens.)

PC-TCS Version 4.5 or later also can display a large format display using the F7 “Hot key” on the keyboard. This will overlay the normal Runtime display with a simpler display in the 40 column mode of the PC video display. This display should NOT be used for long periods of time or for general system monitoring since it does not display error or system menus.

2.4.1 Runtime Display Elements

UT DATE: Displays the Current Universal Date in the form MM/DD/YYYY.

UT: Displays the Universal Time in the form HH:MM:SS.

LST: Displays the Local Sidereal Time or Right Ascension currently on the meridian.

LHA: Displays Local Hour Angle of the telescope. Local Hour Angle is simply $LST - RA$ for any given telescope position, therefore, Hour Angle is positive west and negative east of the meridian.

JD: Displays the Julian Date in days since January 1,4713 B.C.

Tube: Displays the current side of the mount the telescope tube is residing. For german equatorial systems this item is critical in making sure PC-TCS’s limit system is properly initialized. The observer should verify that this field agrees with the current state of the telescope tube position.

ACCESS: Displays the level of security access to PC-TCS functions. Depending upon the security level, certain PC-TCS functions are not available. See section 2.8 for more information.

RA: The Right Ascension label – denotes the column used to display the Right Ascension component of all positions, vectors and rates. The Right Ascension axis label is also used as a motion annunciator. If the RA label begins to flash, the system is indicating the telescope is more than one arcsecond off position in Right Ascension and is trying to correct the error. The annunciator will normally lash whenever a telescope motion to a new object or position is commanded, a Right Ascension paddle button is

pressed, or anytime the telescope is not able to maintain positional accuracy to better than one arcsecond in Right Ascension.

Limit Status: This field is found between the RA and DEC label fields and is activated whenever a software or hardware positional limit has been violated. The field will display a flashing **LIMIT** annunciator along with an arrow (<or>) pointing to the offending axis. If the violation is due to a horizontal limit, both arrows will be visible.

DEC: The Declination label – denotes the column used to display the Declination component of all positions, vectors and rates. The Declination axis label, like the Right Ascension axis label is used as a motion annunciator. If the Declination label begins to flash, the system is indicating the telescope is more than one arcsecond off position in Declination and is trying to correct the error. The annunciator normally will flash whenever a telescope motion to a new object or position is commanded, a Declination paddle button is pressed or any time the telescope is not able to maintain Declination position to better than one arcsecond.

Pole: This field is visible when the current position of the telescope is below the local celestial pole. When the telescope is above the local pole this field is blank.

Epoch: The **Display Epoch** for coordinates being displayed in the “boxes” showing the **Current** telescope coordinates as well as the **Commanded**, **Next** and **Reference** position lines. All Display Epochs are referenced to the Julian System (specifically the IAU standard J2000.0 system) – Besselian coordinate Epochs are not supported by PC-TCS. The display epoch may be changed by the observer using the **Display** subsubsubmenu. (See Reference section, Chapter 3.) A value of “TOD” in this field indicates that the display epoch is “True of Date” (i.e. Today). Before this field is the deconvolution acronym – which indicates which deconvolutions are being applied to the coordinates currently being displayed in the RA and DEC columns. The letters stand for the following: A – Aberration; R – Refraction; F – Flexure. If the deconvolution is disabled for any one of these items an underline place holder is displayed instead. Note that with all deconvolution corrections turned on the RA and DEC columns display the astronomical coordinates of the telescope for the Epoch selected. (See reference section under the **Display** subsubsubmenu.)

Current RA: The Right Ascension of the telescope is displayed inside a “boxed” field for emphasis. Depending on the value of **Epoch** and the **Motion Status** of the telescope the value found in this field may change with time even though tracking is turned on. The field display has a resolution of 1/100 of a second of a time – the resolution of the internal clock of the DCS Servo Control Card. This resolution is limited to the display only. The true tracking and positional resolution of the telescope is limited to the mechanical resolution of the drive train and/or encoder hardware. This field displays the Coordinates of the telescope deconvolved for the effects of Precession and Nutation and those deconvolutions selected in the **Display** subsubsubmenu.

Motion Status: This field is unlabelled and is found between the current position “boxes”. The field is blank if tracking and/or bias rates are turned off. The field will flash **Disabled** or **PANIC** if certain error conditions exist or the observer has disabled the drive via the **Motion** submenu or pushed the PANIC button on the hardware paddle. See the reference section under the **Motion** submenu for more information concerning this field.

Current DEC: The current Declination of the telescope is displayed inside a “boxed” field for emphasis. Depending on the value of **Epoch** and the **Motion Status** of the telescope, the value found in this field may change with time even though tracking is turned off. The field display has the resolution of 1/10 of a second of arc. This field displays the astrometric Declination Coordinates of the telescope deconvolved for the effects of Precession, Nutation and those deconvolutions selected in the **Display** subsubmenu.

El: The current elevation in degrees above the horizon the telescope is pointing.

Az: The current azimuth of the telescope – measured positive from the north to the east. Negative values indicate westerly azimuths.

Commanded: This line displays the last commanded Right Ascension and declination of the telescope. This field changes whenever a move command is issued from the **Move** submenu or via the Remote command channels or from a command file. Paddle, motion, bias, tracking and other rates have no effect on the Command position fields. A **P** preceding the **Commanded:** label indicates that the commanded position is below the local celestial pole. The Commanded Position displays the astrometric commanded coordinates of the telescope for the Epoch displayed in the Epoch line described above. Deconvolution for Annual Aberration, Atmospheric Refraction and Telescope Flexure are also included if these deconvolution have been enabled via the **Display** subsubmenu.

SLEW PERMIT: This field found between the RA and DEC fields of the Commanded position line, displays the current status of the Slew Permit hardware button (if implemented) or the Slew Permit Toggle activated/deactivated by the F8 function key (if implemented). The field acts as an indicator of whether or not the telescope will be allowed to assume high speeds during moves from one object to the next. For more information about Slewing see section 2.5 of this manual.

Secz: The current Secant of the Zenith angle (09° - *Elevation*). Useful as an estimate of the airmass the telescope is looking through. This label will change color in the event that a Secant Z limit violation is detected. (See Chapter 3 under **Limits** subsubmenu.)

PA: The Parallax Angle – the apparent angle made between the Zenith, the current telescope coordinates on the sky and the North Celestial Pole.

Next: Displays the Right Ascension, Declination and file information (if Appropriate) for the Next programmed position. The line contains five fields – the first (**SP**) immediately precedes the **Next:** label and displays the spectacular type of stars retrieved from system catalog files. The fourth field (#) follows the **Next:** label and shows the object number if the coordinates were retrieved from a catalog file. The **Label** field between the Right Ascension and Declination columns displays the first ten characters of any label information retrieved from user and system catalog files. The **Next:** label will change color if the next position violates a telescope limit condition. The next Position field displays the astrometric Next coordinates of the telescope for the Precessional Epoch displayed in the Epoch line (described above). Deconvolutions for Annual Aberration, Atmospheric Refraction and Telescope Flexure are included in this line if they have been enabled via the **Display** subsubsubmenu.

Mag: Displays the Magnitude of the objects retrieved from catalog files, the SUN, Planets and Solar System Objects other than the Moon. The fields is blanked for all other coordinate input.

Reference: Displays the Right Ascension, Declination information for the stored **Reference** position. The **Reference** position is updated via the **Declare** submenu. (See Chapter 3 for more information.) The **Reference:** label will change color if the reference position violates a telescope limit condition. The **Reference** position field displays the Reference coordinates of the telescope for the Precessional Epoch displayed in the Epoch line (described above). Deconvolution for Annual Aberration, Atmospheric Refraction and Telescope Flexure are included in this line if they are enabled via the **Display** subsubsubmenu.

Offset Vector: Displays the Right Ascension and Declination components of a user definable equatorial vector that can be used to offset from the current position of the telescope. This vector differs from the Wobble Vector (See below) in that it has only one direction and does not change magnitude based on declination. The **Offset label** field display the label of the destination star if the offset was computed by PC-TCS from either a user or PCF catalog file. (See Chapter 3 under **Position** and **Move** submenus.)

Wobble Vector: Displays the Right Ascension and Declination components of a user definable vector that can be used to move back and forth between two positions on the sky. This vector differs from the Offset Vector in that it is bi-directional and can maintain the same magnitude (true sky distance) regardless of commanded declination. The field between the Right Ascension and Declination columns will display the current “beam” ID when the telescope is within 3 arcseconds of the commanded wobble beam position. The field will be preceded with the δ symbol if correction for the cosine of declination is currently enabled and the telescope can be made to wobble under external TTL control. (See reference section under **Declare**, **Position** and **Input – Remote** submenus for more information.)

Difference: Displays the difference in Right Ascension and Declination the telescope has been moved since the last time these fields were zeroed. (See Chapter 3 under the **Declare**, submenu.) The fields displays the accumulated motion of the telescope less the effects of sidereal tracking, bias and object/correction rates. This field can display the accumulated differences in RA in either Time or seconds of arc on the sky. If seconds of arc are displayed – the δ symbol will precede the Difference label and indicate that the effects of the cosine of the Declination is being included in the difference coordinates. (See Chapter 3 under the **Globals-Display** submenu.)

IIS: (Optional) If your installation has the ability to encode the Instrument Interface System used to align the position angle of the instrument on the sky, this field will display the current setting of the encoded instrument rotation angle.

Foc: (Optional) If your installation has the ability to encode the focus position this field will display the current focus position in encoder counts.

Dome: (Optional) If your installation has the ability to encode the dome position this field will display the current error in dome azimuth. Useful as an indication of possible obscuration of the telescope optical aperture. This label will change color if the dome azimuth does not match the telescope azimuth to within a user specified error window.

Remote: A window displayed if the remote serial access to PC-TCS is active. (See reference section under **Remote** subsubsubmenu.) This window contains ten fields that display . . .

- **Commands:** - The number of valid commands received from the remote serial lines since the last activation. Below the number of valid commands is a field shown in red that displays the number of “bad” commands received.
- **1:** - The last value of the command channel buffer received from the primary (COM1) serial port after the terminating character was received.
- **2:** - The last value of the command channel buffer received from the secondary (COM2) serial port after the terminating character was received.
- **3:** - The last value of the command channel buffer received from the primary (COM3) serial port after the terminating character was received.
- **4:** - The last value of the command channel buffer received from the secondary (COM4) serial port after the terminating character was received.
- **5:** - The last value of the command channel buffer received from the primary (COM5) serial port after the terminating character was received.

- **6:** - The last value of the command channel buffer received from the secondary (COM6) serial port after the terminating character was received.
- **7:** - The last value of the command channel buffer received from the primary (COM7) serial port after the terminating character was received.
- **8:** - The last value of the command channel buffer received from the secondary (COM8) serial port after terminating character was received.
- **Checksum:** - The status of the check sum error checking routine for serial communications – indicating whether the commands are being check summed or not before execution.

The window may be “hidden” by using the *Alt H* hotkey – in the event the window will be replaced by a flashing “Remote” icon. (See Chapter 2 – **Input – Remote** subsubsubmenu documentation.)

Also see Appendices B, C and D for more detailed information about Remote access to PC-TCS.

Bias: Displays the Right Ascension and Declination components in arcseconds per second of the user definable rates that can replace or be added to the default sidereal tracking rate. When nonzero and activated the **Motion Status** (see above) field will display either **BIAS** or **Track+b** depending upon whether or not sidereal tracking has been turned on. The field will be preceded with the δ symbol if the correction for the cosine of declination is currently enabled. This line is also used to display joy-stick status and bias lock (labeled Just “Stat” in the figure) status. (See reference section under the **Rates** submenu and **Motion-Joy-stick** for more information.)

Guide: Displays the current Right Ascension and Declination component values of the user definable guide rates activated by the paddle buttons when the Guide/Drift switch is in the Guide position. When a paddle button is pushed, the field immediately to the left of the Right Ascension or Declination column (labeled Guide Stat in the figure) will flash with a message indicating the direction of the current paddle motion being commanded. The field will be preceded with the δ symbol if correction for the cosine of declination is currently enabled. (See reference section under the **Rates** submenu.)

Drift: Displays the current Right Ascension and Declination component values of the user definable drift rates activated by the paddle buttons when the Guide/Drift switch is in the Drift position. When a paddle button is pushed the field immediately to the left of the Right Ascension or Declination column (labeled Drift Stat in the Figure) will flash with a message indicating the direction of the current paddle motion being commanded. The field will be preceded with the δ symbol if correction for the cosine of declination is currently enabled. (See reference section under the **Rates** submenu.)

Coordinate Mode: Displays the current source of new object coordinate input. The field can take on the value of either **File** or **Keyboard**. If file mode is indicated, the catalog file name will be displayed in the **Coordinate File** line described below.

Coordinate File: Displays the name of the disk based catalog file currently in use (if any).

Coordinate Epoch: Displays the current epoch of input coordinates. The input epoch refers to the precessional epoch upon which the coordinates are referenced. The input coordinates are then processed from this epoch to the current “true of date” epoch for all telescope motions and then to the display epoch (see **Epoch** above). Following this field is a display of the currently active corrections used to process the **NEXT** position. The first letter of each of the corrections is displayed else a underline place holder is displayed if the correction is inactive. The sequence of letters is as follows: M – proper Motion; P – Precession; N – Nutation; A – Aberration; R – Refraction; F – Flexure (followed by the flexure class - * means the default flexure map); P – Parallax. (See the reference section under the **Corrections** subsubsubmenu.)

Command File: The name of the command file (if any) that has been loaded and processed from disk (see reference section under **Command** subsubsubmenu). If the file is currently active and running the field will flash on and off.

Current Command: The command string from the command file (see above) that is currently being executed. Commands are executed only if the command file is active. (See reference section under the **Command** subsubsubmenu.)

Command Count: The current number of commands that have been executed from the file named in the **Command File** line. The count will roll over past 9999 executions.

Runtime Operations Menu: The level 1 user interface menu. (See Chapter 3.)

2.5 The PC-TCS Paddle

Common to every good telescope control system is the external “paddle” which allows control of the telescope’s motion via simple buttons – one for each of the four possible directions. PC-TCS however, has added a few more features to its paddle. These include...

- **TWO USER DEFINABLE RATES OF TELESCOPE PADDLE MOTION.** A guide/drift switch allows the observer to quickly select the two rates. Each rate is completely definable by the user from the PC-TCS keyboard. The Guide Position of the switch is usually defined to be a rate of motion useful in precisely maintaining telescope position for astrophotographic or data gathering modes of operation.

Drift is useful for times when faster movement of the telescope is required while still maintaining visual steadiness, such scanning for a faint comet or asteroid. As mentioned before however, each of these rates is user definable and can assume values from 0.001 arcseconds per second to full slew speed.

- A SLEW/ SLEW PERMIT BUTTON. For user and instrument safety, PC-TCS will restrict telescope maximum speed to no more than 500.0 arcseconds per second unless given permission by the observer to slew. Permission is obtained from the observer by pressing – *and holding* – the “ Slew/ Slew Permit” button located on the paddle. Alternately, the F8 function key on the PC-TCS keyboard may used to toggle the state of Slew Permit. The Slew Permit feature *will not slew the telescope* unless a new motion is actively being commanded – either from the PC-TCS keyboard or from the paddle. Slew Permit is simply *permission* to move at maximum speed – if and only if required. Optionally (see Appendix E) the Slew Permit button can be made to function as a “slew on demand” button when used in conjunction with one or more paddle direction buttons. This feature, if implemented, can be used to manually slew the telescope around the sky at maximum speed.

Note: Some installations do not have this “Slew Permit” button on the paddle and Slew Permit is toggled using the F8 function key on the PC-TCS keyboard.

- A PANIC STOP BUTTON. For those emergencies when any telescope motion is undesirable, a single button is provided that will discontinue all drive signals from PC-TCS to the motor hardware. If PC-TCS detects a PANIC input, it will shut down the telescope drives and flash a **PANIC** warning on the Runtime Display screen. Once detected, the PANIC condition must be manually reset from the keyboard before further telescope motion can resume. (See the reference section under the **Motion** submenu for further details.)

For special operations, Slew Permit (as well as any other paddle input) may be permanently enabled or disabled by using the paddle input software masks accessible via the **External I/O** subsubsubmenu. These masks are also useful in recovery from paddle and a few benign hardware I/O problems do not need immediate repair. For further details see the Reference section.

>> *Some Notes on Slewing with the Paddle* <<

Because of the way PC-TCS handles rate input, large rates produce the standard telescope “ramp-up-to-speed” and “ramp-down-to-position” type action. Therefore, when a slew paddle button is pressed, the telescope will not immediately assume full slew speed. Depending upon the tuning of your drive system, PC-TCS may take as much as 8 seconds to ramp up to maximum velocity. Consequently, high rate assignments, such as slew, tend to produce a somewhat ponderous ramping action.

For this reason, you may find it frustrating to try and accurately position the telescope using high paddle rates since the telescope will not immediately assume the high velocity you expect. If you must use the paddle in this manner, it may be easier to think of the buttons as digital “hoses” that squirt telescope commands to move the set number of arcseconds for every second you hold the button. (e.g. If a rate of 3600 arcseconds per second is assigned to the Declination buttons, holding the North button down for 10 seconds will ultimately move the telescope ten degrees north of its starting position.)

If this mode of operation is unacceptable – the only alternative is to have COMSOFT “retune” the system for high acceleration applications.

2.6 The PC-TCS Joy-Stick

Version 6.X has the capability of driving the telescope via either a standard PC Game type joy-stick or a custom joy-stick supplied by COMSOFT. The rate at which the telescope is moved is tied to the “Drift” rate assigned to the PC-TCS paddle. Maximum deflection of the joy-stick in either X or Y axis will move the telescope at the drift rates displayed on the **Drift** line of the Runtime display. Changing the drift rate via the **Rates** menu will also change the maximum rate of the joy-stick. Deflections less than maximum will invoke a rate following a logarithmic scale so that fine control (such as guiding operations) may be accomplished with moderate deflections of the joy-stick. If the joy-stick is commanding rates to the control system the **BIAS** line of the Runtime display will show a flashing “Jyst” icon in the appropriate column.

Two buttons are associated with the joy-stick and provide control of rates useful for advanced tracking operations such as satellite tracking. The “Slew Shift” button overrides the default max rate of the joy-stick associated with the Drift rate of the paddle. While the Slew Shift button is pressed the maximum rate associated with the maximum joy-stick deflection is 3600 arcseconds per second. This button is useful in slewing the telescope via the joy-stick or for “catching up” to a fast moving satellite during satellite tracking operations. (Note that although the Slew Shift button is active the telescope will not be allowed to assume this rate unless the Slew Permit button is also pressed (or the F8 key has activated the Slew Permit.)

The “Bias –Lock” button will “hold” or “lock-in” the rate currently being commanded from the joy-stick. Once a deflection of the joy-stick is commanded, pressing the Bias-Lock button will “lock-in” the rate associated with the current deflection. The joy-stick will not work again until the joy-stick has been recentered. Once recentered further deflection of the joy-stick will subsequently allow the user to joy-stick around the “locked-in” rate. The user may lock-in further biases with subsequent bias-lock button. This will clear out all bias-lock rates and return the joy-stick to normal operation. If a bias lock is currently active, the **BIAS** line of the Runtime display will show a “Lock” icon immediately following the **BIAS:** label.

The user should note that the joy-stick can be enabled/disabled via the **Motion** submenu. (See Reference section – chapter 3). Each time the joy-stick is enabled via this menu option PC-TCS auto-zeros the joy-stick to be in the centered position.

2.7 Starting PC-TCS

Since the installation of PC-TCS involves different procedures depending on the options purchased – the instructions for installing PC-TCS can be obtained from the file README.DOC found on the PC-TCS distribution Master diskette. Once the hardware and software have been installed – PC-TCS may be started by changing to the appropriate subdirectory (e.g. C:\TCS) and executing the program file called TCS.EXE.

— IMPORTANT —

PC-TCS Version 6.X/DCS should ALWAYS be executed BEFORE enabling the power to the servo system. This ensures that PC-TCS is in control of the telescope before power is applied to the electronics. If this recommendation is not allowed – the telescope may “drift” in position if any DC offsets are present in the amplifier/servo system. The reverse of this procedure should be followed for PC-TCS shutdown, - power should be turned off to the servo system BEFORE exiting the PC-TCS software.

One the TCS.EXE program is started the screen will clear and fill with the copyright preamble and the instruction. . .

→ **Hit any key to start system check-out and initialization** ←

— A word About Telescope “Stow” —

At start-up, PC-TCS assumes the telescope is in the “zenith stowed” position. Unless changed by the user at a later time (see below) this position is defined as pointing straight up toward the zenith and not moving in either axis. This is the state PC-TCS always expects the telescope to be at cold start. If the telescope is pointed elsewhere, or in motion at PC-TCS start-up – the system will inaccurately display telescope coordinates and the software limits will function improperly – leading to possible **damage to your telescope**.

Version 6.X of PC-TCS allows the user to modify this zenith default stow position by using the **Stow** item of the **Declare** submenu. Once a new stow position is defined, PC-TCS will assume this new position to be stow after a cold start.

At this point any key except the *ESC* key will begin PC-TCS's start up sequence. (*ESC* will abort PC-TCS start up and return to the operating system level.) Several messages will print in the screen indicating the programs progress. If all system checks are found to be favorable, preamble windows will disappear and the PC-TCS Runtime Display will become visible. (See section 2.3.)

As a final check before turning power on to the servo system and moving the telescope, the display time and the date should be carefully scrutinized. PC-TCS uses Universal Time exclusively and the observer should familiarize himself with the correct conversion factor to convert local time to UT for this verification. To obtain the starting UT time PC-TCS reads the internal PC clock – and assumes this to be the correct UT time. If your PC is set to local time (and you don't live in England) PC-TCS will not function properly. Errors in the time and date as little as a few seconds can produce large pointing errors while observing and make finding object almost impossible. If the time and or date need to be modified refer to the **Time** submenu in the Reference section (Chapter 3) of this manual.

Before observing begins we *highly recommend* initializing the telescope coordinate system. Although this is not absolutely necessary, systems that use encoders typically maintain position to a very coarse level when the drive electronics are powered off.

Therefore, before attempting observations, sidereal tracking should be activated and the telescope commanded to move to a target for which precise coordinates are known (such as a visible bright star from one of PC-TCS's system catalog files). After the move is complete the observer should center the target in the telescope's main field of view using the paddle buttons or optional joy-stick. Once centered PC-TCS should then be instructed to assume the coordinates of the selected object, processed by PC-TCS and displayed in the **Commanded:** position line of the Runtime Display (see section 2.3.1) are the true coordinates at which the telescope is pointed. This is accomplished using the **Initial** item of the **Declare** submenu. (see Chapter 3.)

PC-TCS should now be ready to help you explore the wonders of the universe.

If you are starting PC-TCS from a floppy disk or have used the computer with another program and wish to start up PC-TCS from scratch, return to your computer's \TCS subdirectory (C:\TCS for hard disk drives —A:\ for Floppy disk drives) and simply type. . .

TCS ←

From here PC-TCS operation should be the same as described previously.

2.7.1 Starting PC-TCS with a Command File

As will be described in Chapter 4 of this manual, PC-TCS has the ability to read and process a number of commands from a disk file – these files are called Command Files and contain one or more commands outlined in Appendix A.

Command Files enable PC-TCS to operate the telescope in a semi-autonomous way that is useful in automating observing sessions of highly repetitive nature. PC-TCS may be commanded to load and execute a single Command File at PC-TCS start-up as follows. . .

TCS commandfile ←

Where *commandfile* is the file name of the desired Command File to load and execute. Starting PC-TCS in this manner will bypass the preamble window and immediately proceed with the system checkout. If the system check is successful, PC-TCS will load the Command File specification and begin executing its commands without further user intervention.

For installations with multiple observers/ operators PC-TCS comes with a Command File called DEFAULTS.TCS that resets select PC-TCS environment parameters such as drift and guide rates, coordinate corrections and display deconvolutions.

2.7.2 Start Up Errors

If you should receive a PC-TCS error window during PC-TCS start up – refer to Appendix G for interpretation. Other problematic symptoms encountered in novice PC-TCS operation include – poor telescope pointing, Warning and Error messages from DOS and/or PC-TCS – are (rarely) complete System “crashes” . Such problems are usually due to one of three things. . .

- **SYSTEM ERRORS** – the inability of the Operating System (in the case MS-DOS or PC-DOS) to locate the PC-TCS program and/or its associated files. Make sure you are in the directory path that contains the PC-TCS program and data files. A list of necessary files for correct PC-TCS operation are. . .

TCS.EXE The PC-TCS executable program file.

TCS.PF The PC-TCS start up file containing critical system parameters.

PCTCSD.FLX A optional file created by the ANALYZE program that holds the Declination mechanical flexure correction data.

PCTCSR.FLX A optional file created by the ANALYZE program that holds the Right Ascension mechanical flexure correction data.

In most cases these files will be found in the \TCS or (C:\TCS) subdirectory of your hard disk. If not, check the Master floppy diskettes distributed with your system for any documentation updates (README.DOC files) and/or copy the PC-TCS program, data and initialization files from the Master diskettes to your hard disk's \TCS subdirectory.

The user should also take care not to operate PC-TCS from RAM Disks. RAM Disk driver software (e.g. VDISK.SYS, RAMDRIVE.SYS, etc.) usually disable interrupts while accessing files stored in extended or expanded memory. Since PC-TCS utilizes interrupts for servo control, system time keeping and remote communications, operation of PC-TCS may be seriously compromised if executed from RAM Disks.

Hard System errors such as complete computer “crashes” may be result of insufficient memory available to support certain PC-TCS options. These options are usually associated with copying files from floppy disk while PC-TCS is resident in RAM (see the **Import** option under the **Utilities** subsubsubmenu in Chapter 3). Under certain conditions, trying to copy files from floppy to hard disk while PC-TCS is active in memory may cause the computer to completely stop talking to the outside world. Your only recourse in this event is to reset the computer and restart PC-TCS.

- **HARDWARE ERRORS** – These errors are usually caught by PC-TCS's start up sequence. If a bad Counter Timer or Servo Card is present in the computer, compulsory system checks made by PC-TCS will display an appropriate error message (See Appendix G.) and abort PC-TCS. More serious hardware problems such as bad disk drives or errors indicating PC-TCS's inability to read/write to its data files should be handled by your nearest computer repair facility.
- **SET UP ERRORS** – As mentioned earlier pointing problems can usually be traced to earlier improper telescope position on start up, inaccurate time in the PC clock, or poor coordinate system initialization. Check the Reference section of this manual concerning the **Declare** and **Time** submenus for further assistance.

2.8 PC-TCS Security

To better facilitate observatory installations with operators and since some of the functions in PC-TCS are applicable to some mounting types and not others (e.g. German Flip) a security system has been implemented in PC-TCS that restricts access to certain

functions. Menu items that are “unavailable” to a given security level will appear as green letters on a white background and the user will be unable to move the highlighting menu cursor to the item.

Currently there are four levels of security available. Level 0 or “NOVICE” access is the lowest and default level of security. The NOVICE level allows access to only those functions of PC-TCS that the casual user will need or use during the course of operation. Access to functions that modify limits; edit Hardware I/O mappings and telescope stow position; command Hard Stow and German Flips; perform satellite tracking and engineering functions, are typically not available to this level.

Level 1 or “EXPERT” access should be reserved for experienced operators and users that need access to lower level functions that change the way PC-TCS interprets hardware inputs, defines telescope stow position, and corrects calculated positions. This level should restrict access to most telescope limit alteration; engineering and proprietary operations such as servo tuning; and satellite tracking.

Level 2 or “ALPHA” clearance should allow access to all but proprietary functions. This level should be made available only to designated users responsible for site maintenance, engineering or observer training.

Level 3 or “AUTHOR” access allows unrestricted access to PC-TCS operations and is typically unavailable to all but the most trusted engineering and supervisory personnel.

While running PC-TCS the security level may be altered by entering a password appropriate to the level desired. The password entry window may be accessed by hitting the CTRL and Backspace keys simultaneously at anytime while PC-TCS is running.

Level 0 security clearance requires no password. Level 1 security clearance is obtained by entering the password “OPERATOR”. The level 2 security password may be obtained only from your designated site supervisor. The level 3 password is proprietary and may be obtained from COMSOFT only under extraordinary circumstances.

>> A Note to Site Supervisors <<

PC-TCS version 6.X comes with a program editor that will allow modification of security levels for each PC-TCS menu item and site specific information. This program is called PFEDIT and conforms to the same security restrictions as PC-TCS. (i.e. To modify the security level for any particular menu item, you must have the same or higher security clearance as the menu item you wish to modify.) Authorized users may use this program to “customize” PC-TCS for their site specific observing needs. Contact COMSOFT for further information and the level 2 password.

Chapter 3

PC-TCS Reference Guide

The following chapter deals with the menu system of PC-TCS. Each section covers a specific menu and the options found under each item.

Each option is described on a single page with information as to how to access the option, the option's function and suggested uses, along with visual and auditory indicators and possible error messages.

It is suggested that the user run PC-TCS and step through each option while referring to the manual page that describes the function. It should be noted that the option can be invoked either via the PC-TCS menu of which it is a part or by its designated direct access key. Once a menu item is invoked by one or the other of these methods it can not be accessed recursively by the alternate method.

3.1 The Motion Submenu

Cancel Tracking Bias δ anti-back Lash
Scan
Joystick
Disable

The **Motion** submenu is used to control/override telescope motion. Activation of tracking and bias rates, implementation of rate oriented Scan and Joy-stick input, cancellation of commanded moves and telescope drive disables are handled via this menu. The submenu consists of seven executable items. All options except **Cancel** indicate an active state via sideways chevron symbol (>>).

Motion - Cancel

Direct Key Access: F9

Functionality: Makes the current position of the telescope the commanded position. Has no effect on tracking, bias or other “rate” oriented motions. If a slew to a new coordinate has been commanded, this option will override the slew attempt and stop on whatever coordinates the telescope was at when the Cancel command was executed.

Uses: Useful in halting telescope motion to new objects. Also may be of use in stopping drive system oscillations in a poorly tuned or mechanically “open” servo system.

Indicators: The current telescope position will appear in the **Commanded** position line of the Runtime Display.

Errors: No error messages should be produced by this option.

Motion - tracking

Direct Key Access: *Alt T*

Functionality: Toggles sidereal tracking on and off. Rate is engaged/disengaged regardless of bias, paddle, or other motions in progress.

Note: Tracking should be turned on before moves to new coordinates are attempted. If tracking is turned off and a move to a new object is commanded – the telescope will *not* come to the coordinates displayed in the **Commanded** line. In this case the telescope will be off by the amount the sky has apparently moved in the time it took to make the move.

Uses: Used to keep a given star field from drifting out of view due to the earth's diurnal rotation.

Indicators: The current telescope coordinates will stop changing in Right Ascension when tracking is turned on – and on paddle or bias rates are active. In addition the **Motion Status** display will show **TRACKING** or **TRACK +???** when sidereal tracking is engaged. The question marks in this text represent place holders for the rate corrections available via the **Rates** submenu. A lower case “r” on the Runtime Display indicates that refraction rate correction is also active. Lower case “b” indicates that a bias rate refraction rate correction is active, a lower case “f” indicates that flexure correction rates are active, a lower case “o” indicates that an Object (e.g. Moon, Planet, Sun, Comet, Asteroid, etc.) rate correction is active. (See the **Rates** submenu for more information on refraction, flexure, bias and object rates.)

Errors: If the drive system is enabled or the telescope is unable to physically move – the **RA** annunciator may flash or the **Motion Status** display field may show something other than published above.

Motion - Bias

Direct Key Access: *Alt B*

Functionality: Activates/ deactivates any programmed bias rates displayed in the **Bias** line of the Runtime Display. Bias rates are added to the sidereal rate (if active). (See the **Rates** submenu for more information on bias rates.)

Uses: Useful in correcting the tracking rate of the telescope for non-sidereal objects such as the asteroid, planets, comets, etc. Also can be used to set up a scan rate for trailing observations.

Indicators: **Motion status** field will show either **Track+b** or **Bias** when active.

Errors: An error window will appear if no bias rates have been defined.

Motion – δ anti-backLash

Direct Key Access: None

Functionality: Activates/ deactivates a “preload” for declination guide operations. This function when active detects declination guide input from the paddle and when a command motion in declination is opposite in direction from the last commanded guide operation, adds the programmed number of encoded counts to the commanded position. This effectively removes any hysteresis or “backlash” in the gear train controlling dec motion. When invoke, a query window will appear asking the user for the number of encoded counts to add to the commanded position after guiding input of opposite direction. A count of zero will disable this function.

Uses: useful in removing the effects of gear backlash in the declination axis during guiding operations.

Indication: A sideways chevron symbol (>>) will precede the anti-backlash menu option when active.

Errors: None are associated with this menu option.

Motion - Scan

Direct Key Access: None

Functionality: Setup and Activation/ deactivation of a repetitive back and forth “scan” of the sky using a user definable rate and duration for the scan.

When the Scan menu item is first invoked the user will be promoted for a rate in arcseconds per second for each axis long with a duration for a single pass of the scan in seconds of time. Once this data entry is completed the telescope will immediately begin execution of the assigned scan. The telescope will move at the previously assigned RA and Declination rates for the period entered and then reverse direction at the negative of the entered rates for the same period. The process is repeated until the Scan option is invoked a second time – at which point the repetitive scan will be terminated.

Uses: Useful in trailing stars or extended objects back and forth along a slit during spectroscopic observations.

Indicators: The **Scan** menu item will be preceded by a sideways chevron >> when active.

Errors: None associated with this menu item.

Motion – Joy-stick

Direct Key Access: *Alt J*

Functionality: Activation/ deactivation of the optional analog joy-stick interface. In some custom installations of PC-TCS a joy-stick is available to supplement the paddle input for telescope guiding and positional control. This menu item allows the user to activate or deactivate the reading of the joy-stick analog inputs. When selected, this function reads the current ADC values from the joy-stick and assumes these values to be the “zero-point” for the joy-stick inputs. Therefore, this menu item may be used to “re-zero” the joy-stick interface.

Note: the range of rates associated with the joy-stick are determined by the value of the RA and DEC drift rates settings – that is the rate generated when the joy-stick is maximally deflected will be the set Drift Rate for the axis. Intermediate deflections of the joy-stick produce rates that are logarithmically scaled to produce very fine control to slow speeds.

Uses: A joy-stick is useful in providing a more intuitive interface to telescope control during such operations as objects and guide star centering and satellite tracking.

Indicators: The **Joy-stick** menu item will be preceded by a sideways chevron >> when active. If the joy-stick inputs are nonzero – indicating a rate is being commanded – the field immediately to the left of the RA and Declination columns will flash “Jyst” in the approximate axis column. The user also has the capability of “locking in” a joy-stick rate by using the Bias Lock button found on the joy-stick control box. If a joy-stick bias lock is in effect – the field immediately to the right of the BIAS label on the Runtime display will show “BLK.” To cancel the Bias lock – the user should zero (recenter) the joy-stick and press the Bias lock button a second time.

Errors: None associated with this menu item.

Motion - Disable

Direct Key Access: *F10*

Functionality: Terminates PC-TCS output to the drive system hardware. This will effectively stop the telescope regardless of speed or position. Because the immediacy of this command, the user should use this command only as an “emergency stop” command as it may stress mechanical and electrical drive system hardware.

Note: This command acts only on the drive outputs and does not affect the commanded position or rates. Therefore the telescope will try to “catch up” to the commanded coordinates when and if the drives are enable again. After disabling the drives with this command it is recommended that all tracking and bias rates be terminated and the *Cancel* Command be executed. (See **Motion** submenu for more details.)

Uses: Emergency stop when at the PC-TCS console. It is also used to clear out any previously encountered Disable or PANIC condition when position limits have been exceeded or paddle PANIC has been activated. The option must be engaged and then disengaged to clear such conditions.

Indicators: The **Motion Status** field will flash **DISABLED** when active, also the menu item will be prefixed with the sideways chevron symbol.

Errors: No error messages should be produced by this option.

3.2 The Declare Submenu

Reference Beam 1 Beam 2 Initial
Stow
Dome East Zero Diff

The **Declare** submenu has up to seven items, depending on installed options, pertaining to the PC-TCS update of storable coordinate pairs. The menu allows the user to alter the **Reference** position and **Wobble** vector as well as the system wide coordinate zero point (**Initial**), stow position **Stow**, Dome Position **Dome East**, and Difference **Zero Diff** read-out.

As a descriptive aid in remembering the type of action invoked by each option, each item may be thought of as having the phrase “Declare as” in front of its name.

Declare - Reference

Direct Key Access: *Ctrl R*

Functionality: Saves the current telescope coordinates as the Reference position for later move-to-reference command. (See **Move** submenu section.) The position is remembered only as long as the PC-TCS remains in memory. Exiting PC-TCS will clear this field. For more permanent logging of telescope position see the **POSITION – SAVE** and **POSITION – RECALL** menu items.

Uses: Used to save the coordinates of an interesting object so that a later move can be made back to these coordinates. Coordinates displayed are those of the display epoch indicated in the **Epoch** field.

Indicators: When invoked, the current coordinates of the telescope will be copied into the reference line of the Runtime Display.

Errors: No error messages should be produced by this option.

Declare – Beam 1

Direct Key Access: *Ctrl F1*

Functionality: Internally saves the current position of the telescope in preparation for computing a vector for later use in wobbling. The wobble vector is a transportable true sky distance vector that is dynamically definable by the observer either by declaring both “beams” or directly inputting the vector. (See **Position** submenu.) The vector can be made transportable in that it maintains true “sky magnitude” regardless of the declination the telescope is move to. (i.e. the cosine of declination is taken to account whenever the telescope is commanded to a new (**Next**) object via the **Move** submenu.) In order for the vector to be computed, both Beam 1 and Beam 2 must be declared. The correction for the cosine of the declination can be enable and disable via the **Cosine δ** subsubsubmenu. (See the **Parameters** submenu under **Globals**.)

Uses: Wobble vectors are useful when repeated move between two coordinate pairs will be made. Also useful in “measuring” the equatorial distance between objects since the wobble vector display line always shows the equatorial magnitude of the vector.

Indicators: If tracking is active the beam ID field, located between the RA and DEC columns of the Wobble Vector line, will flash **Beam 1** and remain flashing until the telescope is moved ore than 1.5 arcsecond from this position. If beam 2 has been previously declared, the equatorial components of the vector will be displayed in the Wobble Vector line of the Runtime Display. Each time a **Move** to Next, Reference or Stow command is issued, a new value for the wobble vector is computed and displayed.

Errors: No error messages should be produced by this option.

Declare – Beam 2

Direct Key Access: *Ctrl F2*

Functionality: Internally saves the current position of the telescope in preparation for computing a vector for later use in wobbling. The wobble vector is a transportable true sky distance vector that is dynamically definable by the observer either by declaring both “beams” or directly inputting the vector. (See **Position** submenu.) The vector can be made transportable in that it can maintain true “sky magnitude” regardless of the declination the telescope is moved to. (i.e. the cosine of declination is taken into account whenever the telescope is commanded to a new (**Next**) object via the **Move** submenu.) In order for the vector to be computed, both Beam 1 and Beam 2 must be declared. The correction for the cosine of the declination can be enabled or disabled via the **Cosine δ** subsubmenu. (See the **Parameters** submenu under **Globals**.)

Uses: Wobble vectors are useful when repeated moves between two coordinate pairs will be made. Also useful in “measuring” the equatorial distance between objects since the wobble vector display line always shows the equatorial magnitude of the vector.

Indicators: If tracking is active the beam ID field, located between the RA and DEC columns of the Wobble line, will flash **Beam 2** and remain flashing until the telescope is moved more than 1.5 arcseconds from this position. If beam 1 has been previously declared, the equatorial components of the vector will be displayed in the Wobble Vector line of the Runtime Display. Each time a **Move** to Next, Reference or Stow command is issued, a new value for the wobble vector is computed and displayed.

Errors: No error messages should be produced by this option.

Declare - Initial

Direct Key Access: *Alt =*

Functionality: A powerful command that should be used with great caution since it modifies how PC-TCS interprets telescope position. The command is used to initialize the coordinate system of PC-TCS by making it assume that whatever coordinates are displayed in the **Commanded** object display line are the true coordinates that the telescope is pointing. If the telescope is not accurately positioned at the actual **Commanded** coordinates, PC-TCS will function improperly. The displayed elevation, azimuth and position of the telescope will be in error and the software limits will function improperly – allowing the telescope to assume positions that may be hazardous to the instrumentation and/or the observer himself.

The procedure for initialization is described in Chapter 2 under **Starting PC-TCS**.

Uses: Used to initialize PC-TCS's coordinate system and therefore the true position of the telescope. Used whenever PC-TCS is started, after the telescope motors or drivers has been powered off, or accurate registration (pointing) on the sky has been lost.

Indicators: The **Commanded** position coordinates will replace the current position coordinates on the Runtime Display.

Errors: If no **Commanded** object has been declared, (See **Position** submenu) an error window will appear and/or a warning bell will be heard.

Declare - Stow

Direct Key Access: None

Functionality: A powerful command that should be used with great caution since it modifies the position PC-TCS will assume the telescope is at during system startup. After this command is invoked the current Hour Angle and Declination is memorized by PC-TCS. All subsequent commands that move the telescope to “Stow” will then move the telescope to this position.

Uses: Used changed the position of the telescope stow attitude. Allows the user to pick the best telescope position for instrument changes and system shutdown.

Indicators: A confirming window will be displayed at invocation of this menu item – the user may abort the operation by hitting any key except “Y”. The “Y” key will complete the redefinition.

Errors: None.

Declare – Dome East

Direct Key Access: None

Functionality: This Optional menu item is used to initialize the dome position difference read-out on the Runtime display.

For those users with this option, after the dome is moved to a reference fiducial point – this menu item will instruct PC-TCS to re-index its internal calibration of dome position. PC-TCS uses two input ports to read an external dome position encoder which gives the current azimuth with the telescope azimuth and displays the difference on the Runtime display.

Uses: Recalibration of Dome position display.

Indicators: Dome position display may be updated on the Runtime Display.

Errors: None are associated with this item.

Declare – Zero Diff

Direct Key Access: *F5*

Functionality: PC-TCS can track telescope motion relative to an indexed position. Movement of the telescope using the **Move** commands or using the paddle/joystick inputs will cause the telescope Difference display lines to accumulate position differences from the indexed position. Telescope sidereal tracking and bias rates do not accumulate differences. This menu item is used to zero the difference accumulators and thus re-index the difference reference.

Uses: Useful in keeping track of telescope movement from a user defined fiducial.

Indicators: The RA and DEC field on the Difference line of the Runtime display will be zeroed.

Errors: None are associated with this item.

3.3 The Move Submenu

Next Reference Beam 1 Beam 2 Cur+offset Stow Az-El
German Flip
Hard Stow

The **Move** submenu contains nine items that allow the user to move to previously declared or defined coordinate pairs. A facility to perform an automated axial “flip” to position the telescope tube on the opposite side of the mount is provided for German Equatorial systems, along with a hardware oriented zenith stowing procedures for those telescopes equipped with an optional Zenith finder.

Move - Next

Direct Key Access: *Alt N*

Functionality: Commands the telescope to move to the coordinates specified in the **Next** line of the Runtime Display. Most **POSITION** menu items will update the **NEXT** coordinate line on the Runtime display. The user can use this menu item to move the telescope to the updated **NEXT** position.

Uses: Used to move to the coordinates in the **Next** line of the Runtime Display – may be used to return to initial coordinates after using paddle buttons or bias rates to drift off the position.

Indicators: The coordinates specified in the **Next** line of the Runtime Display will be copied into the **Commanded** line of the Runtime Display. If the coordinates are sufficiently far from the current position the RA and DEC labels may also begin to flash while the telescope is in motion.

Errors: If no **Next** position has been an error window may appear and/or a bell may sound.

Move - Reference

Direct Key Access: *Alt R*

Functionality: Used to move the telescope to the position specified in the **Reference** line of the Runtime Display. The **REFERENCE** coordinates are updated using the **Declare – REFERENCE** menu item.

Uses: Used to move the telescope to the coordinates displayed in the **Reference** line of the Runtime Display.

Indicators: The coordinates specified in the **Reference** line will be copied into the **Commanded** line of the Runtime Display. If the coordinates are sufficiently far from the current position, the RA and DEC labels may also begin to flash while the telescope is in motion.

Errors: If no **Reference** position has been an error window may appear and a bell will sound.

Move – Cur+offset

Direct Key Access: Moves the telescope to the Current position plus the value of the offset vector displayed in the **Offset** line of the Runtime Display. The offset vector is a unidirectional vector that is not compensated for changing declination, and thus maintains a fixed equatorial magnitude.

Uses: Useful when moving to a target for which only a position offset from a visible object is known or would make location easier.

Indicators: The **Commanded** line of the Runtime display will be updated with the sum of the current telescope position and the vector specified in the **Offset** line. If the motion is sufficiently large the RA and DEC labels may also begin to flash while the telescope is in motion.

Errors: If no Offset Vector has been specified, an error window may appear and a bell will sound.

Move – Beam 1

Direct Key Access: *F1*

Functionality: Moves the telescope to coordinates specified by the current telescope position *minus* the vector shown in the **Wobble** line of the Runtime Display. The Wobble vector is a bi-directional vector that can be made transportable – maintaining a true “sky” magnitude in RA – if the **COSINE** δ option for the vector is enabled.

Uses: Useful in moving a user specified amount of “sky” distance and/or moving back and forth between two objects of known or computable separation.

Indicators: The **Commanded** position line of the Runtime Display will be updated with the value of the telescopes current position less the value of the vector displayed in the **Wobble** line. If tracking is enabled, the Beam ID annunciator will begin to flash.

Errors: If no wobble vector has not been set up, an error may be displayed and a bell will sound.

Move – Beam 2

Direct Key Access: Moves the telescope to coordinates specified by the current telescope position *plus* the vector shown in the **Wobble** line of the Runtime Display. The Wobble vector is a bi-directional vector that can be made transportable – maintaining a true “sky” magnitude in RA – if the **COSINE** δ option for the vector is enabled.

Uses: Useful in moving a user specified amount of “sky” distance and/or moving back and forth between two objects of known or computable separation.

Indicators: The **Commanded** position line of the Runtime Display will be updated with the value of the telescopes current position plus the value of the vector displayed in the **Wobble** line. If tracking is enabled the Beam ID will flash with the Beam 2 annunciator.

Errors: If no wobble vector has been set up, an error window may be displayed and a bell will sound.

Move - Stow

Direct Key Access: *Alt S*

Functionality: Moves the telescope to the defined “stow” position. This motion, when completed, will leave the telescope in a position that PC –TCS expects the telescope to be when the system is “cold started”. Sidereal Tracking and Bias Rates will automatically be turned off upon execution of this command.

Note: True stow can only be achieved if all bias rates and joy-stick bias lock are disabled when this command is executed. The user may modify the default stow position using the **Declare** submenu.

Uses: If all rates have been disabled, the command will move the telescope to the “resting” position for shut down.

Indicators: The **Commanded** line of the Runtime Display will be updated with the coordinates of the zenith at the time of command execution. If the motion is sufficiently large, the RA and DEC labels will blink while the telescope is in motion.

Errors: No errors are associated with this command.

Move – Az-El

Direct Key Access: None

Functionality: Moves the telescope to a user defined Azimuth and Elevation. When selected this option will prompt the user for the Azimuth and Elevation and subsequently move the telescope to the specified position.

Uses: It is sometimes useful to have the telescope in a position more easily defined by an Elevation and Azimuth where coordinates are not subject to time of day and date of the year. The Az-El option provides this utility for such things as loading instrumentation, testing telescope balance, determining horizon limits, etc. Azimuth by convention is measured in degrees along the horizon starting from the true north and increasing toward the east. Due north is then 0 degrees; due east is 90 degrees; due south is 180 degrees; due west is 270 degrees. Negative azimuth are allowed and indicate direction measured along the horizon and decreasing toward the west. (e.g. -90 degrees azimuth is the same as 270 degrees azimuth which is due west.) Elevation is measured from the horizon and orthogonal toward the zenith, therefore 0 degrees elevation is on the horizon and 90 elevation is at the zenith.

Indicators: A window will be opened to allow the user to enter the desired Azimuth and Elevation for the move. Upon completion of the Azimuth, elevation data entry, the **Commanded** line of the Runtime Display will be updated with the equatorial coordinates of the defined Azimuth and Elevation at the time of command execution. If the motion is sufficiently large, the RA and DEC labels will blink while the telescope is in motion.

Errors: If the desired azimuth/ elevation combinations is outside the positional limits of the telescope – an error window will be displayed and no move will be commanded.

Move – German Flip

Direct Key Access: None

Functionality: Performs an automated slew that will ultimately move the telescope tube from one side of the mount to the other and return the telescope to the original coordinates. The operation involves turning of all commanded rates; moving the telescope to the meridian in hour angle then to the north celestial pole in declination; rotating the hour angle axis 180 degrees, tube high; reenabling the rates and then a slew to the original coordinates. (Also see the **Limits** submenu.)

Note: Since telescope flexure is likely to be different with the telescope tube on the opposite side of the mount – it is suggested that the user reinitialize the telescope coordinate system after this operation. (See the **Declare** submenu.)

Warning: This operation may not be appropriate for some mounts where control or instrument cabling may be too short for the required declination and hour angle motions of this routine (more than 180 degrees). Use caution and make sure the cabling associated with an axial flip will not compromise control cables and telescope instrumentation.

Uses: For German equatorial mounted telescope, the option provides perhaps the only way of acquiring all positions without having the tube below level of the mount counter weights.

Indicators: A warning/ confirmation window will be displayed, and upon confirmation a progress window will appear.

Errors: If PC-TCS determines that the destination coordinates after the flip will be outside the limit range as set by the user – an error window will appear and the Flip will be terminated before it is started.

Move – Hard Stow

Direct Key Access: None

Functionality: moves the telescope to a hardware defined zenith position.

WARNING!!! This option will function properly only if the optional Zenith finder hardware has been installed and is working properly. Users without this option should NEVER invoke this procedure since all coordinate reference will be lost and the telescope may be damaged in the resulting slew.

Uses: If in the event of a power or either catastrophic failure during which all coordinate reference are lost to PC-TCS, this option can be used to grossly recover the coordinate reference before performing a startup coordinate reinitialization. (See the **Declare** submenu.)

Indicators: A warning/ confirmation window will be displayed.

Errors: None.

3.4 The Position Submenu

Next Offset Wobble
Object
saVe Recall

The **position** submenu contains three items pertain to the modification of the three user definable positions and vectors. The **Object** item will allow the user to compute the positions of the Sun, Moon and Planets as well as a comets and asteroids provided an orbital parameter file has been created for the objects. The **SAVE** option allows the user to store the current position of the telescope for later retrieval by the **RECALL** menu item.

Position - Next

Direct Key Access: *Ctrl N*

Functionality: Requests information about Next position coordinates. If the **Coordinate Mode** is set to **File**, this item will request the object number to retrieve from the file specified in the **Coordinate File** line of the Runtime Display. In the File mode a window will be opened showing the partial contents of the catalog file selected. The user may select a given object by entering the catalog number and hitting ← twice. In addition the user may use the *PGUP* and *PGDN* keys to view the contents of the file. The arrows keys ↑ and ↓ can be used to point at the desired object. A subsequent ← will select the arrowed object. Coordinate data is displayed in columnar format starting with the arrowed object. Coordinate data is displayed in columnar format starting with the object Right Ascension, followed by it's Declination, Coordinate Epoch, Magnitude and object label. Objects highlighted in black-on-red are currently outside the range of telescope pointing limits; objects displayed in black-on-white are currently visible and within telescope pointing limits.

If the mode is set to **Keyboard** this item will query the user for coordinate Epoch and position and (optionally) Proper Motion (see **Corrections** subsubsubmenu) information about the object.

Uses: Used to set the Next Position information into the **Next** position line of the Runtime Display. This information can then used to move to new coordinates via the Move Next command (see **Move** submenu).

Indicators: The **Next** Position line of the Runtime Display will be updated with the processed coordinates specified by the user.

Errors: Illegal Data in any of the data fields will sound a bell. Values out of range of the equatorial coordinate system will cause an error window and/or a bell to sound.

Position - Offset

Direct Key Access: *Ctrl O*

Functionality: Requests positional information about the Offset vector from the keyboard. The Offset vector is a unidirectional vector used to move a specified amount of Right Ascension and Declination from the current position of the telescope. Vector may either be positive or negative. The vector is taken as literal input and is not “transportable” like the Wobble vector in that the correction for declination is not taken into account.

Uses: Useful in finding an object when an offset from a previous coordinate is known.

Indicators: The **Offset Vector** line of the Runtime Display is updated with the positional information by the user.

Errors: If the equatorial data is out of range an error window and/or a bell will sound. If unrecognizable data input is attempted a bell will sound.

Position - Wobble

Direct Key Access: *Ctrl W*

Functionality: Requests positional information about the wobble vector from the keyboard. The vector information is in the form of an equatorial coordinate pair where the magnitude of Right Ascension component is assumed to be the magnitude on the sky taken at zero degrees (0°) declination. The vector is then corrected for the current declination (divided by $\cos(\delta)$) (if option is enabled) to maintain accurate hour angle displacements on the sky. A new value for the Right Ascension component is also computed each time a Move to Next Command is executed if the cosine declination correction is enabled via the **Cosine δ** subsubsubmenu. (See **Move** submenu and the **Parameters** submenu under **Globals**.)

Uses: Useful in setting up a bi-directional vector on the sky. The vector will always maintain a true “sky” magnitude if Move to Next command are used to move from object to object.

Indicators: The **Wobble Vector** line will be updated with the equatorial projection of the vector input by the user. Moves made via the Move to Next command will also update this vector line on the Runtime Display if it is nonzero.

Errors: If nonnumeric data is typed a bell will sound. If data is out of range for equatorial coordinates, an error window and/or a bell will sound.

Position - Object

Direct Key Access: none

Functionality: Computes the True of Date position of the Sun, Moon, Planets, or comets and asteroids for which an orbital parameter file has been previously created. (See Chapter 4 – Appendix H.)

The equations used to compute the object's position are taken from Jean Meeus's book *Astronomical Formulae For Calculators*.

See File Window Documentation in Chapter 2.

Uses: Used to compute the current position of solar system objects.

Indicators: A window will be displayed showing a list of Sun, Moon, Planets and solar system objects for which orbital parameter files are currently available. The user selects the desired object using the arrow keys and confirms the selection using the ← key. The **Next** line of the Runtime display will be updated with the object's coordinates and the object file name will be displayed in the label field along with the objects magnitude in the magnitude field.

Errors: If an error in computing the coordinates is detected, an error window and/or a bell will sound denoting the problem.

Position - saVe

Direct Key Access: None

Functionality: The **saVe** option will allow the user to store the current position of the telescope for later reference or reacquisition. When selected, this option will bring up a window showing 15 lines of possible positions. If a position has been previously saved in one of the 15 available lines, the line will contain an RA and DEC and an optional label. The user may select any one of the lines for storing the current telescope position using the up and down arrow keys of the keyboard. Hitting the Enter key will copy the current telescope position in RA and DEC as displayed on the Runtime Display into the line selected. PC-TCS will then allow the user to enter a label to be associated with this saved position.

If a line already contains data – a save to that line will overwrite that position with the new position and the old position will be lost. Data stored in the save window is recommended by PC-TCS even after performing shutdowns and restarts. The saved options are maintained in a file on a disk – and therefore a more permanent method of storing telescope position data than the **Declare – Reference** menu item. These stored position are available even after a PC-TCS cold start. The saved position may later be recalled using the **Recall** option described next.

Uses: Useful in storing the coordinates of an interesting object for later recall and moves back to the position. This is a good way small distances within extended object for taking data or moving repeatedly between a number of objects.

Indicators: the line designated by the cursor in the Save window will be updated with the current telescope coordinates.

Errors: Not applicable.

Position - Recall

Direct key Access; None

Functionality: The **Recall** option will allow the user to recall a previously stored position saved using a **saVe** option described on the previous page. When selected, this option will bring up a window showing 15 lines of possible positions. If a position has been previously saved in one of the 15 available lines, the line will contain an RA and DEC and an optional label. The user may select any one of the lines that contain data for recall using the up and down arrow keys of selected line into the **NEXT** coordinate line of the Runtime Display. If the position displayed in the line of the recall window is currently outside of the telescope motion limits – the line will be displayed in Red.

Uses: Useful in restoring the coordinates of an previously saved interesting object. This option along with the move to Next command will allow moving repeatedly between a number of objects.

Indicators: The line designated by the cursor in the Recall window will be copied into the **NEXT** position line of the Runtime display.

Errors: Selecting a line that has not been initialized with a saved position will cause the bell to sound.

3.5 The Rates Submenu

Bias Guide Drift
Object
Refraction Flexure PEC

This submenu contains seven items pertaining to the three user modifiable sources of rate generation (**Bias, Guide, and Drift** rates), and the rates associated with solar system objects such as the Sun, Moon, Planets, and other objects for which there is a PC-TCS orbital parameter file.

Corrective rates applied to the default sidereal tracking rate may also be activated (atmospheric **refraction**, mechanical **Flexure**, and Periodic Error Correction **PEC**). These rates may be used in conjunction with the **Object** rates to provide extended tracking accuracy for a wide variety of sidereal and nonsidereal objects.

Rates - Bias

Direct Key Access: *Ctrl B*

Functionality: Allows the user to modify the background ‘bias’ rate which is added to any rate set by tracking or paddle input. The rate does not become active, however, until a **Bias** active command is executed from the **Motion** submenu. If a bias rate is already active and the user uses this item to modify the rate, the new rate will become active immediately upon completion of the input window. A correction for the cosine of the declination can be imposed upon the Right Ascension component of the rate via the **Cosine δ** subsubsubmenu. (See the **Parameters** submenu under **Globals**.)

Uses: Used to set a background rate for tracking nonsidereal objects (Asteroid, planets, satellites, etc.) or scan drifting when searching for comets, etc.

Indicators: The Right Ascension and Declination components of the rate will be displayed in the **Bias** line of the Runtime Display.

Errors: Nonnumeric values will cause a bell to sound.

Rates - Guide

Direct key Access: *Ctrl G*

Functionality: Allows the user to modify the rate associated with the guide button input on the paddle. The rate becomes active immediately upon the normal close of the guide rate window. The rate associated with Right Ascension is corrected for the cosine of the Declination. A correction for the cosine of the declination can be imposed upon the Right Ascension component of the rate via the **Cosine δ** subsubsubmenu. (See the **Parameters** submenu under **Globals**.)

Uses: Used to change the guide rate file user's needs.

Indicators: the value entered by the user will be displayed on the **Guide** line of the Runtime Display.

Errors: Nonnumeric values entered will cause a bell to sound. Values greater than the maximum slew rate will produce an error window and/or cause a bell to sound.

Rates - Object

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction to the standard sidereal rate necessary to track a solar system object for like the Sun, Moon, Planets, or Asteroids and Comets for which there is a PC-TCS compatible orbit parameter file. The rate becomes active only when sidereal tracking is on. (See the **Motion** submenu.)

See File Window Documentation in Chapter 2.

Uses: Useful in maintaining tracking integrity over extended observing periods while observing the non-sidereal objects.

Indicators: The **Object** label in the **Rates** menu will be preceded with a sideways chevron symbol when active. In addition, if sidereal tracking is active, the **Motion Status** field will hold **TRACK+o**, or **TRACK+ro**, or **TRACK+BIAS**, or **Tro+BIAS** if a bias and/or refraction rate is also active.

Error: No errors are associated with this option.

Rates - Refraction

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction to the standard sidereal rate necessary to correct for differential atmospheric refraction. The rate becomes active only when sidereal tracking is on. (See the **Motion** submenu.)

Uses: Useful in maintaining tracking integrity over extended observing periods where atmospheric refraction changes significantly with time.

Indicators: The **Refraction** label in the **Rates** menu will be preceded with a sidereal chevron symbol when active. In addition, if sidereal tracking is active, the **Motion Status** field will hold **TRACK+r**.

Errors: No errors are associated with this option.

Rates - Flexure

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction to the standard sidereal rate necessary to correct for differential flexure of the telescope. Flexure can be associated with the mechanical “bending” of the telescope support system and/or polar misalignment of the mount. This error is determined in Chapter 4. This menu item is available provided that Flexure Data collection has been accomplished and that the Flexure data files are available on disk. The rate becomes active only when sidereal tracking is on. (See the **Motion** submenu.)

Uses: Useful in maintaining tracking integrity over extended observing periods where telescope mechanical flexure and polar misalignment significantly effect tracking.
Note: This menu item will NOT correct field rotation due to polar misalignment.

Indicators: The **Flexure** menu item in the **Rates** menu will be preceded with a sideways chevron symbol when active. In addition, if sidereal tracking is active, the **Motion Status** field will hold **TRACK+f**.

Errors: No errors are associated with this option.

Rates - PEC

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction to the standard sidereal rate necessary to correct for periodic error in gear driven systems. This function requires special custom hardware to be installed on the worm gears of the telescope drive system and is therefore optional. This item is only functional when sidereal tracking is turned on.

As of version 4.0 this option is still under development and is unavailable to all users except those with Level 4 access.

Uses: Useful in maintaining tracking integrity over extended observing periods where periodic in the drive system may cause significant errors in tracking.

Indicators: The **PEC** menu item in the **Rates** menu will be preceded with a sideways chevron symbol when active.

Errors: No errors are associated with this option.

3.6 The Source Submenu

Keyboard PCF Catalog User Catalog

The **Source** Submenu contains three options pertaining to the source of Next object coordinates. The user has a choice of manual (**Keyboard**) data entry, or data stored in various system catalogs, or a User generated file created using the PC-TCS Catalog Generation Utility.

Source - Keyboard

Direct Key Access: *Alt K*

Functionality: Sets the source of **Next** object coordinates to be the user keyboard. With this option set, all **Next** position data (see **Position** submenu) will be taken from the keyboard as opposed to a file. The observer should also take care to set the proper precessional epoch for the coordinates entered in this mode.

Uses: Used to set the source of **Next** object coordinates to be from the user keyboard. Useful when manual entry of coordinates is necessary.

Indicators: The **Coordinate Mode** field on the Runtime Display will show **Keyboard** and the **Coordinate File** field will be blanked.

Errors: None are associated with this menu item.

Source – PCF Catalog

Direct Key Access: *Alt P*

Functionality: Sets the source of **Next** position coordinates to be a System Catalog file on disk. When selected, a window containing the file names of the available system catalogs will appear. The user selects the desired file using the arrow keys and confirms the selection using the ← key.

See File Window Documentation in Chapter 2.

In this other file oriented source modes the user has only to enter the object number when using the **POSITION – NEXT** menu item. When data is processed after a Next Position request, the object number will appear immediately to the right of the **NEXT** label. The processed coordinates will be displayed in their appropriate columns, along with the magnitude/ spectral class in the label field of the Runtime display.

Uses: Used to set the source of retrieved coordinate data from one of the System catalog files stars.

Indicators: The **Coordinate Mode** field of the Runtime Display will show **File** and the **Coordinate File** field will display the name of the PCF catalog as the file name.

Errors: If there are no system catalog files found on disk or if there is an error in opening the selected catalog file, an error window will appear and/or a bell will sound.

Source – User Catalog

Direct Key Access: *Alt U*

Functionality: Sets the source of **Next** position coordinates to be a user specified catalog on disk. In this and other file oriented source modes the user has only to enter the object number to set a new **Next** position.

See File Window Documentation in Chapter 2.

Uses: Used to retrieve coordinate data from the User generated catalogs star catalogs. (See **Utilities** subsubsubmenu and Chapter 4 under Generating Catalogs.)

Indicators: A window will be displayed showing a list of all user catalog files currently available. The user selects the desired file using the arrow keys and confirms the selection using the ← key. The **Coordinate Mode** field of the Runtime Display will show **File** and the **Coordinate File** field will display ?????????.CAT (where ???????? is the file name selected by the user). When data is processed after a **POSITION – NEXT** declaration, the object number will appear immediately to the right of the **NEXT** label. The processed coordinates will be displayed in their appropriate columns, along with the magnitude and a label in the appropriate fields on the Runtime display.

Errors: If the user specified file can not be found, an error window will appear and/or a bell will sound.

3.7 The Time Submenu

UT Date UT Time
UTC Disable UTC resync

The **Time** submenu contains two items used to modify the system **UT Date** and **UT Time**. PC-TCS uses universal Time (Greenwich Time) exclusively and the user should take care in making sure that the system date and time are correct at system startup. Failure to do so will result in poor pointing performance and may compromise the software limits detection system of PC-TCS.

Optionally, this Submenu may contain two additional menu items – pertaining to the operation of the external UTC Clock provided as an option to PC-TCS users.

Time – UT Date

Direct Key Access: None

Functionality: Allows the user to set the Universal Date that PC-TCS will use to compute Local Sidereal time and telescope position. It is up to the user to know/set this date if proper system operation is to be maintained. Upon startup, PC-TCS will retrieve the Date and Time from the PC system clock and the optional UTC Clock if installed. If this date is correct it may be corrected (for PC-TCS use only) via this menu item. (Ultimately the Time and Date for the PC system clock should be correctly set. See your DOS and/or system hardware/software reference manual for how to do this.)

Uses: Useful in correcting the displayed PC-TCS date to the correct Universal Date.

Indicators: The **UT DATE** field will be updated with the user given UT date. The Current Position and LST fields may also change to reflect the new information.

Errors: Nonnumeric data entry will cause a bell to sound. Illegal Dates will display an error window and/or cause a bell to sound.

Time – UT Time

Direct Key Access: None

Functionality: Allows the user to get the Universal time that PC-TCS will use to compute Local Sidereal Time and telescope position. It is up to the user to know/set this time if proper system operation is to be maintained. Upon boot PC-TCS will retrieve the Date and Time from the PC system clock or the optional UTC Clock if installed. If this time is incorrect it may be corrected (for PC-TCS use only) via this menu item. (Ultimately the Time and Date for the PC system clock should be correctly set. See your DOS and/or system hardware/software reference manual for instructions on how to do this.)

Note: Failure to set the correct UT Time will have the effect of producing poor telescope pointing and may cause inaccuracies in the software limit detection algorithms – leading to possible DAMAGE to your telescope.

Uses: Useful in correcting the displayed PC-TCS time to the correct Universal Time.

Indicators: The UT time field will be updated with the user given UT time. The Current Position and LST fields may also change to reflect the new information.

Errors: Nonnumeric data entry will cause a bell to sound. Illegal times will display an error window and/or cause a bell to sound.

Time – UTC Disable

Direct Key Access: None

Functionality: Disables/ Enables communications with the optional external UT Clock. PC-TCS can communicate with an external UT clock capable of keeping Universal Time to the nearest millisecond. When installed, this option allows PC-TCS to automatically “resync” its internal time based to an accurate time source. In normal operation PC-TCS attempts resync to this clock at program start and every 10 minutes of elapsed run time. The communication is accomplished through one of the serial ports available to most PC’s. If this option is not available – or communications are not possible because of hardware failure this item can be used to disable this failure.

Uses: Used to enable/ disable the optional external UT Clock communications.

Indicators: If menu item is enabled and the UT Clock communications are not functioning properly – The UTC label on the Runtime Display will flash red. If the menu item is disabled or enabled and communications are successful the UTC label will be green in color.

Errors: The UTC label on the Runtime display will flash red.

Time – UTC Resync

Direct Key Access: None

Functionality: As described in the previous menu item – PC-TCS can communicate to an optional external UT clock that provides time accurate to 1 millisecond. If enabled via the previous menu item, PC-TCS will attempt to resync to this clock at start-up and every 10 minutes of PC-TCS run time. This menu item can be used to manually “force” a resync.

Uses: Useful in forcing PC-TCS to resync to the optional external UT Clock in the event of a communication failure during a previous automatic resync attempt or after a bad start-up resync attempt.

Indicators: If the **UTC Disable** menu item is disabled and the UT Clock communications are not functioning properly – The UTC label on the Runtime Display will flash red. If **UTC Disable** menu item is disabled or enabled and communications are successful the UTC label will be green in color.

Errors: The UTC label on the Runtime display will flash red.

3.8 The Parameters Submenu

Input Globals Utilities

The **Parameters** submenu has no directly executable items, instead contains a potpourri of subsubsubmenus that allow the user access to the more intricate and powerful options and utilities of PC-TCS.

The three subsubsubmenus are: **Input** – used to modify the source and content of input into PC-TCS, **Globals** – used to change operational attributes of the drive system and user interface, and **Utilities** – used to gain access to the powerful subprograms and functions of PC-TCS.

3.8.1 The Input Subsubmenu

Corrections Remote Command I/O Externals
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The **Input** subsubmenu, contains four subsubsubmenus. The subsubsubmenus provide user access to PC-TCS's coordinate processing functions, as well as control of the remote serial and command file functions and Hardware input and output control functions.

The Corrections Subsubsubmenus

Proper Motion
Precession
Nutation
Aberration
Refraction
Flexure
Parallax

The **Corrections** subsubsubmenu allows the user to selectively enable or disable additive corrections made to the coordinate retrieved from the coordinate source specified in the **Source** subsubsubmenu. The corrections compensate for perturbations in the coordinates ranging from the earth's rotational axis precession to the refraction of light as it through the earth's atmosphere.

Corrections – Proper Motion

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction associated with proper motion during processing of **Next** coordinates. This correction and others (if enabled) will be added to the **Next** objects coordinates when defined. The proper motion of the objects is obtained from the catalog disk file if the **Coordinate Mode** is set to **File**, or optionally retrieved from the user if the mode is set to **Keyboard**. Proper Motion is defined as the apparent motion of a star with respect to the True of Date equatorial coordinate system due to the star's motion through space relative to the sun; usually measured in arcseconds per year century.

Uses: Useful in disabling the Proper Motion PC-TCS query and correction while in **Keyboard** mode if this correction is determined to be necessary for accurate pointing or is unknown to the user.

Indicators: The sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition the anagram “M” will be displayed after the File Coordinate Epoch field.

Errors: No error windows are associated with this menu item.

Corrections – Precession

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction associated with precession during processing of **Next** coordinates. This correction and others (if enabled) will be added to the **Next** objects coordinates when entered to produce “telescope relative” coordinates for accurate telescope pointing. The precession , the object is computed from the epoch information stored in the catalog disk file if the **Coordinate Mode** is set to **File** or must be entered manually by the user when processing a **POSITION – NEXT** menu item if the mode is set to **Keyboard**. Precession is defined as the apparent motion of the coordinate system on the sky due to gyroscopic tumbling of the earth’s axis of rotation. The effect can be very significant for coordinate epoch differences spanning more than a few years. (Note: the correction for nutation is also disabled if the correction for precession is turned off.)

Uses: Useful in disabling the Precessional correction computations if coordinate epochs are not known by the user in **Keyboard** mode.

Indicators: The sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition the anagram “P” will be displayed after the File Coordinate Epoch field.

Errors: No error windows are associated with this menu item.

Corrections – Nutation

Direct Key Access: None

Functionality: Allows the user to enable / disable the correction associated with nutation during processing of **Next** coordinates. This correction and others (if enabled) will be added to the **Next** objects coordinates when entered to produce “telescope relative” coordinates for accurate telescope pointing. The nutation correction for an object is computed from the date/time information available to PC-TCS. Nutation is the perturbation of the equatorial coordinate system by the moon’s gravitation influence over a 19 year period and has a maximum deviating effect of 9.23 arseconds. (Note: This correction can only be turned on if the correction for precession is also enabled.)

Uses: useful in disabling the Nutation correction computations if this computation is determined to be unnecessary by the user.

Indicators: The sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition the anagram “N” will be displayed after the File Coordination Epoch field.

Errors: No error windows are associated with this menu item.

Corrections – Aberration

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction associated with aberration during processing of **Next** coordinates. This correction and others (if enabled) will be added to the **Next** objects coordinates when entered to produce “telescope relative” coordinates for accurate telescope pointing. The aberration correction for an object is derived from the position/time information computed motion of the earth as it revolves about the sun (also known as *annual aberration*). The effect can be as great as 20.48 arcseconds for any given position in the sky at any given time.

Uses: Useful in disabling the Aberration correction computation if its computation is determined to be unnecessary by the user.

Indicators: the sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition the anagram “A” will be displayed after the File Coordinate Epoch field.

Errors: No error windows are associated with this menu item.

Corrections – Refraction

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction with Atmospheric Refraction during processing of **Next** coordinates. This correction and others (if enabled) will be added to the **Next** objects coordinates when entered to produce “telescope relative” coordinates for accurate telescope pointing. The refraction correction for an object is computed from the elevation/ position information available to PC-TCS. Refraction tends to make an object appear higher in the sky than its true elevation. The effect can be as 12 arcminutes within the operating limits assigned by PC-TCS. The closer to the horizon an object is the greater the effect of refraction.

Uses: Useful in disabling the Refraction correction computations if this computation is determined to be unnecessary by the user.

Indicators: The sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition the anagram “R” will be displayed after the File Coordination Epoch field.

Errors: No error windows are associated with this item.

Corrections – Flexure

Direct Key Access: None

Functionality: Allows the user to enable/ disable the correction associated with telescope flexure during process of **Next** coordinates. This correction and others (if enabled) will be added to the **Next** objects coordinates when entered to produce “telescope relative” coordinates for accurate telescope pointing. The flexure correction for an object is derived from the position information available to PC-TCS and read from a flexure disk file called PCTCS.FLX. Flexure is the deviation of the true optical path of the telescope due to flexure in the materials that make up its support structure. The magnitudes flexure may attain are different from one telescope to the next and typically change with time. Therefore, flexure (as used by PC-TCS) is determined empirically via analysis of data collected specifically for its purpose. (See Chapter 4 under Flexure Data Collection.)

Uses: Useful in disabling the Flexure correction computations if this computation is determined to be necessary or inaccurate by the user, or the flexure file becomes unreadable by PC-TCS.

Indicators: The sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition, the anagram “F” will be displayed along with the active map mnemonic (*,a,b,c, or d) after the File Coordinate Epoch field on the Runtime Display.

Errors: If PC-TCS is unable to find file “PCTCS.FLX” on disk, an error window and/or a bell will sound.

Corrections – Parallax

Direct Key Access: None

Functionality: Allows the user to enable/ disable correction associated with parallax during processing of **Next** coordinates associated with solar system objects. This correction and others (if enabled) will be added to the **Next** objects coordinates when entered to produce “telescope relative” coordinates for accurate telescope pointing. The parallax correction for an object is derived from the position information available to PC-TCS and computed based on the object’s distance from earth.

Uses: Useful in correcting the computed positions of solar system objects for parallax of the telescope topocentric observing site.

Indicators: The sideways chevron symbol (>>) will prefix the menu item if the correction is enabled. In addition, the anagram “p” will be displayed after the File Coordination Epoch field.

Errors: None are associated with this menu item.

The Remote Subsubsubmenu

Active
COM1 COM2 COM3 COM4 COM5 COM6 COM7 COM8
Checksum Baudrate Error log
Level Wobble

This subsubsubmenu allows the user to control the Remote communications function which utilizes the optional RS232 serial communications hardware found on most PC class machines. Currently, up to eight channels of external communications are supported by PC-TCS. Via the twelve serial port related menu items found here, the user can enable and disable remote access to PC-TCS; enable and disable the security checksum and error logging features, and change the speed at which serial communications are carried out. The fourteenth menu item allows the user to enable/ disable the hardware wobble command input. This input is used to allow external hardware to wobble the telescope with a simple TTL signal.

Remote - Active

Direct Key Access: *Alt I* and *Alt H* (hide/ unhide)

Functionality: Option “activates” the Remote serial interface channels selected by the user in the menu items COM1 through COM8. The serial interface provides remote control capabilities via up to 8 optional RS232 serial ports available to the PC (see Chapter 4-**Remote Serial Access**). When activated the selected serial ports will begin to accept command input and transmit a telemetry stream containing the most critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection to PC-TCS. Up to 8 simultaneous channels can be active if the appropriate hardware is installed.

Indicators: A window will appear when this menu item is activated which displays pertinent Remote information (See section 2.2- The PC-TCS Runtime Display.) In addition, a sideways chevron will prefix this item when active. When active the communications status window may be “hidden” by using the *Alt H* hotkey toggle. An additional *Alt H* will “unhide” the window. Hiding the window has now effect on the operation of remote serial access.

Errors: If no serial ports have been selected an error window will appear and/or a bell will sound.

Remote – COM1

Direct Key Access: None

Functionality: Option enables/ disables the COM1 Remote serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS232 serial ports available to the PC (see Chapter 4 –**Remote Serial Access**). When activated the COM1 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated the COM1 serial port will begin to accept serial commands and broadcast a telemetry stream containing the critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Note: operational baud rate may be selected using the **Baudrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM1. Up to eight channels can be active simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window displays pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.) In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/ or an error window will be displayed describing the problem.

Remote - COM2

Direct Key Access: None

Functionality: Option enables/ disables the COM2 Remotes serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS232 serial ports available to the PC (see Chapter 4 –**Remote Serial Access**). When activated the COM2 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated the COM2 serial port will begin to accept serial commands and broadcast a telemetry stream containing the critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Note: operational baud rate may be selected using the **Baudrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM2. Up to eight channels can be active simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window displays pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.) In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/ or an error window will be displayed describing the problem.

Remote – COM3

Direct Key Access: None

Functionality: Option enables/ disables the COM3 remote serial interface lines. The serial provides remote control capabilities via up to eight optional RS232 serial ports available to the PC (see Chapter 4 – **Remote Serial Access**). When activated, the COM3 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated, the COM3 serial port will begin to accept serial commands and broadcast a telemetry stream containing the critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Note: Operational baud rate may be selected using the **Baudrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM3. Up to eight channels can be activated simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window display pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.). In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/or an error window will be displayed describing the problem.

Remote – COM4

Direct Key Access: None

Functionality: Option enables/ disables the COM4 Remote serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS 232 serial port available to the PC (see Chapter 4 – **Remote Serial Access**). When activated, the COM4 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated, the COM4 serial port will begin to accept serial commands and broadcast a telemetry stream containing critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Not: Operational baud rate may be selected using the **Buadrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM4. Up to eight channels can be activate simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window display pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.). In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/or an error window will be displayed describing the problem.

Remote – COM5

Direct Key Access: None

Functionality: Option enables/ disables the COM5 Remote serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS 232 serial port available to the PC (see Chapter 4 – **Remote Serial Access**). When activated, the COM5 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated, the COM5 serial port will begin to accept serial commands and broadcast a telemetry stream containing critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Not: Operational baud rate may be selected using the **Buadrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM5. Up to eight channels can be activate simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window display pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.). In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/or an error window will be displayed describing the problem.

Remote – COM6

Direct Key Access: None

Functionality: Option enables/ disables the COM6 Remote serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS 232 serial port available to the PC (see Chapter 4 – **Remote Serial Access**). When activated, the COM6 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated, the COM6 serial port will begin to accept serial commands and broadcast a telemetry stream containing critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Not: Operational baud rate may be selected using the **Buadrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM6. Up to eight channels can be activate simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window display pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.). In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/or an error window will be displayed describing the problem.

Remote – COM7

Direct Key Access: None

Functionality: Option enables/ disables the COM7 Remote serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS 232 serial port available to the PC (see Chapter 4 – **Remote Serial Access**). When activated, the COM7 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated, the COM7 serial port will begin to accept serial commands and broadcast a telemetry stream containing critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Not: Operational baud rate may be selected using the **Buadrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM7. Up to eight channels can be activate simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window display pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.). In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/or an error window will be displayed describing the problem.

Remote – COM8

Direct Key Access: None

Functionality: Option enables/ disables the COM8 Remote serial interface lines. The serial interface provides remote control capabilities via up to eight optional RS 232 serial port available to the PC (see Chapter 4 – **Remote Serial Access**). When activated, the COM8 serial port will be selected for active remote serial access. Activation of selected ports is accomplished with the **Active** menu item. Once activated, the COM8 serial port will begin to accept serial commands and broadcast a telemetry stream containing critical information found on the Runtime Display. Remote command mnemonics are documented in Appendix B and the layout of the serial telemetry stream in Appendix C.

Not: Operational baud rate may be selected using the **Buadrate** menu item found in this subsubsubmenu.

Uses: Used to allow remote access to PC-TCS via modem or direct serial connection on COM8. Up to eight channels can be activate simultaneously if the appropriate hardware is installed.

Indicators: A window will appear when remote communications are activated via the **Active** menu item. The Window display pertinent Remote communications information (See section 2.2 – The PC-TCS Runtime Display.). In addition, a sideways chevron will prefix this item when selected.

Errors: If the serial port is currently in use by another device or is not physically present – a bell will sound and/or an error window will be displayed describing the problem.

Remote - Checksum

Direct Key Access: None

Functionality: Allows the user to enable/ disable the checksumming of remote serial command mnemonics. Serial commands are all terminated with a ← character (13 decimal). When this character is detected, PC-TCS will perform a sum check on previously received serial data before processing the command. If the computed sum differs from the sum received with the command, processing of the command is terminated without further examination.
A sample program for remote command checksum generation is outlined in Appendix D.

Uses: Checksumming is useful in protecting PC-TCS from illegal access or noisy communication hardware that may generate unwanted commands.

Indicators: A sideways chevron will prefix the menu item when active. In addition, if the remote communication have been enabled via the **Remote** subsubsubmenu **Active** command, the **Checksum status** field will indicate active by displaying **YES**.

Errors: No errors are associated with this menu item.

Remote - Baudrate

Direct Key Access: None

Functionality: Allows the user to modify the communication “speed” of the remote serial interface lines. Baudrate is a computer term used to indicate the number of bits per second the hardware is set to communicate. (e.g. 1200 baud is equivalent to 120 characters per second receive and transmit speed.) Speeds of up to 19200 baud can be selected via this menu option. Note: Setting a speed while the remote is active is possible but may disrupt modem carrier signals.

Uses: Used to match PC-TCS communication speed with the modem or remote device.

Indicators: When selected a window will appear with a selection of possible baud rates.

Errors: If an option number is selected that is not listed on the data window an error window and/or a bell will sound.

Remote – Error Log

Direct Key Access: None

Functionality: Allows the user to enable/ disable error logging of remote serial communications. When enabled PC-TCS will log all communications errors to a file on disk called “ERROR LOG”. This file may be examined later for the causes of communications errors. The file contains one line per error event per channel with the following format ...

Time COM# Errorcode Command

Uses: Useful in debugging remote serial communications problems with PC-TCS.

Indicators: When enabled a sideways chevron (>>) will appear to the left of the Error log menu item.

Uses: If the error logging file can be opened – an error will appear and/or a bell will sound.

Remote - Level

Direct Key Access: None

Functionality: This and the following menu item pertain to the remote Hardware Wobbling function available through PC-TCS. The user can “wobble” (see the **Wobble** menu item in the **Declare** submenu) the telescope in one of four ways – via the keyboard using the “Move beam” commands as documented in the **Move** submenu; the command **MOVB1** and **MOVB2** in either the remote serial or Command File modes; or via a TTL hardware signal. The hardware signal can either be a TTL “pulse” or a TTL level change.

This menu item when selected will enable the TTL “level” operation of the hardware wobble. In this mode a change from TTL high to TTL low or a change from TTL low to TTL high will cause the telescope to move to the next consecutive “beam” of the wobble vector previously declared via the **Declare-Wobble** menu item.

When this menu item is deactivated – a single TTL pulse will move the telescope to the next consecutive “beam”

Note: This menu item only selects the hardware wobble “mode” and will not “activate” the hardware wobble functionality. Activation of the hardware wobble functionality must be accomplished using the **Remote-Wobble** menu item documented on the following page of this manual.

Uses: Useful in automating telescope wobbling via instrument or remote computer signals without having to implement the full serial communications software/hardware.

Indicators: A bracketed “L” will appear immediately to the left of the **Wobble Vector** label field on the runtime display when the hardware level wobble feature is enabled – otherwise a “P” indicates the “pulse” wobble mode. A sideways chevron symbol (>>) will prefix this menu item when level mode is enabled.

Errors: None are associated with this option.

Remote - Wobble

Direct Key Access: *Alt W*

Functionality: Allows the user to enable/ disable the operation of the hardware wobble input. Bit 7 of the limit port is used as a level trigger for telescope wobbling. An external device that can supply a simple TTL level to this input can alternately command the telescope to move to consecutive beams. Enabling this feature does *not* affect the operation of keyboard, serial remote or command file wobble commands. The user should be careful not to mix the command source since the hardware wobble has no way of determining which beam is currently selected.

Uses: Useful in automating telescope wobbling via instrument or remote computer signals without having to implement the full serial communications software/hardware.

Indicators: A bracketed “L” or “P” will appear immediately to the left of the **Wobble Vector** label field depending on which mode of hardware wobble operation has been selected (see **Remote-Level** menu item described on the previous page) when the hardware wobble feature is enabled. A sideways chevron symbol (>>) will prefix the menu item when enabled.

Errors: If no wobble vector has been setup – an error will appear and/or a bell will sound.

The Command Subsubsubmenu

File Active

The **Command** subsubsubmenu allows the user to control the operation of the command file utility available to PC-TCS. Command files are disk files created by the user, containing special mnemonics interpreted by PC-TCS in much the same way as keyboard commands are interpreted. The files can be used to automate activity that would be tedious or repetitive by hand. An entire night's observing can even be "programmed" in advance.

Options within this menu include an item to activate/ deactivate (**Active**) and execution of command file commands, and an item to change the command file (**File**) name.

Note: PC-TCS normally looks for Command files on the default disk directory from which PC-TCS was executed. If the user wishes to use a different disk or subdirectory he may do so by selecting the TCSCFP environment variable B E F O R E PC-TCS execution.

The command syntax is as follows ...

At the DOS prompt type SET TCSCFP= *[path]* where *[path]* is the desired drive/ path combination where PC-TCS should look for all Command files. For example, if SET TCSCFP= *A* :

Note: No spaces or tabs should be inserted between the equal sign, the variable name and the drive/ path specifier.

Command - File

Direct Key Access: None

Functionality: Allows the user to specify the file name to be used in Command file processing. Command files are a powerful PC-TCS tool for automating parts of, or an entire observing session via user created disk files. Command files must conform to the command syntax and structure described in Chapter 4 and Appendix A of this manual. In addition, the file name must have the three character extension of “TCS”.

Command Files are read in their entirety from disk into PC-TCS memory – no attempt is made to verify command syntax until the Command File is activated. Therefore, the user should verify that enough memory is available to PC-TCS to load the file and that the file command syntax is correct.

The user may set a different disk or path from which to read Command Files by setting the TCSCFP environment variable – see the introductory page on this subsubsubmenu for details on defining a different path for Command files.

Uses: To specify/ change and preprocess a user command file.

Indicators: A window will be displayed showing a list of all command files currently available. The user selects the desired command file using the arrow keys and confirms the selection using the ← key. The **Command File** line of the Runtime display will be updated with the file name.

Indicators: The user selected file name will appear in the **Command File** field of the Runtime Display.

Errors: If the file specified can not be found or read properly, an error window will appear and/or a bell will sound.

Command - File

Direct Key Access: *Ctrl A*

Functionality: Option “activates/ deactivates” the execution of Command File processing. Command Files provide automated control of PC-TCS via a user created file containing one or more of the PC-TCS commands outlined in Appendix A.

Uses: Useful in automating PC-TCS functions that might be tedious or repetitive if execution from the keyboard.

Indicators: The file name specified in the **Command File** field will begin to flash when PC-TCS is processing commands and the **Command Count** will increment after each command is executed.

Errors: If no command file has been specified, an error window will appear and/or a bell will sound.

The I/O Externals Subsubsubmenu

Pad/Lim Input
HS/Extra Input
TTL1 Output
TTL2 Output
Enable ADC
Current ADC
Joy-stick ADC

The **I/O Externals** subsubsubmenu contains two items which allow the user to examine and modify the states of the digital inputs associated with the paddle and limits hardware. The two items are **Pad/Lim Input** which is used to debug digital problems with the limits and paddle related input hardware and **HS/Extra Input** which relate to the Hardware Stow and Extra TTL input hardware.

TTL and **TTL2 Output** menu items allow the user to modify the TTL output signals which control various custom installation functions such as Servo Amplifier control lines; focus motors or dome control lines, etc.

The three item relating to **ADC** (Analog to Digital) functions control access and monitoring of various custom analog input functions such as servo and encoder current, analog joy-stick input, etc.

I / O Externals – Pad / Lim Input

Direct Key Access: Note

Functionality: Allows the user to view and modify the PC-TCS interpretation of the digital inputs associated with the paddle and limits TTL hardware signals. Two windows are displayed, the upper showing the signal states of the individual “bits” being read by PC-TCS. Vertical arrows (↓ or ↑) in this left window indicate the logical value of the input — ↑ for high or true, and ↓ for low or false. The values displayed are those after the logical masks displayed in the lower window have been applied. The user can set the masks to any binary bit pattern to “xor” (invert), “and” and “or” the input into whatever state is desired.

Note: If your version of PC-TCS is setup for a “Slew Permit” button and is not activated by the F8 hotkey, this menu item may be used to permanently enable or disable the Slew Permit capability by simply “or-ing” on or “and-ing” off the “SIP” bit.

Uses: Useful in disabling bad or intermittent paddle/ limit input or turn on/ off an input permanently. The “xor” function can be used to invert the sense of the inputs for inverted external logic.

Indicators: The logic “arrows” may assume certain states depending upon the masks set. Masks do not go into effect until all masks are defined and a terminating← has been entered.

Errors: Nonnumeric data will cause a bell to sound. Nonbinary data will also cause a bell to sound at data entry completion.

I / O Externals – NS / Extra Input

Direct Key Access: None

Functionality: Allows the user to view and modify the PC-TCS interpretation of the digital inputs associated with the Hardware Stow and Extra hardware inputs. Two windows are displayed, the upper window displays the signal states of the individual “bits” being read by PC-TCS. Vertical arrows (↓ or ↑) in this left window indicate the logical value of the input — ↑ for high or true, and ↓ for low or false. The values displayed are those *after* the logical masks displayed in the lower window have been applied. The user can set the masks to any binary bit pattern to “xor” (invert), “and” and “or” the input into whatever state is desired.

Uses: Useful in disabling bad or intermittent input or turn on/ off an input permanently.

Indicators: The logic “arrows” may assume certain states depending upon the masks set. Masks do not go into effect until all masks fields are defined and a terminating ← has been entered.

Errors: Nonnumeric data will cause a bell to sound. Nonbinary data will also cause a bell to sound at data entry completion.

I / O Externals – TTL1 Output

Direct Key Access: None

Functionality: Allows the user to view and modify the PC-TCS interpretation of the digital outputs associated with the TTL1 Port Hardware. Two windows are displayed, the upper window displays the signal states of the individual “bits” being output by PC-TCS. Vertical arrows (↓ or ↑) in this left window indicate the logical value of the output — ↑ for high or true, and ↓ for low or false. The values displayed are those *after* the logical masks displayed in the lower window have been applied. The user can set the masks to any binary bit pattern to “xor” (invert), “and” and “or” the output into whatever state is desired.

Uses: Useful in modifying output values associated with TTL1.

Indicators: The logic “arrows” may assume certain states depending upon the masks set. Masks do not go into effect until all masks fields are defined and a terminating ← has been entered.

Errors: Nonnumeric data will cause a bell to sound. Nonbinary data will also cause a bell to sound at the entry completion.

I / O Externals – TTL2 Output

Direct Key Access: None

Functionality: Allows the user to view and modify the PC-TCS interpretation of the digital outputs associated with the TTL2 Port Hardware. Two windows are displayed, the upper window displays the signal states of the individual “bits” being output by PC-TCS. Vertical arrows (↓ or ↑) in this left window indicate the logical value of the output — ↑ for high or true, and ↓ for low or false. The values displayed are those *after* the logical masks displayed in the lower window have been applied. The user can set the masks to any binary bit pattern to “xor” (invert), “and” and “or” the output into whatever state is desired.

Uses: Useful in modifying output values associated with TTL2.

Indicators: The logic “arrows” may assume certain states depending upon the masks set. Masks do not go into effect until all masks fields are defined and terminating ← has been entered.

Errors: Nonnumeric data will cause a bell to sound. Nonbinary data will also cause a bell to sound at data entry completion.

I /O Externals – Enable ADC

Direct Key Access: None

Functionality: Enables/ disables the sensing and decoding of the optional Analog to Digital circuitry on the PC-TCS hardware interface card. This optional hardware, when installed, can be used to interface to external analog devices such as a joy-stick, servo and encoder lamp current monitors, etc.

Uses: Useful in enabling/ disabling PC-TCS sensing of ADC input. Problems involving the sensing of optional joy-stick and encoder lamp current sensing hardware may be masked by disabling this function.

Indicators: A sideways chevron (>>) will prefix the menu item when enabled.

Errors: None are associated with this menu item.

I / O Externals – Current ADC

Direct Key Access: *Alt C*

Functionality: Allows the user to monitor the ADC values associated with the optional Servo Drive current and Encoder Lamp Current monitors.

Uses: Useful in monitoring the servo current while balancing the telescope and diagnosis of encoder lamp failures.

Indicators: A window will appear displaying the currents associated with the servo drive system and the Encoder lamps.

Errors: None are associated with this menu item.

I / O Externals – Joy-stick ADC

Direct Key Access: None

Functionality: Allows the user to monitor the ADC values associated with the optional Joy-stick available to PC-TCS.

Uses: Allows the user to monitor the ADC values associated with the optional Joy-stick available to PC-TCS.

Indicators: A window will appear displaying the currents associated with the Joy-stick analog input.

Errors: None are associated with this menu item.

3.8.2 The Globals Subsubmenu

Display Cosine δ Limits Flexclass Paths Servo
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The **Globals** menu contains six items, four of which are subsubsubmenus, that allow the user to modify system data or parameters that have a global effect upon PC-TCS operation. The three subsubsubmenus are **Display**, which can be used to change the Display epoch and deconvolutions and the way errors are reported. The **Cosine δ** subsubsubmenu is used to enable and disable the correction for declination upon various user definable rates and the wobble vector. The **Limits** subsubsubmenu is used to modify the positional limits associated with telescope motion. The Flexclass subsubsubmenu is used to change the flexure maps sets – useful in setting up multiple maps for various instrument/ secondary configurations that alter telescope flexure.

The **Paths** item is used to change the data paths to catalog; command and track files. The **Servo** item is used to perform low level tuning of the servo algorithm internal to PC-TCS. This item is “locked out” from access by all but authorized users via the security system of PC-TCS. Special permission is necessary to use this utility since random modification of the servo parameters can result in damage to the telescope drive system hardware.

The Display Subsubsubmenu

ePoch
Aberration
Refraction
Flexure
Diff Sec
rIse/set
rates
Errors

The **Display** subsubsubmenu contains four items pertaining to the modification of the coordinate display on the runtime menu. **Epoch**, **Aberration Refraction and Flexure** are used to enable/ disable the deconvolution of raw telescope mount coordinates into astrometric coordinate display.

The **Diff Sec** menu item modifies the way the RA Difference field is displayed on the Runtime Display. **rIse / set** toggles the display the time till rise/ set of the **Current**, **Next**, and **Reference** targets. **raTes** will toggle the display of correction rates currently being added to the default sidereal tracking rate.

The **Errors** display modifies the way errors are presented to the user.

Display – epoch

Direct Key Access: *Ctrl E*

Functionality: Allows the user to change the display epoch denoted by the **Epoch** field of the Runtime Display. The display epoch is the coordinate epoch date used to precess coordinates for *display only*. Modifying this epoch has no effect on actual telescope position. The value is used to precess the current position, Next position, and Reference position and updates to these fields are made anytime a new Display epoch value is entered.

Uses: Useful in displaying the epoch coordinates of the telescope where semi-astrometric coordinates are required. (e.g Charting the position of a comet on a 1950 star atlas.) Note: PC-TCS Version 3.x can display true “astrometric” coordinates if all deconvolution options are turned on – otherwise the current RA and DEC display fields will show telescope-relative coordinates.
Note: The value “TOD” in the Epoch field of the Runtime display indicates an epoch value of “True Of Date” – the epoch based on the current date and time.

Indicators: The **Epoch** field of the Runtime Display will be updated with the data entered. The current, Next and Reference position will be updated to reflect the new display epoch.

Errors: Illegal dates will show an error window and/ or a bell will sound. Nonnumeric data will also cause a bell to sound.

Display – Aberration

Direct Key Access: None

Functionality: Enables/ disables the deconvolution for Aberration of the Current, Next and Reference RA and DEC displays on the Runtime Display.
This item must be “active” for astrometric coordinate display.

Uses: Useful enabling/ disabling astrometric telescope coordinate display of telescope position.

Indicators: The sideways chevron (>>) will prefix the menu item when active. In addition the anagram “A” will be displayed immediately preceeding Current Display Epoch field.

Errors: No errors are associated with this option.

Display – Refraction

Direct Key Access: None

Functionality: Enables/ disables the deconvolution for atmospheric Refraction of the Current, Next and Reference RA and DEC displays on the Runtime Display.
This item must be “active” for astrometric coordinate display.

Uses: Useful enabling/ disabling astrometric telescope coordinate display of telescope position.

Indicators: The sideways chevron (>>) will prefix the menu item when active. In addition the anagram “R” will be displayed immediately preceeding Current Display Epoch field.

Errors: No errors are associated with this option.

Display – Flexure

Direct Keys Access: None

Functionality: Enables/ disables the deconvolution for Flexure of the Current, Next and Reference RA and DEC displays on the Runtime Display. This item must be “active” for astrometric coordinate display.

Uses: Useful enabling/ disabling astrometric telescope coordinate display of telescope position.

Indicators: The sideways chevron (>>) will prefix the menu item when active. In addition the anagram “F” will be displayed immediately preceeding Current Display Epoch field.

Errors: If the flexure data files can not be opened or read – an error window will appear and / or a bell will sound

Display – Diff Sec

Direct Key Access: None

Functionality: Allows the user to modify the way PC-TCS displays the RA difference field of telescope motion. If the item is “active” the RA difference field on the Runtime display will display RA differences in “time” the form of HH:MM:SS.ss. If “deactivated” the RA difference display will show differences in arcseconds on the sky (accounting for the cosine of Declination) in the form SSSSS.s.

Uses: Useful in display “sky” relative displacement in RA of the telescope.

Indicators: The sideways chevron (>>) will prefix the menu item when active.

Errors: No errors are associated with this option.

Display – rIse / set

Direct Key Access: *F4*

Functionality: Enables/ disables the display of “ time to Rise/ Set” for the telescope’s Current, Next and Reference positions. If the item is activated – a window will be opened displaying the rise and or set delta times on the Runtime display.

Uses: Useful in determining the time remaining to observe an object or the time until an object becomes visible.

Indicators: The sideways chevron (>>) will prefix the menu item when active, and a window will be opened on the runtime display showing the rise set delta times.

Errors: No errors are associated with this option.

Display – rates

Direct Key Access: *F3*

Functionality: Enables/ disables the display of correction rates related to refraction, flexure and any object rate being added to the default sidereal track rate.

Uses: Useful in determining the additive rates to the default sidereal rate. All rates are in arcseconds per second.

Indicators: The sideways chevron (>>) will prefix the menu item when active, and a window will be opened on the runtime display overlaying the bias, drift and guide rates.

Errors: No errors are associated with this option.

Display – Errors

Direct Key Access: None

Functionality: Allows the user to modify the way PC-TCS errors are reported. If the item is activated – errors will be reported both visually and audibly via an error window and a bell. If deactivated, only the bell sound will designate errors.

Note: As of version 3.5 – PC-TCS will **NOT** report errors visually if the remote serial access is active.

Uses: Useful in setting an “expert” or “no help” mode of PC-TCS operation.

Indicators: The sideways chevron (>>) will prefix the menu item when active.

Errors: No errors are associated with this option.

The Cosine δ Subsubsubmenu

Wobble Bias Guide Drift

The **Cosine** δ subsubsubmenu contains four items. Each item indicates the rate or vector that can be corrected for changes in declination. As the declination of the telescope increases (or decreases) from zero, the rate or distance in Right Ascension must be modified to maintain the same magnitude on the sky. This is accomplished by dividing the associated Right Ascension Component of the selected item by the cosine of the declination.

Cosine δ – Wobble

Direct Key Access: None

Functionality: Whenever a new position is commanded via a **Move Next** command, the wobble vector is recomputed and if this option is active the cosine of the declination is taken into account.

Uses: Used to enable/ disable the correction for cosine declination upon the wobble vector.

Indicators: A sideways chevron symbol immediately preceding the menu item indicates cosine declination correction is active. If active, the symbol “ δ ” will precede the **Wobble Vector** label on the Runtime Display.

Errors: None are associated with this item.

Cosine δ – Bias

Direct Key Access: None

Functionality: Enables the correction for cosine of declination for the user defined Bias rate in Right Ascension. If a bias rate is already active, this option will have no effect until the rate is reactivated.

Uses: Used to enable/ disable the correction for cosine declination on the Bias Rate in Right Ascension.

Indicators: A sideways chevron symbol immediately preceding the menu item indicates cosine declination correction is active. If active, the symbol “ δ ” will precede the **Bias** label on the Runtime Display.

Errors: None are associated with this item.

Cosine δ – Guide

Direct Key Access: None

Functionality: Enables the correction for cosine of declination for the user defined Paddle Guide rate in Right Ascension. If an Right Ascension guide paddle operation is active during this function call – the option will have no effect until the next paddle operation.

Uses: Used to enable/ disable the correction for cosine declination on the paddle Guide Rate in Right Ascension.

Indicators: A sideways chevron symbol immediately preceding the menu item indicates cosine declination correction is active. If active, the symbol “ δ ” will precede the **Guide** label on the Runtime Display.

Errors: None are associated with this item.

Cosine δ - Drift

Direct Key Access: None

Functionality: Enables the correction for cosine of declination for the user defined Paddle Drift rate in Right Ascension. If the Right Ascension Drift paddle button is active during this function call – the option will have no effect until the next paddle operation.

Uses: Used to enable/ disable the correction for cosine declination on the paddle Drift Rate in Right Ascension.

Indicators: A sideways chevron symbol immediately preceding the menu item indicates cosine declination correction is active. If active, the symbol “ δ ” will precede the **Drift** label on the Runtime Display.

Errors: None are associated with item.

The Limits Subsubsubmenu

Hour Angle Declination Horizon Sec Z Hardware
German HA
Override

The **Limits** submenu contains seven items pertaining to software and hardware motion limit detection. The menu allows the user to modify the range of motion the telescope is physically allow to attain.

The hardware limits are different from the software limits in that the user can not modify them, only enable or disable their function. Hardware limits are meant to be a backup to the PC-TCS software limit system in the event PC-TCS is initialized incorrectly. In most cases the hardware limits are optional as they require special “on mount” hardware to be installed.

Limits – Hour Angle

Direct Key Access: None

Functionality: Allows the user to modify the software limit associated with the “Hour Angle” or Right Ascension axis. When shipped the limit is set to a ± 9 hours – making for a maximum rotation of 135 degrees from the meridian in both directions. If for any reason this limit is too large (cable wrap is comprised, camera will fall off telescope, etc.) it may be modified via this menu item. If a Hour angle software limit is encountered PC-TCS will limit the maximum speed to approximately $1/10^{\text{th}}$ of a degree per second in all modes and directions in hour angle. This speed limit will be maintained until the final software limit is reached (see below) or the initial limit condition is removed.

Note: The maximum software limit allowed is ± 9.5 hours. If the telescope will be immobilized – in both hour angle directions. It is suggested that you NEVER set the value of this limit to this maximum. Recovery from a limit induced drive shut down is difficult and dangerous (see Appendix G).

Remember: The PC-TCS has the ability to safely point the telescope to objects located under the pole. You need not set the hour angle limit to its absolute maximum to achieve subpolar coordinates.

Uses: May be used to prevent telescope slews into hour angle positions that may be harmful to the telescope or its instruments.

Indicators: The **Limit Status** field will flash **LIMIT** and an arrow (<) will point to the **RA** label indicating the offending axis. Slew speed will be limited to 500 arcseconds per second in Right Ascension until the limit condition is removed. Coordinates labels for the Next and Reference fields will turn red if there associated value are outside the set Hour Angle limit.

Uses: Nonnumeric data will cause a bell to sound. Attempts to set the limit beyond the absolute maximum will produce an error window and/or cause a bell to sound.

Limits – Declination

Direct Key Access: None

Functionality: Allows the user to modify the software limit associated with the Declination axis. When shipped, the north is set to 90 degrees. (This is actually not true – it is more than 90 since objects under the pole are internally considered to be at declinations greater than 90 by PC-TCS and therefore 90 degrees actually indicates no restriction since values greater than 90 can not be entered by the user.) The default south limit is the declination of the horizon at the meridian plus 10 degrees. If for any reason these limits are unsatisfactory (instrument may hit yoke, etc.) they may be modified via this menu item. If a Declination software limit is encountered, PC-TCS will limit the maximum speed to approximately $1/10^{\text{th}}$ of a degree per second in all modes and directions in declination. This speed limit will be maintained until the absolute limit is reached (see below) or the initial condition is removed.

Note: the absolute maximum southern declination is considered by PC-TCS to be the declination of the meridian horizon plus 2.5 degrees. If the telescope reaches this maximum limit, the drives will be shut down and the telescope will be immobilized – in both declination directions. It is suggested that you NEVER set the value of the southern limit to this maximum. Recovery from a limit induced drive shut down is difficult and dangerous (see Appendix G).

Uses: May be used to prevent telescope slews into declination positions that may be harmful to the telescope and/or its instruments.

Indicators: The **Limit Status** field will flash **LIMIT** and an arrow (>) will point to the **DEC** label indicating the offending axis. Slew speed will be limited to 500 arcseconds per second in Declination until the limit condition is removed. Coordinates labels for the Next and Reference fields will turn red if there associated values are outside the set Declination limit.

Errors: Nonnumeric data will cause a bell to sound. Attempts to set the limit beyond the absolute maximum will produce an error window and/or cause a bell to sound.

Limits – Horizon

Direct Key Access: None

Functionality: Allows the user to modify the software horizon limit in 15 degrees azimuth intervals. When shipped the horizon limit is set to +10 degrees. If for any reason this limit is unsatisfactory (observer needs to look at Mercury, an interesting insect in the next country, etc.) it may be modified via this menu item. If a horizon software limit is encountered, PC-TCS will limit the maximum speed to 500 arcseconds per second in all modes and axes. This speed limit will be maintained until the absolute limit is reached (see below) or the initial limit condition is removed.

Note: The absolute maximum horizon limit is considered by PC-TCS to be +2.5 degrees. If the telescope reaches this limit, the drives will be shut down and the telescope will be immobilized – in all directions and axes. It is suggested that you NEVER set the value of the horizon limit to this maximum. Recovery from a limit induced drive shut down is difficult and dangerous (see Appendix G).

Uses: May be used to prevent telescope slews into elevations that are blocked from view or may be harmful to the telescope and/or its instrumentation.

Indicators: The **Limit Status** field will flash **LIMIT** and both arrows (<and>) will point to the **DEC** and **RA** labels indicating the horizon limit is offending. Slew speed will be limited to 500 arcseconds per second in all axes until the limit condition is removed. Coordinates labels for the Next and Reference fields will turn red if there associated value are outside the set horizon limit.

Errors: Nonnumeric data will cause a bell to sound. Attempts to set the limit below the absolute minimum will produce an error window and/or cause a bell to sound.

Limits – Sec Z

Direct key Access: None

Functionality: Allows the user to modify the software limit associated with the computed target, or current, Secant of the Zenith distance of the telescope.

Uses: May be used to prevent telescope from moving to coordinates based on an airmass greater than the limit set.

Indicators: A window will appear allowing the user to modify the SecZ limit. If the current telescope position is outside this limit the Sec Z label on the Runtime Display will turn red. Coordinates labels for the Next and Reference fields will also turn red if there associated value are outside the set Secant Z limit.

Errors: Nonnumeric data will cause a bell to sound.

Limits – Hardware

Direct Key Access: None

Functionality: Allows the user to enable/ disable the hardware limit detection circuitry. The hardware limits act exactly the same way as the software axial and horizon limits. The hardware limits are meant to be a backup to the software limits in the event PC-TCS has been initialized improperly.

Uses: Useful in overriding hardware limit detection if hardware limits are interfering with desired operation.

Indicators: **LIMIT** field will also flashing if a hardware induced limit has been reached and this option is used to disable it. In addition the sideways chevron (») will be present if hardware limit detection circuitry is enabled. Hardware limits may also be monitored via the **External I/O** subsubsubmenu.

Errors: No errors are associated with this menu.

Limits – German HA

Direct Key Access: None

Functionality: Allows the user to modify the software Hour Angle limit associated with German Equatorial mounted telescopes. The limit relates to the Hour Angle (East for tube EAST conditions and West for tube WEST conditions) at which no further motion should be permitted until a “German Flip” is performed. (See the **Move** submenu.)

When shipped the limit is set to 9 hours – making for a maximum rotation of 135 degrees from the meridian in both directions – effectively disabling the limit as one would for a fork the mounted telescope. If the user has a German Equatorial mount – it will be necessary to set this limit to prevent the telescope tube from getting below the counter weight level or physically hitting the telescope mount. Note: The maximum German HA software limit allowed is ± 0.5 hours from whatever limit is entered for this option. If the telescope reaches this maximum limit the drives will be shut down and the telescope will be immobilized – in both hour angle directions. It is suggested that you never allow the telescope to continue to track or move beyond the German HA limit. Recovery from a limit induced drive shut down will require the user to set a higher German HA limit that may permit telescope motion into dangerous attitudes.

Uses: May be used to prevent telescope slews into hour angle positions that may be harmful to German Equatorial mounted telescope.

Indicators: The **Limit Status** field will flash **LIMIT** and an arrow (<) will point to the **RA** label indicating the offending axis. Telescope Slew speed will be limited to 500 arcseconds per second in Right Ascension until the limit condition is removed or the final German HA limit is reached. Coordinates labels for the Next and Reference fields will turn red if there associated values are outside the set German HA limit.

Errors: Nonnumeric data will cause a bell to sound. Attempts to set the limit beyond the absolute maximum of 9.5 hours will produce an error window and/or cause a bell to sound.

Limits – Override

Direct Key Access: None

Functionality: Allows the user to override software and hardware limit shutdown of the drive system. If the final software limits are reached during normal tracking, PC-TCS will nominally attempt to disable the drive system to prevent damage to your telescope and related hardware. This option will disable this shutdown and allow the telescope to continue **BELOW THE FINAL SOFTWARE LIMITS!!!** therefore, this option should be used with great caution since it will allow the telescope to assume positions that may be hazardous. If a limit has been reached and this option is active – the telescope will be allowed to move no faster than 500 arcseconds per second.

This function also allows moves to objects at or above the final software limits. (i.e. >3.0 degrees above the horizon; < declination north limit; > hour angle limit + 0.5 hours; no secant Z limit restriction; no declination south limit other than the horizon limit described above.)

Uses: Useful in recovering from limit induced drive system shutdowns and allowing the telescope to move to and object that is above the final software limits. To recover from a final software limit drive shutdown the user should first invoke this option then use the **Disable** option of the **Motion** submenu to reenable the drive system.

Indicators: The Limit Status field will be replaced by a red flashing “LIMIT OVERRIDE” message when this option is active. In addition, the sideways chevron (») will prefix the **Override** option in the manu if the option is active.

Errors: No errors are associated with this item.

The Flexclass Subsubsubmenu

class A
class B
class C
class D

The **Flexclass** submenu contains four items pertaining to flexure correction data files. Several sets of flexure correction data files may be read by PC-TCS for the purpose of correcting telescope position and tracking for the effects of mechanical flexure and other systematic misalignments.

The **Flexclass** will allow the user to change the flexure data sets used to correct telescope pointing and tracking. In all five sets are supported. Class A through D and a default data set selected when no class is set. (This is provided to maintain compatibility with previous versions of PC-TCS.) Note: Class A is automatically associated with “Tube East” orientation of the telescope and Class B is associated with “Tubes West” orientation. This allows PC-TCS to automatically and accurately correct for telescope flexure with different flexure maps for german equatorial telescopes which require axial “flip” to acquire objects on opposite sides of the meridian.

Separate data sets are useful for providing accurate pointing and tracking with different telescope configurations such as Tube EAST and WEST, different secondaries and instruments, etc.

Flexclass – class A

Direct Key Access: None

Functionality: Selects the class A flexure data set. Data for this set must first be taken using the **FDC** option of the **Utilities** subsubmenu. Once data is taken and analyzed with ANALYZE program, two files can be created with the names PCTCSDA.FLX and PCTCSRA.FLX by copying the files PCTCSD.FLX and PCTCSR.FLX to the two files above. The class A maps are automatically selected for the “Tube East” configuration of PC–TCS.

Note: this menu item only **SELECTS** a data set. To correct telescope pointing and tracking for flexure effects, the user must enable these corrections via the **Corrections – Flexure**, **Rates – Flexure**, and **Display –Flexure** menu items.

Uses: Useful in selecting the flexure data sets for correcting telescope pointing and tracking for a user determined telescope mechanical configurations.

Indicators: The **class A** menu item will be prefixed by a sideways chevron symbol (>>) when this class is selected.

Errors: If the data set can not be opened, flexure corrections for pointing and tracking as well as coordinate display may be disabled.

Flexclass – class B

Direct Key Access: None

Functionality: Selects the class Bflexure data set. Data for this set must first be taken using the **FDC** option of the **Utilities** subsubmenu. Once data is taken and analyzed with ANALYZE program, two files can be created with the names PCTCSDB.FLX and PCTCSRB.FLX by copying the files PCTCSD.FLX and PCTCSR.FLX to the two files above. The class A maps are automatically selected for the “Tube West” configuration of PC–TCS.

Note: this menu item only **SELECTS** a data set. To correct telescope pointing and tracking for flexure effects, the user must enable these corrections via the **Corrections – Flexure**, **Rates – Flexure**, and **Display –Flexure** menu items.

Uses: Useful in selecting the flexure data sets for correcting telescope pointing and tracking for a user determined telescope mechanical configurations.

Indicators: The **class B** menu item will be prefixed by a sideways chevron symbol (>>) when this class is selected.

Errors: If the data set can not be opened, flexure corrections for pointing and tracking as well as coordinate display may be disabled.

Flexclass – class C

Direct Key Access: None

Functionality: Selects the class C flexure data set. Data for this set must first be taken using the **FDC** option of the **Utilities** subsubmenu. Once data is taken and analyzed with ANALYZE program, two files can be created with the names PCTCSDC.FLX and PCTCSRC.FLX by copying the files PCTCSD.FLX and PCTCSR.FLX to the two files above.

Note: this menu item only SELECTS a data set. To correct telescope pointing and tracking for flexure effects, the user must enable these corrections via the **Corrections – Flexure**, **Rates – Flexure**, and **Display –Flexure** menu items.

Uses: Useful in selecting the flexure data sets for correcting telescope pointing and tracking for a user determined telescope mechanical configurations.

Indicators: The **class C** menu item will be prefixed by a sideways chevron symbol (>>) when this class is selected.

Errors: If the data set can not be opened, flexure corrections for pointing and tracking as well as coordinate display may be disabled.

Flexclass – class D

Direct Key Access: None

Functionality: Selects the class D flexure data set. Data for this set must first be taken using the **FDC** option of the **Utilities** subsubmenu. Once data is taken and analyzed with ANALYZE program, two files can be created with the names PCTCSDD.FLX and PCTCSR.D.FLX by copying the files PCTCSD.FLX and PCTCSR.FLX to the two files above.

Note: this menu item only SELECTS a data set. To correct telescope pointing and tracking for flexure effects, the user must enable these corrections via the **Corrections – Flexure**, **Rates – Flexure**, and **Display –Flexure** menu items.

Uses: Useful in selecting the flexure data sets for correcting telescope pointing and tracking for a user determined telescope mechanical configurations.

Indicators: The **class D** menu item will be prefixed by a sideways chevron symbol (>>) when this class is selected.

Errors: If the data set can not be opened, flexure corrections for pointing and tracking as well as coordinate display may be disabled.

Globals – Paths

Direct Key Access: None

Functionality: Allows the user to modify the data subdirectory paths associated with user and permanent catalog files, command files, orbit and satellite tracking files.

Uses: Used to alter the search path that PC–TCS uses to access various files of objects, stars, command files, orbit files, and satellite track files.

Indicators: A window will be opened displaying the data paths for each of the 5 file types PC–TCS recognizes. The paths must end in a \ to be valid.

Errors: None are associated with this item.

Globals – Servo

Direct Key Access: None

Functionality: Allows the user to modify the constants and parameters associated with the servo algorithm. The function can also provide graphical output of position error, commanded velocity and commanded drive frequency as a function of time.

Uses: Used to tune the servo system for optimum operational characteristics.

Indicators: Two windows will be displayed showing servo related variables for each axis and their assigned values. Modifications do not take effect until Ctrl ↵ is entered. Temporal display of servo response for a variety of system parameters may be displayed graphically. Graphics hardware options supported are CGA, HERC and EGA.

Errors: A bell will sound if nonnumeric data is entered.

3.8.3 The Utilities Subsubmenu

Raster Spiral FDC Orbit Import saTrak

The **Utilities** subsubmenu contains four items that allow the user access to the **FDC** (Flexure Data Collection), **Raster**, **Spiral** and **Orbit** Parameter Generation internal utilities.

Utilities – Raster

Direct Key Access: *Ctrl X*

Functionality: Allows user to setup and execute a “Raster” type motion with the telescope. The user will be asked to enter the Right Ascension and Declination step size of the raster, along with the number of elements per axis and dwell time per step to use in the scan. The user is able to terminate Raster scanning once activated by hitting the **ESC** key. (For more information on this utility, see Chapter 4 – **Raster scan Utility**.)

Uses: Useful in setting up a raster search pattern for an elusive object, or mapping of regions on the sky.

Indicators: A window will appear with the user queries along with a status field indicating progress once execution of the raster is started.

Errors: Nonnumeric data entry will cause a bell to sound.

Utilities – Spiral

Direct Key Access: *Ctrl K*

Functionality: Allows the user to setup and execute a “Spiral” type motion with the telescope. The user will be asked to enter the step size of the spiral, along with the total number of “loops” or passes for the scan as well as the dwell time after each step is completed. The user is able to terminate Spiral scanning once activated by hitting **ESC** key. (For more information on this utility see Chapter 4 – **Spiral Scan Utility**.)

Uses: Useful in setting up a spiral search pattern for an elusive object, or mapping of regions on the sky.

Indicators: A window will appear with the user queries. Once execution of the spiral is started, no further indications are obvious except for telescope motion.

Errors: Nonnumeric data entry will cause a bell to sound.

Utilities – FDC

Direct Key Access: None

Functionality: Invokes the FDC pointing data collection routines which will automatically point the telescope to a number of FK5 stars, collect data on pointing errors and construct the file POINT.DAT that can be later reduced by the ANALYZE program and ultimately create the files PCTCSR.FLX and PCTCSD.FLX. These files are then used by PC-TCS to correct mechanical flexure in telescope pointing and tracking. The user should refer to Chapter 4 for operational information on this utility.

Uses: Useful in creating the files necessary to correct telescope flexure related pointing and tracking problems.

Indicators: The FDC subsubsubmenu will appear.

Errors: No errors are associated with invocation of this option.

The Orbit Subsubsubmenu

New Edit

The **Orbit** subsubsubmenu contains two items pertaining to the creation and editing of PC-TCS orbit parameter files. These files contain orbital data of various objects such as comets and asteroids which PC-TCS can use to compute the True of Date Position of such objects. These menu item allow the user to create new object parameter files and edit files already on disk.

All PC-TCS readable orbit files conform to the structure documented in Appendix H of this manual. The user should familiarize themselves with the information in Appendix H before attempting to edit or construct orbit files using these menu options. By default all orbit parameter files have the DOS extension .ORB.

Orbit – New

Direct Key Access: None

Functionality: Allows the user to create the new file containing heliocentric orbital data for a comet or asteroid which can later read by PC-TCS to compute the current positions of the object. Before invoking this option the user should consult Appendix H on the structure of the Orbit files and review the information necessary to construct an orbit file.

When invoked, this menu option will open a window and prompt the user for all necessary orbital information needed to compute the position of the object. Once all data has been entered PC-TCS will prompt the user for a file name to store the data.

Uses: Used to create orbit files of comets and asteroid for later position computations and moves to these objects using **Object** option of the **Position** submenu.

Indicators: A window will be opened and the user will be prompted for the orbital data and file name for the object.

Errors: Nonnumeric data in fields requiring this type of data will cause a bell to sound. Embedded spaces in the file name will cause a bell to sound.

Orbit – Edit

Direct Key Access: None

Functionality: Allows the user to an orbit parameter containing heliocentric orbital data for a comet or asteroid already resident on disk. Before invoking this option the user should consult Appendix H on the structure of the Orbit files and review the information necessary to construct an orbit file.

See File Window Documentation in Chapter 2.

When invoked this menu option will open a window displaying the available file names of objects already on disk. The user then can select the desired file to edit by using the arrow keys or mouse to highlight the filename and hitting the enter key to confirm the selection. Upon selection, PC-TCS will display an edit window containing the orbital data previously entered and allow the user to modify this information.

Uses: Used to edit an orbit file containing orbital data of comets and asteroids for later position computation and moves to these objects using the **Object** option of the **Position** submenu.

Indicators: A window will be opened and the user will be asked to select a file from a list of orbit files already on disk. Once selection has been made the user will be show the orbital data for the selected object and allowed to modify the data.

Errors: Nonnumeric data in fields requiring this type of data will cause a bell to sound. Embedded spaces in the file name will cause a bell to sound.

The Import Subsubsubmenu

From A: From B:

The **Import** subsubsubmenu contains two items pertaining to the duplication of files from floppy drives. Currently (version 6.X) PC-TCS expects all user catalog files to be in default PC-TCS working directory on the Hard Drive. If a catalog file is available on floppy it must be “copied” to the default working directory of PC-TCS before PC-TCS can use it.

Import – From A:

Direct key Access: None

Functionality: Allows the user to “import” (i.e. copy) files from the A: external floppy drive.

Uses: Used to copy user catalog (.CAT) or command files (.TCS) from the A: floppy drive.

Indicators: A window will appear showing the file names available on the A: drive. The user may use the mouse or arrow keys on the keyboard to select the file to import. Highlighting the file and then hitting the ↵ key on the keyboard or the left mouse button will copy the selected file to the PC–TCS working directory.

Errors: If the file does not exist or there is insufficient memory available for the copy operations – an error window will appear and/or a bell will sound.

Import – From B:

Direct key Access: None

Functionality: Allows the user to “import” (i.e. copy) files from the B: external floppy drive.

Uses: Use to copy user catalog (.CAT) or command (.TCS) files from the B: floppy drive.

Indicators: A window will appear showing the file names available on the B: drive. The user may use the mouse or arrow keys on the keyboard to select the file to import. Highlighting the file and then hitting the ↵ key on the keyboard or the left mouse button will copy the selected file to the PC–TCS working directory.

Errors: If the file does not exist or there is insufficient memory available for the copy operation – an error window will appear and/or a bell will sound.

Utilities – saTrak

Direct Key Access: None

Functionality: Allows the user to perform earth orbiting satellite tracking. This function requires optional software from COMSOFT to generate the satellite ephemerides read by PC-TCS to implement this function.

See Chapter 4 section 4.8 for further information.

Uses: Used to track earth orbiting satellites.

Indicators: Not Documented.

Errors: Not Documented.

3.9 The Shutdown Submenu

Auto Manual

The **Shutdown** submenu contains three items used to control the termination of PC-TCS at the end of an observing session. The options are **Auto** for automatic shutdown which includes shutdown of all telescope motions before stowing and termination of all PC-TCS file/remote activities, **Manual** for manual activation of the same operation as for **Auto**.

Shutdown - Auto

Direct Key Access: *Alt Z*

Functionality: Performs an automatic shut down of PCTCS operations including disabling all background rates and scans in progress, terminating of all command file executions and remote serial access, and stowing the telescope. PCTCS will not, however, be removed from memory until all operations are completed and the user types an acknowledging ↵.

Uses: Used to perform an automated shut down of PCTCS.

Indicators: A shut down window will appear and messages as to shut down progress will be displayed.

Errors: Errors encountered during shut down will cause an error window to appear and/or a bell will sound.

Shutdown - Manual

Direct Key Access: *Alt Q*

Functionality: Performs a manual shut down of PCTCS operations including disabling all background rates and scans in progress, termination of all command file executions and remote serial access, and stowing the telescope. PCTCS will ask for user confirmation at each step of the shut down. Hitting an *ESC* key at any time during this process will abort the operation being performed. As with the auto shut down option, the user will be asked for an acknowledging ↵ before PCTCS will be removed from memory after shut operations are completed.

Uses: Used to perform a manual shut down of PCTCS.

Indicators: A shut down window will appear and messages as to shut down progress will be displayed.

Errors: Errors encountered during shut down will cause an error window to appear and/or a bell will sound.

3.10 Direct (Hot) Key Summary

Motion- Cancel: *F9*

Motion- Tracking: *Alt T*

Motion- Bias: *Alt B*

Motion- Disable: *F10*

Declare- Reference: *Ctrl R*

Declare- Beam 1: *Ctrl F1*

Declare- Beam 2: *Ctrl F2*

Declare- Initial: *Alt =*

Declare- Zero Diff: *F5*

Move- Next: *Alt N*

Move- Reference: *Alt R*

Move- Cur+Offset: *Alt O*

Move- Beam 1: *F1*

Move- Beam 2: *F2*

Move- Stow: *Alt S*

Position- Next: *Ctrl N*

Position- Offset: *Ctrl O*

Position- Wobble: *Ctrl W*

Rates- Bias: *Ctrl B*

Rates- Guide: *Ctrl G*

Rates- Drift: *Ctrl D*

Source- Keyboard: *Alt K*

Source- PCF: *Alt P*

Source- User: *Alt U*

Parameters- Input- Remote- Active: *Alt I*
Parameters- Input- Command file- Active: *Ctrl A*
Parameters- Input- I/O Externals- Current ADC: *Alt C*
Parameters- Globals- Display- Epoch: *Ctrl E*
Parameters- Globals- Display- rIse/set: *F4*
Parameters- Globals- Display- rates: *F3*
Parameters- Utilities- Raster: *Ctrl X*
Parameters- Utilities- Spiral: *Ctrl K*
Shutdown- Auto: *Alt Z*
Shutdown- Manual: *Alt Q*
Hide/Unhide Remote Status Window: *Alt H*
Password entry: *Ctrl Backspace*
Large Format Runtime Display: *F7*
Slew Permit Toggle: *F8* (For keyboard slew enabled systems only).

Chapter 4

Advanced PC-TCS Utilities

The following chapter deals with the features available to the user via the **Input** and **Utilities** subsubsubmenus. These features are intended to provide the advanced user with a means to automate and remote PC-TCS operations and create/maintain custom PC-TCS object catalog files.

4.1 Command Files

PC-TCS has the ability to accept commands from a file can move and control the telescope in much the same manner as a user sitting at the console. Command files allow the user to perform “hands off” operation of the telescope and even automate an entire nights observing from first object acquisition to final shutdown.

Command files are simple ASCII text files, stored on disk, that contain one or more PC-TCS command mnemonics documented in appendix A. The files can be created with any text editor capable of writing “nondocumented” files free of text formatting and control codes. All files used for this purpose must be named in the standard DOS style and have the extension “.TCS”. In addition, PC-TCS will recognize only those commands documented in appendix A. Unrecognized commands will cause premature termination of the commands execution process. (Commands are *not* prescanned by PC-TCS for correct syntax.)

The repertoire of commands available include Mnemonics to define the full range of NEXT telescope positions including objects from standard and user catalog files, and the eight major planets. WOBBLE and OFFSET vectors and telescope tracking and bias rates are also settable from within command files.

Command files may be paused internally, waiting for a specific Universal, Local Sidereal or user defined time delay to occur. Four external TTL inputs may also be used to control command file flow and execution with pausing and branching capabilities. Facility to

unconditionally branch (GOTO) and call repetitive routines (GOSUB) from within the command file is also provided.

Motions to NEXT, BEAM 1, BEAM 2, OFFSET and STOW may be commanded with automatic pause for motion completion before the next command is executed. Slew enable and disable mnemonics are provided for user/instrumentation safety.

A sample file called DEMO.TCS is included with the PC-TCS software that demonstrates the power of this unique facility. The DEMO file can be loaded via the **Command** subsubsubmenu or at PC-TCS start up from command line as follows.

C:\ > TCS DEMO ↵

Activation of the command file is automatic is defined on the command line, else execution can be accomplished via the **Command** subsubsubmenu or the hot key *Alt A*.

When activated the DEMO file will first prompt for continuance from the user then define a 10 degree declination and 1 hour Right Ascension wobble vector, turn slew permit on, move the telescope to stow and pause for 10 seconds. A subroutine is then called that will move to BEAM 2 (10 degrees north and 1 hour east), turn the tracking on and wait for 5 seconds, turn slew permit off then move to BEAM 1, turn tracking off and return. The program will then turn tracking back on, wait for tracking stability, turn any active bias off, set a south bias rate of 1000 arcseconds per second and turn slew permit on then activate the bias rate for 18.0 seconds, wait for the telescope to become stable, set a west bias rate of 2000 arcseconds per seconds and activate it, pause for 10 seconds, deactivate the bias rate and wait for telescope to become stable. The program will then prompt the user for continuance and finally perform a stowing shutdown that will terminate the tracking and bias rates, move the telescope back to the stow position and disable the slew permit. The final instruction loops back to the beginning.

A Note on Command File Size

Currently, PC-TCS is set up to accommodate command files with up to 1000 command lines, or to the limit of available memory to store the command lines, which ever comes first. In addition, individual Command File lines must contain only one mnemonic per line.

4.2 Remote Serial Access

For those users wishing to gain remote access to PC-TCS, a facility to communicate via up to eight RS-232 serial channels is provided. This remote access feature can be used to command PC-TCS from a remote terminal or another computer connected either directly to the PC-TCS computer or via external modems.

The remote subsubsubmenu documented in chapter 2, is used to activate/ deactivate this utility. Included in this subsubsubmenu are menu options to change baud rates (transmission/ reception speeds) and enable / disable the serial channels. When the remote serial access is activated, a small window will appear on the left center of the Run Time display showing the number of commands executed and the command activity on each of the eight possible channels. This window can be used to verify command reception once all hardware connections are in place.

Hardware connections necessary to complete this interface are documented in Appendix E. Communications Protocols and sample software subroutines are documented in appendices B,C and D. To use this feature, the user must have one or more optional serial channels installed in the PC-TCS computer, a cable providing the four wire interface to the remote terminal/ modem, and a terminal/ modem or auxiliary computer capable of standard RS-232 communications.

The remote access channels operate somewhat differently than the menu or command file interface, but provide similar capabilities. Commands sent to PC-TCS must conform to the syntax and handshake protocol documented in Appendix B. Information sent from PC-TCS is in the form of a continuously broadcast telemetry stream consisting of 100 characters as documented in Appendix C. The user can inspect this telemetry to obtain current information as to telescope position, motion status, time, external inputs and command activity.

Commands are all simple uppercase ASCII text and may directly on a terminal, however because of the strict syntax employed, the user may find it easier to program a remote computer to build these commands based on a more user friendly interface of his own design. To aid the user in this task, subroutines to send commands and receive telemetry information from the PC-TCS computer's remote serial channels are provided in Appendix D.

This facility is perhaps the most difficult to utilize since it requires external hardware and options not commonly associated with Personal Computers along with some programming by the user. However, it provides a unique method with which to control your telescope even if you are hundred miles from your site.

Some Notes on Remote Access via Modems

Those implementing remote access with modems should take care to disable the textual modem responses broadcast by their hardware. The responses, such as CONNECT, RING and OK are not recognized by PC-TCS and may cause PC-TCS to ignore the first command sent to it over remote serial lines.

These modem responses can usually be disabled with a hardware switch inside the modem or with ASCII commands that reprogram the internal nonvolatile RAM that hold modem setup parameters. Consult your modem reference manual for further information.

The CTS pulse generated by PC-TCS as a command acknowledgment is not propagated through the modem lines and therefore is not available to the commanding device. It is suggested that the user utilize the telemetry stream Channel Status (see Appendix B, C and E) information to confirm command reception. Valid command acknowledgment is propagated in the Channel Status columns in the form of a uppercase or lowercase "E" (or "e"). The case is alternated so that the user can differentiate between two consecutive "executed" commands on the same channel. In addition it may be necessary to disconnect the RTS and CTS lines on the Computer side of the connector to allow proper modem operations with PC-TCS.

4.3 User Catalog Generation

User catalogs are files containing coordinate and other information about astronomical objects (defined by the user) for digestion by PC-TCS. The data contained in these files is used by PC-TCS to declare the NEXT position that the telescope should be moved to saving the user the time and effort of manually entering coordinate data during valuable observing time.

User Catalogs files must be constructed manually by using either a word processing program capable of producing “nondocument” files, or the CATALOG utility program provided with PC-TCS. It is much easier to use the later method since the data files have a very strict structure and format that word processors do not inherently understand without some sophisticated setup. The CATALOG program, on the other hand, already knows about this structure and will take care of the tedious task of generating the file for you. If however, you would like to better understand the catalog data file format, the record structure is outlined in Appendix F.

4.3.1 Starting CATALOG

The CATALOG programs is a separate program that may be executed from the DOS Operating System level of your PC.

To execute CATALOG from DOS, at the **C:\TCS>** prompt type . . .

```
C:\TCS>CATALOG ↵
```

Optionally, a catalog file name may be specified on the command line as follows . . .

```
C:\TCS>CATALOG mydata.cat↵
```

Where “*mydata.cat*” is the name of the file you wish to either create, modify or examine using the CATALOG program. (Note: All PC-TCS user catalog file names must have the extension “.CAT” to be readable by PC-TCS. If you forget and accidentally create a catalog file without this extension, simply rename the file after exiting CATALOG with the DOS rename (REN) command.)

If the CATALOG program is found by DOS the screen will clear and the following menu will appear at the top center of the screen . . .

Scan/ Edit Append Create Print

At this point you must decide which operation you wish to perform. If you wish to exit CATALOG the *ESC* key will return you to the previous level of operations – in this case DOS.

The following sections will give a brief overview of each of the four possible catalog operations displayed in the above menu.

4.3.2 Scan/ Edit

The **Scan/ Edit** option is useful in modifying and/ or scanning catalogs previously created with CATALOG or distributed with PC-TCS. You *can not* scan/ edit a catalog file that does not already exist. If you specify a file that CATALOG can not find in this mode an error window will appear informing you of the error.

The **Scan/ Edit** option is invoked by highlighting the option using the left and right arrow keys (*← or →*) and hitting *Enter* (↵). At this point the program will ask for the file name of the catalog to edit/ scan. If you entered the file name at the command line, the name will appear as the default answer to this question – you need only to hit *Enter* again to begin editing, otherwise type the correct name and hit *Enter*.

If CATALOG is able to open and read the file specified the screen will fill with the Scan/ Edit window and display the information associated with the first record in the file. The fields displayed are as follows . . .

Object Number: The record number of the information stored in the catalog file. This number is the number used to retrieve the object information in PC-TCS.

Magnitude: The visual astronomical brightness of the object.

Epoch: The coordinate epoch for the Right Ascension and Declination data.

Right Ascension: The object RA in the format HH:MM:SS.ss

Declination: The object declination in the format ±DD:MM:SS.s

RA Proper Motion: The object's proper motion in RA in seconds of time per century.

δ Proper Motion: The object's proper motion in Declination in arcseconds per century.

Label: A label field that contains a user defined description of the objects – such as the spectral class, name, etc. The catalogs included with PC-TCS use this field to hold the objects spectroscopic classification and magnitude. The Underline below the

filed shows the extent of the label that will be displayed on the PC-TCS runtime display when retrieved from the file.

The active keys associated with Scan/ Edit can be reviewed by typing *Alt H*. Records may be sequentially displayed by using the *Ctrl N* and *Ctrl P* key combinations to show the “Next” and “Previous” records respectively. If the user wishes to view a specific object record, he needs only to type the record number into the Object Number field and type *Ctrl Enter (Ctrl↵)*..

To modify the information in the record the user can use the up and down arrow keys (↑ or ↓) to move to the field desired, enter the new data and hit *Enter (↵)* and then type *Ctrl Enter (Ctrl↵)* to update the record on disk.

When you are finished Scan/ Edit – hit the *Esc* key to return to the main menu.

4.3.3 Append

The **Append** option is used to add to a file previously created with CATALOG or distributed with PC-TCS. You *can not* Append to a catalog file that does not already exist. If you specify a file that CATALOG can not find in this mode of error window will appear informing you of the error.

The **Append** option is invoked by highlighting the option using the left and right arrow keys (← or →) and hitting *Enter (↵)*. At this point the program will ask for the file name of the catalog to Append to. If you entered the file name at the command line, the name will appear as the default answer to this question – you need only to hit *Enter* again to begin data entry, otherwise type the correct name and hit *Enter*.

If CATALOG is able to open and read the file specified the screen will fill with the Append window and display the same entry screen associated with Scan/ Edit with the record number of the next available index appending to the file.

CATALOG will now accept user input for the following fields . . .

Magnitude: The visual astronomical brightness of the object.

Epoch: The coordinate epoch of the Right Ascension and Declination data.

Right Ascension: The object RA in the format HH:MM:SS.ss

Declination: The object declination in the format $\pm DD:MM:SS.s$ Note: the first character of this field must be a “+”, “-” or a space to be formatted correctly. Any other character will cause an error in data entry at confirmation time.

RA Proper Motion: The object’s proper motion in RA in arcseconds per century.

δ Proper Motion: The object’s proper motion in Declination in arcseconds per century.

Label: A label field that contains a user defined description of the object – such as the spectral class, name, etc. The catalogs included with PC-TCS use this field to hold the objects spectroscopic classification and magnitude. The Underline below the fields shows the extent of the label that will be displayed on the PC-TCS runtime display when retrieved from the file.

To modify the information in the record the user can use the up and down arrow keys ($\uparrow$ or $\downarrow$) to move to the field desired, enter the new data and hit *Enter* ($\downarrow$) and then type *Ctrl Enter* (*Ctrl* $\downarrow$) to update the record on disk.

When you are finished Appending – hit the *ESC* key to return to the main menu.

4.3.4 Create

The **Create** option is used to create new catalogs. If you attempt to Create a catalog that already exists, CATALOG will display a warning message and ask if you wish to OVERWRITE (Destroy) the catalog file.

The **Create** option is invoked by highlighting the option using the left and right arrow keys ($\leftarrow$ or $\rightarrow$) and hitting *Enter* ($\leftarrow$). At this point, the program will ask for the file name of the catalog to create. If you entered the filename at the command line, the name will appear as the default answer to this question – you need only to hit *Enter* again to begin editing, otherwise type the correct name and hit *Enter*.

If CATALOG is able to open the file specified, the screen will fill with the Create window and display the same data entry screen associated with Scan/ Edit with the record number of the first record of the file.

CATALOG will now accept user input for the following fields . . .

Object Number: The record number off the information stored in the catalog file. This number is the number used to retrieve the object information in PC-TCS.

Magnitude: The visual astronomical brightness of the object.

Epoch: The coordinate epoch for the Right Ascension and Declination data.

Right Ascension: The object RA in the format HH:MM:SS.ss

Declination: The object Declination in the format $\pm$ DD:MM:SS.s

RA Proper Motion: The object's proper motion in RA in arcseconds per century.

δ Proper Motion: The object's proper motion in Declination in arcseconds per century.

Label: A label field that contains a user defined description of the object – such as the spectral class, name, etc. The catalogs including with PC-TCS use in this field to hold the objects spectroscopic classification and magnitude. The Underline below the field shows the extent of the label that will be displayed on the PC-TCS runtime display when retrieved from the file.

The active key associated with Create can be reviewed by typing *Alt H*. Records previously entered may be sequentially displayed by using the *Ctrl N* and *Ctrl P* key combinations to show the “Next” and “Previous” records respectively. If the user wishes to view a specific object record, he needs only to type the record number into the Object Number field and type *Ctrl Enter* (*Ctrl ←*).

To modify the information in the record, the user can use the up and down arrow keys ($\uparrow$ or $\downarrow$) to move to the field desired, enter the new data and hit *Enter* and then type *Ctrl Enter* to update the record on disk.

When you are finished Creating – hit the *ESC* key to return to the main menu.

4.3.5 Print

The **Print** option is used to display/ print one or more record from a catalog file previously created with CATALOG or distributed with PC-TCS. You *can not* Print a catalog file that does not already exist. If you specify a file that CATALOG can not find in this mode, an error window will appear informing you of the error.

The **Print** option is invoked by highlighting the option using the left and right arrow keys ($\leftarrow$ or $\rightarrow$) and hitting *Enter*. At this point, the program will ask for the file name of the catalog to print. If you entered the file name at the command line, the name will appear as

the default answer to this question – you need only to hit *Enter* again to begin editing, otherwise type the correct name and hit *Enter*.

If CATALOG is able to open and read the file specified, a window will appear asking the user for the range of records to display and the output device upon which to display them – the screen, the printer or a default file called CATFILE.SRN. If the screen is specified, a new window will appear, the records will be shown in the format displayed in the top line of the window. Each record is displayed as two lines, one for the coordinate, magnitude, epoch and proper motion data, and one for the label data.

Output to the screen may be paused by typing *Ctrl S* and restarted by typing *Ctrl Q*. When complete, the user may return to the main menu by hitting the *ESC* key.

4.4 The FDC Utility

The Flexure Flexure Data Collection Option of the Utilities subsubmenu provides the user with a semi-automated capability of collecting data that can later be used to correct telescope pointing errors induced by repeatable mechanical flexure and alignment problems.

Operationally, the FDC utility divides the majority of the visible sky into a coarse grid map bounded by +80 degrees north declination to -45 degrees south declination (northern hemisphere sites +45 and -80 for southern sites and ± 5 hours in Hour Angle. Individual grid “cells” are 5 degrees in declination by 15 minutes of time in Hour Angle – for a total of 1000 cells in 25 declination “rows” and 40 hour angle “columns”. Figure 4.1 shows the structure of the pointing grid cell map. The user should copy this map and keep a running log of the cells processed during a pointing run. The log can then be used to restart the run if interrupted during the data collection process.

The grid is used to select star from a sorted version of the FK5 Star Catalog called FLEX2000.PCF. Once started, the FDC utility will search the catalog for objects visible within a given cell, slew the telescope to the computed position and prompt the user to center the star in the field of view. After centering, the object has been accomplished, the utility computes the difference between the computed and actual telescope position then stores this information in a disk file called POINT.DAT. The utility then chooses another cell and repeats the search and point process until all cells have been processed.

The POINT.DAT file can later be reduced by the ANALYZE Utility software to construct the Flexure data files PCTSCSD.FLX and PCTCSR.FLX. PC-TCS can then read these files during observing sessions to correct for linear and nonlinear pointing errors caused by mechanical flexure of the telescope mount and tube assembly as well as errors due to improper polar alignment, nonperpendicularity of the Declination and Right Ascension axes, etc.

The user should be aware that the FDC and ANALYZE utilities will allow PC-TCS to correct for only those errors that are *repeatable*. Errors arising from such deficiencies as gear backlash, improper balance or preloading, and mechanical hysteresis are nonrepeatable and therefore uncorrectable by PC-TCS.

As a matter of procedure, the user should always make every effort to complete any data collection run in the space of a single observing session. This will ensure that the registration point for the data set does not change in the course of data acquisition. Spreading data collecting over several nights is highly discouraged since the user will most likely have to define a new “initial position” at the start of the observing session each night, thus changing the implied “zero-point” of the flexure data.

The following sections describe the use of each of the options of the FDC subsubsubmenu as well as the sequence of events the user may expect to follow during a data collection run.

4.4.1 The FDC subsubsubmenu

Incremental Full Map Start δ Purge Data
Execute

The FDC subsubsubmenu contains five options. The first four options are used to control how the FDC utility will collect data. The last option is used to start the semi-automated data collection process.

Once started, the FDC utility will choose a star from the FLEXURE.PCF based on the starting parameters set by user with **Increment**, **Full Map** and **Start δ** menu options. The telescope will automatically be slewed to the star's coordinates, and the user will then be prompted to center the star in the telescope's field of view. Once centered, the user may confirm the data collection process with a simple Carriage Return ($\leftarrow$ key), bypass the star with a backspace (*BS*) key, or abort the FDC run altogether with the *ESC* key.

A confirming $\leftarrow$ will instruct PC-TCS to compute the difference between the commanded position and the current position of the telescope, store this result in the POINT.DAT data file on disk, increment the cell number and repeat the search and point process. During the processing of a given declination "row" of the pointing map, the telescope will always be moved westward. This ensures the telescope tracks. After the last cell of a given declination row is processed, the telescope will be slewed east to the start of the next row in the sequence.

Since not all cells will contain a visible FK5 star during any given run, the pointing map will never be completely filled. In addition, depending upon the operational limits set by the user and the latitude of the telescope, some cells may be outside the pointing limits of the telescope. Therefore, even a Full Map FDC pointing run will never collect data for all 1000 possible cells. Even so, a Full Map data set takes about three or four hours to complete, depending upon how efficient the operator is at moving the dome, centering

stars and confirming FDC operation. With the **Full Map** option disabled, FDC pointing runs can be completed in approximately 2 to 3 hours.

Figure 4.1: The Pointing Grid Cell Map

The Incremental Option

The incremental option of the FDC menu allows the user to take pointing data using a previously computed flexure map. Data taken with this option enabled will be added to the already existing flexure map upon completion and analysis of the pointing data. Data acquisition using the incremental option will typically proceed much faster than a raw pointing run since PC-TCS can use the previously computed flexure map to better center the stars. However, this option can only be used after a “raw” pointing run has been accomplished and the data analyzed.

After a flexure map has been created using the ANALYZE utility, incremental pointing runs can be made that will allow easier acquisition of stars than the “raw” pointing necessary when collecting the first data set.

This option if enabled will instruct PC-TCS to turn on flexure corrections during pointing data acquisition. If disabled, PC-TCS will not correct positions for flexure.

The full Map Option

As mentioned above, FDC divides the sky into a 1000 element (or cell) map made up of 25 declination “rows” and 40 Hour Angle “columns”.

The **Full Map** option is used to modify the sequence of rows that FDC will select to collect data. If the “Full Map” option is enabled, the FDC utility will attempt to collect data for each declination row starting at the row determined by the **Start δ** Option.

If disabled, only every other row in the grid from the starting declination will be selected for data collection. This allows the user to make a truncated pointing run – significantly reducing the time needed to collect a useful data set.

The user will most likely want to disable this option for the first pointing run performed. By disabling the option, the user can collect a data set, spread over the entire map, in half the time it takes to collect a full map data set.

When activated the **Full Map** option will be prefixed by the sideways chevron symbol (>>).

The Start δ Option

This option allows the user to choose the starting “row” of the grid cell map that the FDC uses to select stars from the FLEXURE.PCF catalog.

This option is useful in restarting an aborted data collection run at the approximate row that was last being processed. As a default, FDC assumes that the starting declination is coincident with the northeast corner of the grid cell map (Cell number 1 or declination +80 and hour angle 5 hour east.).

The Purge Data Option

The Purge Data option of the FDC subsubmneu is used to create/ restart the POINT.DAT data file. The option should be used whenever a new pointing data set is going to be collected.

WARNING: This option is destructive.

The option will delete any existing POINT.DAT file. The user should take care when executing this option as it will destroy all previously collected pointing information in the POINT.DAT file. It should only be used when a completely new pointing run is to be undertaken and only *open* prior to execution of the **Execute** option of the FDC subsubmenu.

The Execute Option

This option is used to start the FDC semi-automated pointing run procedure. When selected, this option will display a warning message informing the operator of impending automated operation. The operator may then confirm or abort the FDC procedure with an *Enter* or *ESC* keys respectively.

During the course of the pointing run, the telescope will be automatically commanded to move from one star to the next. This automated motion can be hazardous if the operator is unprepared for the slews and caution should be observed at all times during the pointing run so that people and instrumentation are outside the path of telescope motion.

It should also be noted that the FDC utility does not automatically enable “SLEW PERMIT” and the operator will have to either hold the “SLEW PERMIT” button on the PC-TCS paddle, or mask on the “SLEW PERMIT” bit with the **External I/O** subsubmneu, (See Chapter 3.) to allow true slews to take place.

Once in motion, the telescope may be stopped with either the *F9* (Motion Cancel command) or the *F10* (drive disable command) from the keyboard. The **PANIC** button on the PC-TCS paddle may also be used to effect an emergency stop of the telescope if danger is detected.

4.4.2 FDC Procedure

The following is a step by step procedure on how to perform a pointing run using the FDC Utility. The operator should adhere to the procedure closely as to avoid corruption of the flexure maps with poor data.

1. **Start PC-TCS.** Follow normal power up procedures for PC-TCS as outlined in Chapter 2. Make sure the UT Date and Time are correct. Turn the telescope sidereal Tracking on. Make sure no BIAS rates are active. Make sure that the “Flexure” corrections are *disabled* (for “raw” run) or *enabled* (for “Incremental” run). This can be accomplished with the **Corrections** subsubsubmneu of the **Parameters** subsubmenu. (See Chapter 3.)
2. **Select the FK5 catalog as input and Initialize the Coordinate system.** Select a visible bright star from the catalog near the zenith for this initialization. Move to and center the star and then “initialize” the PC-TCS coordinate system using the initialization procedure documented in Chapter 3 under the **Declare** submenu.

**DO NOT ATTEMPT TO REINITIALIZE
AFTER THIS POINT.**

3. **Mask on the SLEW PERMIT bit.** (Non – keyboard SLEW PERMIT systems only). If completely automated slews are desired, use the **External I/O** subsubsubmenu in the **Globals** subsubmneu to “OR” on the Slew Permit bit with the **Paddle I/O Alteration** utility. Omitting this step will require the operator to press and hold the Slew Permit button on the PC-TCS paddle if the telescope is to assume during star acquisitions.
4. **Use the FDC subsubmenu to select either FULL MAP or HALF MAP FDC Operation.** If a Full Map pointing run is to be completed (all declination rows), enable the **Full Map** option. If an abbreviated run is desired (every other Declination row), disable then **Full Map** option.
5. **Select the default Starting Declination.** Using the **Start Declination** option of the FDC subsubmenu, select +80 degrees (default) as the starting Declination for the pointing run.
6. **Restart the POINT.DAT data file.** Using the **Purge Data File** option of the FDC subsubmenu, restart the POINT.DAT file.
7. **Begin FDC pointing run.** Using the **Execute** option of the FDC subsubmenu, start the pointing run.

As this point the telescope will slew to the first star. Once the telescope motion is stable. The FDC window will prompt the operator to center the star as quickly as possible as to minimize the effects of pointing drifts due to atmospheric refraction. Once centered, confirm the operation to the FDC utility using the *Enter* key.

If, for some reason, the run needs to be terminated, the user can do so by hitting the *ESC* key. Note that aborting a run will NOT destroy or necessarily invalidate any data previously collected. The run may be restarted by repeating steps 5 and 7 – *omitting* step 6. the user should take care to select a declination row near the declination that was last being processed before starting a run.

Once the star has been confirmed or bypassed, the FDC utility will select a new cell, search the catalog for a close star and move the telescope to the new position. In some cases, the *same* star may be selected for this new cell. This is not an error in the utility and the operator should follow the data collection procedures as before.

Repeat this search, point, center, and confirm procedure until FDC has exhausted the cells in the Pointing Grid.

Note: If bogus data is accidentally collected for a given cell of the map – the user should note the cell number for later deletion from the POINT.DAT file. This deletion can be accomplished using any text editing program capable of

nondocument file (program source file) editing. Simply delete the line in the POINT.DAT file for the bad cell collected. (the cell number is given as the first number on each of the data lines). The user must also modify the number on the first line of the file to reflect the new number of data points in the file (plus 2). The file uses fixed length records of 128 bytes and the user should take care when editing this first line, not to insert or delete characters.

8. **Terminate PC-TCS and Analyze the pointing data.** Once the pointing run has been completed, the operator may resume normal PC-TCS operations. (Note that if the Slew Permit bit was masked on, the telescope will continue to slew to new positions without operator permission. The user may wish to disable this auto-slew mode by unmasking the slew permit bit with the **Paddle I/O Alteration** utility in the **Globals** subsubmenu.

The pointing data collected must first be analyzed with the ANALYZE program before the flexure corrections can be used to correct telescope pointing. When the operator is ready to perform this analysis, PC-TCS must be terminated and removed from memory. Perform a normal shutdown of PC-TCS as outlined in Chapter 3 – **Shutdown** submenu.

The operator is now ready to analyze the pointing data and construct the flexure files data files PCTCSR.FLX and PCTCSD.FLX using the ANALYZE Utility described in the next section.

4.5 Analyze Utility

The ANALYZE Utility is a separate program that will reduce the pointing data collected with the FDC utility and produce the flexure correction files PCTCSR.FLX and PCTCSD.FLX. PC-TCS can subsequently use these files to correct telescope pointing errors induced by telescope flexure.

Operationally, ANALYZE reads the POINT.DAT file produced by the FDC utility and performs a mathematical smoothing of the data over a 18,000 element map that covers the same range of hour angle and declination as the pointing grid cell map – but spaced on one (1) degree cell boundaries.

ANALYZE uses an interpolating fifth-degree polynomial in hour angle and declination to define a smooth function which passes through all raw data points and whose first order derivatives are continuous. The algorithm is designed for use with irregularly distributed data sets such as those produced by the FDC Utility.

The program is computationally intensive and may take several minutes to reduce the data set depending upon the number of points in the POINT.DAT file, the speed of the computer and whether or not a math coprocessor is available on the system.

To start the analysis, the user needs only to type the following command at the DOS prompt . . .

```
C:\TCS> ANALYZE POINT.DAT←
```

At this point the screen will clear and subsequently display an informational banner. The program will then attempt to read the POINT.DAT data file and begin reduction of the Hour Angle error surface.

After a time the screen will again clear and the user will be asked if they wish to construct the PCTCSR.FLX file. If the user answers in the affirmative the program will delete any existing PCTCSR.FLX file and write the new file to the disk. The procedure is repeated for the declination error if the flexure correction option in the **Correction** subsubsubmenu is enabled.

It should be noted that if flexure correction are to be used properly, the operator must initialize the PC-TCS coordinate system with the flexure correction option turned on – before selecting and processing the coordinates for the star to be used in the initialization.

The user should consult Appendix G for information on any error messages that might be displayed during execution of the ANALYZE programs or PC-TCS with regards to flexure correction operation.

Figure 4.2: The Raster Utility Scan Pattern

4.6 Raster Scan Utility

The Raster Utility provides the user with the ability to perform an automated scan of the sky in the pattern depicted in figure 4.2. The scan can be used to search for objects whose exact location is uncertain or to precisely register an object in detection equipment with a small visual field of view (such as a photometer).

The Raster Utility is available via the **Utilities** subsubsubmenu or from a Command file. When initiated from the keyboard, the Raster utility will query the user for five (5) parameters related to the scan.

The first query, RA Step Size, refers to the length of the motion made in Right Ascension in arcseconds on the sky. The second question, the number of steps in Right Ascension refers to the number of Right Ascension moves that will be made before a declination motion is attempted. The third question, DEC step size, is the size in arcseconds of the step in declination that will be made in declination before termination of the Raster. The last query asks the user for the length of time in seconds that the Scan should be paused after each move. This “dwell” time is the time delay that Raster will wait before attempting the next step in the scan.

Once started, the Scan will perform each move as requested, pausing between each move for the dwell time specified or until an *ESC* key is hit. If Raster is initiated from inside a Command file (see Appendix A and the section on **Command Files** in this Chapter), the Raster can be caused to terminate from either the keyboard or by a high signal from any one of the four external TTL inputs documented in Appendix E – figure E.2.

Figure 4.3: The Spiral Utility Scan Pattern

4.7 Spiral Scan Utility

The Spiral Utility provides the user with the ability to perform an automated scan of the sky in the pattern depicted in figure 4.3. the scan can be used to search for objects whose exact location is uncertain or to precisely register an object in detection equipment with a small visual field of view (such as a photographer).

The Spiral Utility is available via the **Utilities** subsubmenu or from a Command file. When initiated from the keyboard, the Spiral Utility will query the user for three (3) parameters related to the scan.

The first query, Motion Step Size, refers to the length of the motion made in Right Ascension and Declination in arcseconds on the sky. The second question, the number of loops, refers to the number of full circle motion loops that will be made before termination of the Spiral. The last query asks the user for the length of time in seconds that the Scan should be paused after each move. This “dwell” time is the time delay that Spiral will wait before attempting the next step (not loop) in the scan.

Once started, the Scan will perform each move as requested, pausing between each move for the dwell time specified or until an *ESC* key is hit. If Spiral is initiated from inside a Command file (see Appendix A and the section on **Command Files** in this Chapter), the Spiral can be caused to terminate from either the keyboard or by a high signal from any one of the four external TTL inputs documented in Appendix E – figure E.2.

4.8 saTrak Utility

The Satellite Tracking Utility (**saTrak**) available under PC-TCS Version 6.X requires an additional software package called SATPRO available from COMSOFT. The SATPRO program will allow the user to generate satellite ephemerides files that can be imported and read by PC-TCS for the purposes of tracking Earth Orbiting satellites.

Since this utility will automatically allow the telescope to follow a preprogrammed set of high speed rates, access to this function should be restricted to advanced user via the security system built into PC-TCS. The password for access to this utility may be obtained from your site supervisor.

Note: PC-TCS uses a “Velocity” tracking method and does not track “position” of the satellite on the sky. During a track, rates are feed to the control system from the ephemeris file generated by SATPRO. PC-TCS follows these rates to track the satellite. The only time PC-TCS uses positions from the ephemeris file is when the telescope is pointed at the starting coordinates just prior to the start of an actual track.

Once the track files have been generated with SATPRO and imported to the PC-TCS system area, the following procedure may be used to perform a satellite track.

Enter Password: As mentioned above, access to the satellite tracking utility will usually be restricted. After PC-TCS is up and running the user should bring up the password enter window by hitting the *Cntrl* and *BS* KEYS simultaneously. Enter the password given to you by your site supervisor. After password entry the user should check the **ACCESS** line on the **PC – TCS Runtime Window** to verify that the level of clearance is appropriate to the level necessary to operate this **saTrak** utility. (See chapter 2 under PC – TCS security.)

Track Preflight: Before proceeding with a track the user should initialize the telescope coordinate system as documented in Chapter @ of this Manual. Sidereal Tracking should also be enabled (*ALT T*).

Load Track File: At this point the user should proceed to the **UTILITIES – saTrak** menu item and press *Enter*. A window will be displayed showing the file names of the available track files found on the PC-TCS system disk. These files all have the default extension .TRK (not displayed). Any track files without this extension will NOT be accessible by PC-TCS. The user should select one of the displayed files via the method described in Chapter 2 of this manual under **The PC-TCS File Window**. Once selected, PC-TCS will attempt to load the ephemeris information into PC-TCS memory. The file select window will appear and be replaced by the Satellite Tracking Progress Monitor Window. The Satellite

Tracking Progress Monitor shows information concerning current track start time (UT); start position in RA and Dec and Altitude and Azimuth; track rates in arcseconds per second; acceleration in arcseconds per second; shadow status and remote start hardware input status. The track start time and position may be incremented or decremented using the arrow keys. The ↑ key increments the track start time by one ephemeris entry, the ↓ decrements the track start time by one ephemeris entry, the → key increments the track start by 10 ephemeris entries.

Prelaunch Checklist: Before starting the track the user must command the telescope to point at the starting RA and Dec or Altitude and Azimuth for the track start position currently displayed in the Satellite Tracking Monitor Window. Commanding the telescope to move to the displayed RA and Dec coordinates is accomplished by pressing the “P” key. Care should be taken before using this command with long lead times before a pass is to take place since the RA and Declination of the track start may not be accessible to the telescope. This method should be used only if a “Real” track is to be performed. If a test is going to be performed, pressing the “T” key will move the telescope to the displayed altitude and Azimuth. Test tracks are desirable when a trial of the track is to be performed well in advance of a real pass. Trial tracks give observers a chance to check telescope motion and dome positioning without having to wait for the actual start time of a satellite pass. When ready to perform an actual track, user should use the “P” key to position the telescope at the correct start **RA** and Dec a few minutes before the start time displayed on the Progress Monitor window.

Starting the Track: Once the telescope has moved to the starting position, the observer has three ways of starting the track. Hitting the “A” key will instruct PC-TCS to automatically start the track at the displayed UT start time. When the “A” key is pressed, the Progress Monitor will display a white-on-red field just below the Start Time file showing the remaining until start of the track. This method is useful if the object to be tracked is too faint to be observed visually and the observer is willing to trust the accuracy of the ephemeris generated by the SATPRO. Typically, satellites are “early” or “late” by a few seconds – an effect known as “along-track” error. For slow moving satellites this is usually not a problem since an along track error will still put the satellite in the field of view. However, fastest satellites may be a degree or more from the predicted start position at the predicted start time due to along-track error. In this scenario – the observer would be better off to start the track manually when he sees the satellite enter the field of view. A manual start is performed by pressing the *Enter*. This will immediately start the track. A remote start option is also available. This option allows the manual start a satellite tracking sequence using the wobble hardware wobble TTL input as a trigger. To enable this option, press the “R” key on the

keyboard. Pressing the “R” key a second time will disable this option. Once enabled, the remote option will wait for a high level on the hardware wobble TTL input and start the track on this signal. Once the track started the TTL signal has no more effect on the satellite track unless it is subsequently suspended.

Suspending a Track: Once a track has been started, the track may be temporarily suspended by pressing the space bar on the keyboard. This command PC-TCS to halt the track at the current position and wait for the user to command a restart. Pressing the *Enter* again will restart the track at the next commanded velocity from the ephemeris file. If the remote option has been enabled, the track can be started by providing a high level TTL signal to the Hardware wobble TTL input. This is useful in allowing a satellite to “catch-up” during an auto-started track where the satellite was “late” or a track started prematurely using the *Enter* or remote start option.

Moving Along Track: While a track is in progress the user may command the telescope to move “along-track” using the “+” and “-” keys. This allows searching for satellites in tracks started using the auto-start feature of the saTrak Utility. The “+” key moves telescope forward along the track 0.1 seconds of time, conversely the “-” key moves the telescope backward 0.1 seconds along-track.

Aborting a Track: After a track has been started, the observer may abort the track at any time by pressing the *ESC* key. This will immediately halt the track and return the user to the **Utilities** subsubsubmenu. A track allowed to run to completion will automatically halt in this manner.

Some note on Satellite Tracking with PC-TCS

Open loop satellite tracking like that performed by PC-TCS can be highly rewarding if done properly and with great patience, or tedious frustrating if care is not taken to properly prepare for the track.

To better ensure success the observer should take great care to obtain the very latest orbital elements for propagation with the SATPRO utility software. Orbital order tan a week (especially for low earth orbit targets) are almost useless. SATPRO also requires that you have the proper site coordinates, date and time set for any given propagation to be accurate. Even with these propagations, the satellite is likely to be off the predicted path and time by varying amounts.

To facilitate the observer in tracking these elusive objects, PC-TCS performs several system modifications when the **saTrak** utility is invoked. Because satellite tracking can

produce loads on the servo algorithm incorporated into PC-TCS, PC-TCS will help the observer by automatically performing the following operations . . .

Enable SLEW PERMIT: Since the telescope must be able to slew while tracking fast satellites, PC-TCS will automatically enable slew permit during **saTrak** operation. SLEW PERMIT will be disabled automatically upon aborting or normal termination of a satellite track – provided SLEW PERMIT was not permanently enabled beforehand by the user.

Modification of Drift and Guide Rates: PC-TCS modifies the Drift rates to be the maximum velocity of the track if less than 500 arcseconds per second or 500 arcseconds per second if the max track rate is greater than 500 arcseconds per second. The Guide rates are set to one third of the drift rate set as explained above. This procedure allows for better “guiding” and centering control during satellite tracks. The rates are reset to their previous values upon exit of the **saTrak** utility.

Disable Active Command File Processing: To ensure a track is not interrupted by a Command File command to move the telescope elsewhere, Active command file processing is disabled upon entry to the **saTrak** utility. Command File processing will continue upon exit if previously active.

Appendix A

Command File Mnemonics

ACCESS XXXXXXXX: Allows changing the security level. XXXXXXXX is the appropriate password for the level.

ABERON: Turns the Position Correction for Aberration ON. Used to correct Telescope Pointing.

ABEROFF: Turns the Position Correction OFF.

BIASDEC $\pm$ ss.s: Defines the Declination component of bias rate. Use the BIASRA to define full BIAS rate. Use BIASON and BIASOFF to activate and deactivate, respectively, the rate.

BIASOFF: Deactivates previously declared bias rates. (See Chapter 3 – **Motion** submenu.)

BIASON: Activates previously declared bias rates. (See Chapter 3 – **Motion** – submenu.)

BIASRA $\pm$ ss.s: Defines Right Ascension component of bias rate. Use with BIASDEC to define full BIAS rate. Use BIASON and BIASOFF to activate and deactivate, respectively, the rate.

CDBIASOFF: Turns the Cosine δ correction OFF for any declared BIAS rates.

CDBIASON: Turns the Cosine δ correction ON for any declared BIAS rates.

CDDRIFTOFF: Turns the Cosine δ correction OFF for Drift input.

CDDRIFTON: Turns the Cosine δ correction ON for Drift input.

CDGUIDEOFF: Turns the Cosine δ correction OFF for Guide input.

CDGUIDEON: Turns the Cosine δ correction ON for Guide input.

CDWOBOFF: Turns the Cosine δ correction OFF for any declared WOBBLE Vectors.

CDWOBON: Turns the Cosine δ ON for any declared WOBBLE Vectors.

CLEARDIFF: Performs the same functions as the F5 hotkey and the Clear diff option of the **Declare** submenu.

DECLAREINIT: Instructs PC-TCS to make the NEXT Object coordinates the current coordinates. Used to initialize the telescope coordinate system at the beginning of an observing session. (See Chapter 3 – **Declare** submenu.)

DECLAREREF: Saves the current coordinates as the REFERENCE Object. Used with MOVREF to move to the saved coordinate.

DECLARESTOW: Instructs PC-TCS to make current position in Hour Angle and Declination the New Stow Position of the Telescope. Subsequent MOVSTOW command will move the telescope to this HA and DEC.

DISEPOCH # # # # #: Changes the display epoch for current commanded and reference displays.

DRIFTDEC ±ss.s: Defines Declination component of drift date.

DRIFTRA ±ss.s: Defines Right Ascension component of drift rate.

ELAZ ee.e aaa.a: Instructs PC-TCS to move the telescope to the specified Elevation (ee.e) and Azimuth (aaa.a).

END: Immediately Terminates command file execution.

EPOCH # # # # #: changes the input epoch for manual NEXTRA and NEXTDEC precession computations. (Should not be used to change epoch when catalogs are being used NEXT object input.)

FILEPERM filename: Requests that PC-TCS use the permanent catalog file specified by “filename” for NEXT coordinate input. Use with OBJECT command to read file. Note that the file name specified should not include the “.CAT” extension

FLEXON: Turns the Position Corrections for Telescope Flexure ON. Used to correct Telescope Pointing

FLEX OFF: Turns the Position Correction for Telescope Flexure OFF.

FLEXRATEON: Turns the Rate Correction for Telescope Flexure ON. Used to Correct Telescope tracking for differential flexure

FLEXRATEOFF: Turns the Rate Corrections for the Telescope Flexure OFF.

GOSUB # # #: Branches to line number # # # # in the command file. Execution continues from this new line number until a RETURN command is executed. Execution continues at the original line number plus one (+1) after the RETURN statement is encountered. (Note: subroutines may be nested of a maximum of nine (9) levels.)

GOTO # # #: Unconditionally branches to line number # # # # in the command file.

GUIDEDEC ±ss.s: Defines of Declination component of guide rate.

GUIDERA ±ss.s: Defines Right Ascension component of the guide rate.

IF NEXT ###: Branches to line number #### if NEXT position is within telescope limits.

JUPITER: Computes the position of the planet Jupiter as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

KILL: Disables the servo drive output of PC-TCS to the servo amplifiers. This command is identical to the **Disable** command described in Chapter 2 of this manual. (Also see the “UNKILL” command later in this Chapter.)

LUNARRATEOFF: Turns the lunar rate correction OFF during tracking.

LUNARRATEON: Turns the lunar rate correction ON during tracking.

MANUAL: Resets MANUAL (Keyboard) data entry mode. Note this command is not especially necessary if NEXTRA and NEXTDEC are used but is included for continuity inside command file operation.

MARS: Computes the position of the planet Mars as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

MOVEB1: Moves telescope to the current minus previously declared WOBBLE vector position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVEB2: Moves telescope to the current plus position declared WOBBLE vector position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVNEXT: Move telescope to the previously declared NEXT object. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVOFF: Moves telescope to the current plus previously declared OFFSET vector position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVREF: Moves telescope to the previously declared REFERENCE object. (See DECLAREREF command.) Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVSTOW: Moves telescope to the STOW position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

NEPTUNE: Computes position of the planet Neptune as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

NEXTDEC ±ddmmss.s: Defines NEXT position Declination. Use NEXTRA to define full NEXT position. Use MOVNEXT to move the NEXT position.

NEXTOBJECT: Increments the object number by one and attempts to read that object number from the currently selected catalog file. Object number is modified via the OBJECT command.

NEXTRA hhmmss.s: Defines NEXT position Right Ascension. Use NEXTDEC to define full NEXT position. Use MOVNEXT to move to NEXT position.

NUTON: Turns the Position Correction for Nuton ON. Used to correct Telescope Pointing.

NUTOFF: Turns the Position Correction for Nutation OFF.

OBJECT # # # #: Instructs PC-TCS to retrieve object number # # # # from the previously specified catalog file. See FILEUSER and FILEPERM commands.

OFFDEC ±ddmmss.s: Defines OFFSET vector Declination. Use OFFRA to define full OFFSET vector. Use MOVOFF to move to offset.

ORBIT object: Requests that PC-TCS compute the current position of the object whose orbital parameters are in the file “object”. Note that the file name specified should not include the “.ORB” extension.

PARAOFF: Turns the Position Correction for Topocentric Parallax OFF.

PARAON: Turns the Position Correction for Topocentric Parallax ON. Used to correct Telescope Pointing.

PRECOFF: Turns the Position Correction for Precession OFF.

PRECON: Turns Position Correction for Precession ON. Used to correct Telescope Pointing.

PROPEROFF: Turns the Position Correction for Proper Motion OFF.

PROPERON: Turns the Position Correction for Proper Motion ON. Used to correct Telescope Pointing.

PLUTO: Computes the position of the planet Pluto as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

PRECON: Turns the Position Correction for Precession ON. Used to correct Telescope Pointing.

PRECOFF: Turns the Position Correction for Precession OFF.

PROMPT message: Pauses command file execution for user input with the instructional message “message”. Command file execution will terminate if the ESC key is hit – or continue if any key is hit.

PROPON: Turns the Position Correction for Proper Motion ON. Used to correct Telescope Pointing.

PROPOFF: Turns the Position Correction for Proper Motion OFF.

RASTER rr.r dd.d RR DD tt.t key: Invoke the Raster scan utility with Right Ascension step size of rr.r arcseconds, declination step size of dd.d arcseconds, for RR step in Right Ascension, DD step in Declination, with a dwell time of tt.t seconds. The “key” parameter is used to decide the input upon which the Raster will terminate. Zero (0) will perform the full raster scan with no interruption capability, one (1) will terminate the raster if the TTL S1 input is high, two (2) will terminate the raster if the TTL S2 input is high, three (3) will terminate the raster if the TTL S3 input is high, four (4) will terminate the raster if the TTL S4 input is high, five (5) will terminate the raster if a keyboard character input is detected.

RECALLPOS # #: Recalls a previously saved position (see SAVEPOS command) and copies the coordinates to the NEXT position registers. The telescope can be commanded to move to this position using the MOVNEXT command. # # can have a value between 1 and 15.

REFRACOFF: Turns the Position Correction for Atmospheric Refraction OFF.

REFRACON: Turns the Position Correction for Atmospheric Refraction ON. Used to correct Telescope Pointing.

REFRRATEOFF: Turns the refraction rate correction OFF during tracking.

REFRRATEON: Turns the refraction rate correction ON during tracking.

REMOTECHKOFF: Turns Remote Serial Access command checksumming OFF.

REMOTECHON: Turns Remote Serial Access command checksumming ON.

REMOTEOFF: Turns Remote serial access OFF.

REMOTEON #: Turns Remote Serial access ON with a baudrate selection number of #. The options are 1= 110, 2= 300, 3= 1200, 4= 2400, 5= 4800, and 6= 9600 baud. (See Chapter 3 – **Remote** Subsubsubmenu.)

RETURN: Used to GOSUB to define the end of a subroutine. Execution will continue at the statements immediately after the GOSUB command.

SATURN: Computes the position of the planet Saturn as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

SAVEPOS # #: Saves the current position of the telescope to the save register # #. The position can later be retrieved via the RACALLPOS command.

SLEWOFF: Deactivates slew permit for slow slews from object to object. (See Chapter 3 – **External I/O** subsubsubmenu.) Note: This command is NOT available if your installation has a Slew Permit button that must be pressed to enable slew.

SLEWON: Activates slew permit for fast slews from object to object. (See Chapter 3 – **External I/O** subsubsubmenu.) Note: This command is NOT available if your installation has a Slew Permit button that must be pressed to enable slew.

SOLARRATEOFF: Turns the Solar rate correction OFF during tracking.

SOALRRATEON: Turns the Solar rate correction ON during tracking.

SPIRAL ss.s LL tt.t key: Invokes the Spiral scan utility with Right Ascension and Declination step size of ss.s arcseconds, for LL loops, with a dwell time of tt.t seconds. The “key” parameter is used to decide the input upon which the Spiral will terminate. Zero (0) will perform the full raster scan with no interruption capability, one (1) will terminate the spiral if the TTL S1 input is high, two (2) will terminate the spiral if the TTL S2 input is high, three (3) will terminate the spiral if the TTL S3 input is high, four (4) will terminate the spiral if the TTL S4 input is high, five (5) will terminate the Spiral if a keyboard character input is detected.

SUN: Computes the position of the Sun as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

TRKOFF: Turns the sidereal tracking off. (See Chapter 3 – **Motion** submenu.)

TRKON: Turns the sidereal tracking on. (See Chapter 3 – **Motion** submenu.)

UNABEROFF: Turns the deconvolution for Aberration OFF.

UNABERON: Turns the deconvolution for Aberration ON.

UNFLEXOFF: Turns the deconvolution for Flexure OFF.

UNFLEXON: Turns the deconvolution for Flexure ON.

UNKILL: Reenables the servo drive output of PC-TCS to the servo amplifiers after a hard limit violation or PANIC situation or a disable accomplished with the KILL command. (Also see the “KILL” command earlier in this Chapter.)

UNREFROF: Turns the deconvolution for Refraction OFF.

UNREFRON: Turns the deconvolution for Refraction ON.

URANUS: Computes the position of the planet Uranus as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

VENUS: Computes the position of the planet Venus as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

WAITDL ss.s: Pauses command file processing for ss.s seconds.

WAITLST hhhmmss.s: Pauses command file processing until LST time hhhmmss.s.

WAITMO: Pauses command file processing until telescope motion is stable.

WAITNEXT: Pauses command file processing until NEXT Object is within Telescope pointing limits.

WAITONNEXT: Pauses command file processing until NEXT Object is outside Telescope pointing limits.

WAITNS1: Pauses command file processing until external TTL input S1 goes low. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITNS2: Pauses command file processing until external TTL input S2 goes low. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITNS3: Pauses command file processing until external TTL input S3 goes low. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITNS4: Pauses command file processing until external TTL input S4 goes low. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITREM: Pauses command file processing until a remote command CLIRESUME command is received.

WAITS1: Pauses command file processing until external TTL input S1 goes high. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITS2: Pauses command file processing until external TTL input S2 goes high. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITS3: Pauses command file processing until external TTL input S3 goes high. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITS4: Pauses command file processing until external TTL input S4 goes high. (See Chapter 3 – **External I/O** subsubsubmenu.)

WAITUT hhmmss.s: Pauses command file processing until UT time hhmmss.s.

WOBDEC ±ddmmss.s: Defines WOBBLE vector Declination. Use WOBRA to define full WOBBLE vector. Use MOV B1 and MOV B2 to move to beams.

WOB OFF: Turns the remote hardware wobble sensing off. (See Chapter 3 – under **Remote** subsubsubmenu.)

WOBRA hhmmss.s: Defines WOBBLE vector Right Ascension. Use WOBDEC to define WOBBLE vector. Use MOV B1 and MOV B2 to move to beams.

Appendix B

Remote Mnemonics

All Remote command are received by PC-TCS over one or more of the eight possible serial ports. Remote commands reception is enable via the **Remote** subsubsubmneu (see Chapter 3 – the **Input** subsubsubmenu.) or via Command Files.

Remote commands are a subset of the Command File mnemonics along with a few additions applicable to remote operations. Serial command may be “checksummed” for better command validation.

If the command is determined to be valid and executed by PC-TCS, the command will be processed and an acknowledgment will be returned to the user in the form of a alternating uppercase “E” or lowercase “e” character embedded in the telemetry stream in appropriate channel columns (see Appendix C), along with a hardware handshake in the form of a change of state in the signal on the CTS line of the serial channel used (see Appendix G). Erroneous commands will *not* be acknowledged with the CTS state change and the telemetry stream will hold an error code describing the problem in the appropriate channel column. (See Appendix C.)

A Note on Remote Command Transmission

The user should be careful not to send a new remote command until the previous command has been acknowledged by PC-TCS. The user should wait for the CTS state change and/or the telemetry stream update before sending the next command. Commands are buffered by PC-TCS in a common memory area. Sending a new command before PC-TCS acknowledges receipt of the previous command may cause an overwrite of the command buffer and affect proper command execution.

The following pages describe the Remote commands that PC-TCS will recognize.

ABERON: Turns the Position Correction for Aberration ON. Used to correct Telescope Pointing.

ABEROFF: Turns the Position Correction for Aberration OFF.

ACCESS XXXXXXXX: Allows changing the security level. XXXXXXXX is the approximate password for the level.

BIASDEC $\pm$ ss.s: Defines Declination component of bias rate. Use with BIASRA to define full BIAS rate. Use BIASON and BIASOFF to activate and deactivate respectively.

BIASOFF: Deactivates previously declared bias rates. (Chapter 3 – **Motion** submenu.)

BIASON : Activates previously declared bias rates. (Chapter 3 - **Motin** sub menu.)

BIASRA $\pm$ ss.s: Defines Right Ascension component of bias rate. Use with BIASDEC to define full BIAS rate. Use BIASON and BIASOFF to activate and deactivate, respectively, the rate.

CANCEL: Cancel telescope move command – identical to the Motion Cancel Menuitem documented in Chapter 2.

CDBIASOFF: Turns the Cosine δ correction OFF for any declared BIAS rates.

CDBIASON: Turns the Cosine δ correction ON for any declared BIAS rates.

CDDRIFTOFF: Turns Cosine δ correction OFF for any declared DRIFT rates.

CDDRIFTON: Turns the Cosine δ correction ON for any declared DRIFT rates.

CDGUIDEOFF: Turns Cosine δ correction OFF for any declared GUIDE rates.

CDGUIDEON: Turns Cosine δ correction ON for any declared GUIDE rates.

CDWOBOFF: Turns Cosine δ correction OFF for any declared WOBBLE Vector.

CDWOBON: Turns Cosine δ correction ON for any declared WOBBLE Vector.

CLEARDIFF: performs same function as the F5 hotkey and the Clear diff option of the **Declare** submenu.

CLIRSUME: Enables CLI command file processing in the event a CLI WAITREM command has been executed. (See WAITREM command in the Command File Mnemonics).

COMFILELOAD filename: Requests that PC – TCS attempt to load the Command File named “filename” from disk command input. Use with COMFILEON command to execute the Command File. Note that the file name specified should *not* include the “TCS” extension.

COMFILEOFF: Suspends Command File execution if one is in progress.

COMFILEON: Resumes or begins Command File execution if a Command File has been loaded previously with the **COMFILELOAD** command.

DECLAREINIT: Instructs PC – TCS to declare the NEXT Object coordinates as the current coordinates. Used to initialize the telescope coordinate system at the beginning of an observing session. (See chapter 3 – **Declare** submenu.)

DECLAREREF: Saves the current coordinates as the REFERENCE Object. Used with MOVREF to move to the saved coordinate.

DECLARESTOW: Instructs PC – TCS to make the current position in Hour Angle and declination the New Stow Position of the Telescope. Subsequent MOVSTOW commands will move the telescope to this HA and DEC.

DISEPOCH #####.##: Changes the display epoch for current commanded and reference displays.

DRIFTDEC ±ss.s: Defines Declination component of the drift rate.

DRIFTRA ±ss.s: Defines Right Ascension component of drift rate.

ELAZ ee.e aaa.a: Instructs PC – TCS to move the telescope to the specified Elevation (ee.e) and Azimuth (aaa.a)

EPOCH #####.##: Changes the input epoch for manual NEXTRA and NEXTDEC precession computations. (Should be used to change epoch when catalogs are being used for NEXT object input).

FLEXONL: Turns the Position Correction for the Telescope Flexure ON. Used to correct Telescope Pointing.

FLEXOFF: Turns the Position Correction for the Telescope Flexure OFF.

FLEXRATEON: Turns the Rate Correction for the Telescope Flexure ON. Used to correct Telescope tracking for differential flexure.

FLEXRATEOFF: Turns the Rate Correction for Telescope Flexure OFF.

FILEPERM filename: Requests that PC – TCS use the permanent catalog file specified by “filename” for NEXT coordinate input. Use with OBJECT command to read file. Note that the file name specified should not include the “PCF” extension.

FILEUSER filename: Requests the PC – TCS use the user catalog file specified by “filename” for NEXT coordinate input. Use with OBJECT command to read file. Note that the file name specified should not include “CAT” extension.

GUIDEDEC ±ss.s: Defines Declination component of guide rate.

GUIDEDERA ±ss.s: Defines Right Ascension component of guide rate.

JUPITER: Computes the position of the planet Jupiter as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

KILL: Disables the servo drive output of PC- TCS to the servo amplifiers. This command is identical to **Disable** command desired in Chapter 2 of this manual. (Also see the “UNKILL” command later in this Chapter.)

LIMITENABLE: Used to reenable the limit detection software after a **LIMITOVERRIDE** command has been issued.

LUNARRATEOFF: Turns the lunar rate correction OFF during tracking.

LUNARRATEON: Turns the lunar rate correction ON during tracking.

MANUAL: Resets MANUAL (Keyboard) data entry mode. Note this command is not specifically necessary if NEXTRA and NEXTDEC are use but is included for continuity.

MARS: Computes the position of planet Mars as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

MERCURY: Computes the position of the planet Mercury as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

MOON: Computes the position of the Moon as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

MOVB1: Moves telescope to the current minus previously declared WOBBLE vector position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVB2: Moves telescope to current plus previously declared WOBBLE vector position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVNEXT: Move telescope to the previously declared NEXT object. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVOFF: Moves telescope to the current plus previously declared OFFSET vector position. Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVEREF: Moves telescope to the previously declared REFERENCE object. (See DECLAREREF command.) Command pauses execution until motion is completed (stable). (See Chapter 3 – **Move** submenu.)

MOVSTOW: Moves telescope to the STOW position. Command pauses execution until motion completed (stable). (See Chapter 3 – **Move** submenu.)

NEPTUNE: Computes the position of the planet Neptune as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

NEXTDEC ±ddmmss.s: Defines NEXT position Declination. Use NEXTRA to define full NXT position. Use MOVNEXT to move to NEXT position.

NEXTRA hhmmss.s: Defines NEXT position Right Ascension. Use NEXTDEC to define full NEXT position. Use MOVNEXT to move to NEXT position.

NOP: A “do nothing” (no operation) command that allows the user to test the communications integrity of the serial channel in use. Useful in development and debugging of PC-TCS serial communications software.

NUTON: Turns the Position Correction for Nutation ON. Used to correct Telescope Pointing.

NUTOFF: Turns Position Correction for Nutation OFF.

OBJECT # # # #: Instructs PC-TCS to retrieve objects number # # # # from the previously specified catalog file. See FILEUSER and FILEPERM commands.

OFFDEC ±ddmmss.s: Defines OFFSET vector Declination. Use OFFRA to define full OFFSET vector. Use MOVOFF to move to offset.

OFFRA hhmmss.s: Defines OFFSET vector Right Ascension. Use OFFDEC to define full OFFSET vector. Use MOVOFF to move to offset.

ORBIT object: Requests that PC-TCS compute the current position of the object whose orbital parameters are in the file “object”. Note that the file name specified should *not* include the “.ORB” extension.

PARAOFF: Turns the Position Correction for Topocentric Parallax OFF.

PARAON: Turns the Position Correction for Topocentric Parallax On. Used to correct Telescope Pointing.

PRECOFF: Turns the Position Correction for Precession OFF.

PRECON: Turns the Position Correction for precession ON. Used to correct Telescope Pointing.

PROPEROFF: Turns the Position Correction for Proper Motion OFF.

PROPERON: Turns the Position Correction for Proper Motion ON. Used to correct Telescope Pointing.

PLUTO: Computes the position of the planet Pluto as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

RATSER rr.r dd.d RR DD tt.t key: Invokes the Raster scan utility with Right Ascension step size of rr.r arcseconds, declination step size of dd.d arcseconds, for RR step in Right Ascension, DD steps in Declination, with a dwell time of tt.t seconds. The “key” parameter is used to decide the input upon which the Raster will terminate. Zero (0) will perform the full scan with no interruption capability, one (1) will terminate the raster if the TTL S1 is high, tow (2) will terminate the

raster if the TTL S2 input is high, three (3) will terminate the raster if the TTL S3 is high, four (4) will terminate the raster if the TTL S4 is high, five (5) will terminate the raster if a keyboard character input is detected.

RECALLPOS # #: Recalls a previously saved position (see SAVEPOS command) and copies the coordinates to the NEXT position registers. The telescope can be commanded to move to this position using the MOVNEXT command. # # can have a value between 1 and 15.

REFRACOFF: Turns the Position Correction for Atmospheric Refraction OFF.

REFRACON: Turns the Position Correction for Atmospheric Refraction ON. Used to correct Telescope Pointing.

REFRRATEOFF: Turns the refraction rate correction OFF during tracking.

REFRRATEON: Turns the refraction correction ON during tracking.

REMOTECHKOFF: Turns the Remote Serial Access command checksumming OFF.

REMOTECHKON: Turns the Remote Serial Access command checksumming ON.

SATURN: Computes the position of the planet Saturn as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

SAVEPOS # #: Saves the current position of the telescope to the save register # #. The position can later be retrieved via the RECALLPOS command.

SATDATE mm/dd/yyyy: Sets the UT date for PC-TCS operation. Date string must conform to the syntax described – single digit months and days must have preceeding zeros.

SETTIME hhmmss.s: Sets the current UT time for PC-TCS operation. Time string must conform to the syntax described – single digit minutes and seconds must have preceeding zeros.

SLEWOFF: Deactivates slew permit for slow slews from object to object. (See Chapter 3 – **External I/O** subsubsubmenu.) Note: This command is NOT available if your installation has a Slew Permit button that must be pressed to enable slew.

SLEWON: Activates slew permit for fast slews from object to object. (See Chapter 3 – **External I/O** subsubsubmenu.) Note: This command is NOT available if your installation has a Slew Permit that must be pressed to enable slew.

SOLARRATEOFF: Turns the Solar rate correction OFF during tracking.

SOLARRATEON: Turns the Solar rate correction ON during tracking.

SPIRAL ss.s LL tt.t key: Invokes the Spiral scan utility with Right Ascension and Declination step size of ss.s arcseconds, for LL loops, with a dwell time of tt.t seconds. The “key” parameter is used to decide the input upon which the Raster will terminate. Zero (0) will perform the full scan with no interruption capability,

one (1) will terminate the raster if the TTL S1 is high, two (2) will terminate the raster if the TTL S2 input is high, three (3) will terminate the raster if the TTL S3 is high, four (4) will terminate the raster if the TTL S4 is high, five (5) will terminate the raster if a keyboard character input is detected.

STEPDEC: $\pm\# \# \# \#$ Commands the telescope to move $\# \# \# \#$ “step” in declination. Note: “Steps” are true stepper motor steps and are therefore subarcsecond in size. The command is meant to give “guiding” capabilities to operators or smart instruments using the serial command channels to control the telescope.

STEPRA: $\pm\# \# \# \#$ Commands the telescope to move $\# \# \# \#$ “steps” in Right Ascension. Note: “Steps” are true stepper motor steps and are therefore subarcseconds in size. The command is meant to give “guiding” capabilities to operators or smart instruments using the serial command channels to control the telescope.

SUN: Computes the position of the Sun as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

TRKOFF: Turns the sidereal tracking off. (See Chapter 3 – **Motion** submenu.)

TRKON: Turns the sidereal tracking on. (See Chapter 3 – **Motion** submenu.)

UNABEROFF: Turns the deconvolution for Aberration OFF.

UNABERON: Turns the deconvolution for Aberration ON.

UNFLEXOFF: Turns the deconvolution for Flexure OFF.

UNFLEXON: Turns the deconvolution for Flexure ON.

UNKILL: Reenables the servo drive output of PC-TCS to the servo amplifiers after a hard limit violation or PANIC situation or a disable accomplished with the KILL command. (Also see the “KILL” command earlier in this Chapter.)

UNREFROFF: Turns the deconvolution for Refraction OFF>

UNREFON: Turns the deconvolution for Refraction ON.

URANUS: Computes the position of the planet Uranus as the NEXT object position. Use with **MOVNEXT** Command to move to this computed position.

VENUS: Computes the position of the planet Venus as the NEXT object position. Use the **MOVNEXT** Command to move to this computed position.

WOBDEC $\pm d d m m s s . s$: Defines WOBBLE vector Declination. Use WOBRA to define full WOBBLE vector. Use MOV B1 and MOV B2 to move to beams.

WOB OFF: Turns the remote hardware wobble sensing off. (See Chapter 3 – under **Remote** subsubsubmenu.)

WOB ON: Turns the remote hardware wobble sensing on. (See Chapter 3 –under **Remote** subsubsubmenu.)

WOBRA hhmmss.s: Defines WOBBLE vector Right Ascension. Use WOBDEC to define full WOBBLE vector. Use MOV B1 and MOV B2 to move to beams.

Appendix C

Telemetry Structure

Remote Telemetry is continuously broadcast over both the selected serial channels when remote access is enabled (see Chapter 3 – **Remote** subsubsubmenu, and Chapter 4 – **Remote Serial Access**.)

The telemetry stream is 100 characters long including the terminating Carriage Return – Line Feed (13 and 10 decimal) characters. The contents of the stream are documented in the following table. Note that some columns are not used and rate reserved for future expansion.

Remote Telemetry Structure		
<i>Columns</i>	<i>Format</i>	<i>Description</i>
1	Number	Motion Status of Telescope, 0= stable, 1= Declination motion, 2= Right Ascension Motion, 3= Both axes in motion.
2	Number	Current Wobble Beam, blank= Not on a Beam, 1= at Beam1, 2= at Beam 2.
4 to 12	HHMMSS.ss	Current Telescope Right Ascension.
14 to 22	±DDMMSS.s	Current Telescope Declination.
25 to 33	±HH:MM:SS	Current Telescope local Hour Angle.
35 to 42	HH:MM:SS	Current Telescope Sidereal Time.
44 to 48	±DD.d	Current Telescope Elevation in degrees.
50 to 55	±DDD.d	Current Telescope Azimuth in degrees, - implies west + implies east.
57 to 61	DD.d	Current Secant of the Zenith distance (Secz).
63	Number	COM1 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command.
64	Number	COM2 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
65	Number	COM3 Remote command execution code. E,e= executed, 1= bogus request, 2= Bogus Data, 3= Unrecognized Command.
66	Number	COM4 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
67	Number	COM5 Remote command execution code. E,e= executed, 1= bogus request, 2= Bogus Data, 3= Unrecognized Command.
68	Number	COM6 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
69	Number	COM7 Remote command execution code. E,e= executed, 1= bogus request, 2= Bogus Data, 3= Unrecognized Command.

Table C.1: Remote Telemetry Structure

Remote Telemetry Structure		
<i>Columns</i>	<i>Format</i>	<i>Description</i>
70	Number	COM8 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
72	Number	RA Limit 1= in Limit blank= not int limit.
73	Number	DEC Limit 1= in Limit blank= not int limit.
74	Number	Horizon Limit 1= in Limit blank= not int limit.
75	Number	Disable 1= drives disabled blank= not disabled.
76 to 83	YYYY.YYY	Display Coordinate Epoch.
85 to 93	JJJJJJ.JJ	Julian Date.
94 to 98	—	Reserved.
99 to 100	chars	CR,LF (Carriage Return Line Feed).

Table C.2: Remote Telemetry Structure (con.)

Appendix D

Interface Programs

D.1 Remote Command Output

The following subroutines and functions can be used to send a preformatted Remote command to PC-TCS from an external computer. The example will work on IBM PC's and compatibles via COM1 or COM2 but must be compiled for the appropriate channel after defining either PORT1 or PORT2 to be TRUE.

```
#include <conio.h>
```

```
#define YES 1
```

```
#define NO 0
```

```
#define PORT1 YES /* Set to YES to use COM1 */
```

```
#define PORT2 NO /* Set to NO to use COM2 */
```

```
#if PORT1
```

```
#define LSR 0x3FD /* address of Line Status Register */
```

```
#define MSR 0x3FE /* address of Modem Status Register */
```

```
#define MCR 0x3FC /* address of Modem Control Register */
```

```
#define THR 0x3F8 /* address of Transmitter Holding Register */
```

```
#endif
```

```
#if PORT2
```

```
#define LSR 0x2FD /* address of Line Status Register */
```

```
#define MSR 0x2FE /* address of Modem Status Register */
```

```
#define MCR 0x2FC /* address of Modem Control Register */
```

```
#define THR 0x3F8 /* address of Transmitter Holding Register */
```

```
#endif
```

```

#define TEMASK    0x20    /* Transmitter Empty Mask */
#define RTSHIGH    0x3      /* Request to Send Mask */
#define CTSMASK    0x10    /* Clear to Send Mask */
#define MAXTIME    100000L /* Maximum timeout count */
#define CR        0xD      /* A Carriage Return */
#define MAXCHARS    80     /* Maximum number of characters in
                             a command string */

/* A subroutine callable from MS-DOS which will send a passed PC-TCS
   command
   * out the appropriate serial port. If an error is encountered during
   * transmission - the procedure is aborted and the function returns NO.
   * Subroutine verify's the transmission by waiting for the CTS level change.
   *
   */

int cmdout (cmd)
char *cmd; /* the command string array */
{
    int ioerr;          /* internal I/O error flag for function outchr */
    long timeout;       /* the retry counter */
    int outchr (char);  /* a function to output characters */
    int loop;           /* a loop counter for outputting characters of
                           the command string */

    int level;
    char comm[100];

    strcpy(comm,cmd);
    chksum(comm);

    /* get the current cts level */

    level= inp (MSR & CTSMASK) ;

    for (loop = 0; loop < MAXCHARS; loop++) {
        if (!outchr (comm[loop])) {
            return (NO);
        }
        if ((int) comm[loop] == CR)
            break;
    }
}

```

```
    }

    /* wait for the CTS flag to change state */

    for (timeout = 0; timeout < MAXTIME; timeout++)
        if ((inp (MSR) & CTSMASK) != level)
            break;

    if (timeout >= MAXTIME)
        return(NO);

    return(YES);
}

/* A function to send a character out the telescope communications
 * serial port. The function returns an error only if it could not
 * send the character.
 */

outchr (value)
char value;
{
    int timer;                /* the retry counter */
    int bval;

    bval = value;

    /* wait for transmitter to be emptied */

    for (timer = 0; timer <MAXTIME; timer++)
        if (inp (LSR) & TEMASK) {
            outp (THR, bval); /* its empty - send char /
            return(YES);
        }
```

```
        }  
  
    return (NO);  
}
```

D.2 Remote Telemetry Input

The following functions can be used to retrieve ASCII data from the telemetry system broadcast by PC-TCS. The example will work on IBM PC's compatibles via COM1 or COM2 but must be compiled for the approximate channel after defining either PORT1 or PORT2 to be YES.

Data is retrieved into s passed character buffer via a call to TELEM which requires a coded number telling TELEM which telemetry item to retrieve. The routine fills the buffer passed in the call with the ASCII data requested. The user is responsible for translating this ASCII data into the desired internal format needed by any calling routine.

```
#include <conio.h>  
#include <string.h>  
  
#define YES 1  
#define NO 2  
  
#define PORT1 YES /* Set to YES to use COM1 */  
#define PORT2 NO /* Set to YES to use COM2 */  
  
#if PORT1  
#define LSR 0x3FD /* address of Line Status Register */  
#define RBR 0x3F8 /* address of Receive Register Buffer */  
#define THR 0x3F8  
#endif  
  
#if PORT2  
#define LSR 0x2FD /* address of Line Status Register */  
#define RBR 0x2F8 /* address of Receive Register Buffer */
```



```
#define THR      0x2F8
#endif
```

```
#define MAXTIME 32000      /* maximum number of retries */
#define BFMASK  0x1        /* receive buffer full mask */
```

```
#define CR      13
#define LF      10
```

```
/* This routine will retrieve telemetry from PC-TCS and pass it to
 * the calling routine in the array BUFF. The function returns YES
 * if the telemetry stream could not be read or if an undefined
 * telemetry request was made.
 *
 * Written by: D. A. Harvey
 *
 */
```

```
int telem(element ,buff)
```

```
int element;
```

```
cahr *buff;
```

```
{
```

```
/*
```

```
 *   num   =>   parameter passed in call which
 *               describes a code for the desired
 *               data to be retrieved.
```

```
 *
```

```
 *   buff  =>   parameter passed in call which is
 *               a 25 byte buffer to handle the
 *               characters returned by TELEM. Buffer
 *               should be of length 30 and be filled
 *               with balnks (20 hex).
```

```
*/
```

```
    int comtyp;
    int subindx;
    int length;
    int index;
    int I;
```

```
/*          Conditional branches to proper data description area.
```

```
*
```

```
*
```

```
Data codes are as follows . . .
```

```
*
```

```
*
```

```
          CODE          DESCRIPTION
```

```
*
```

```
          ----          -
```

```
*
```

```
*
```

```
1          Motion Status.
```

```
*
```

```
*
```

```
2          Current Wob Beam.
```

```
*
```

```
*
```

```
3          RA (HHMMSS.ss)
```

```
*
```

```
*
```

```
4          DEC (+DDMMSS.ss)
```

```
*
```

```
*
```

```
5          LHA (HH:MM:SS)
```

```
*
```

```
*
```

```
6          LST (HH:MM:SS)
```

```
*
```

```
*
```

```
7          Elevation (+DD.d)
```

```
*
```

```
*
```

```
8          Azimuth (+DDD.d)
```

```
*
```

```
*
```

```
9          Secant Z (DD.ddd)
```

```
*
```

```
*
```

```
10         Command Exec. Status All 8 cahnnels
```

```
*
```

```
*
```

```
11         Limit Status (RDHd)
```

```
*
```

```
*
```

```
12         Epoch (YYYY.YYY)
```

```
*
```

```
*
```

```
13         JD (JJJJJJ.JJJ)
```

```
*/
```

```
switch (element) {
```

```
case 1: /* motion status */
```

```
length= 1;
```

```
index= 0;
```

```
fetch(length,index,buff;
```

```
break;
```

```
case 2: /* beam */  
length= 1;  
index= 1;  
fetch(length,index,buff);  
break;
```

```
case 3: /* RA hhmmss.ss */  
length= 9;  
index= 3;  
fetch(length,index,buff);  
break;
```

```
case 4: /* DEC +ddmmss.s */  
length= 9;  
index= 13;  
fetch(length,index,buff);  
break;
```

```
case 5: /* HA +hh:mm:ss.ss */  
length= 12;  
index= 24;  
fetch(length,index,buff);  
break;
```

```
case 6: /* LST hh:mm:ss */  
length= 8;  
index= 34;  
fetch(length,index,buff);  
break;
```

```
case 7: /* EL +dd.d */  
length= 5;  
index= 43;  
fetch(length,index,buff);  
break;
```

```
case 8: /* Az +ddd.d */
```

```
length= 6;  
index= 49;  
fetch(length,index,buff);  
break;
```

```
case 9: /* SecZ dd.dd */  
length= 5;  
index= 56;  
fetch(length,index,buff);  
break;
```

```
case 10: /* Comm status 12345678 */  
length= 8;  
index= 62;  
fetch(length,index,buff);  
break;
```

```
case 11: /* Limit - disable status */  
length= 4;  
index= 71;  
fetch(length,index,buff);  
break;
```

```
case 12: /* Epoch eeee.eee */  
length= 8;  
index= 75;  
fetch(length,index,buff);  
break;
```

```
case 13: /* JD dddddddd.ddd */  
length= 9;  
index= 84;  
fetch(length,index,buff);  
break;
```

```
default:  
return(YES);  
break;
```

```
    }  
    buff[length+1]= 0;  
    return(NO);  
}
```

```
/*          Subroutine FETCH will retrieve the desired data field  
*          from the telemetry stream as defined by the TELEM  
*          calling program. The passed parameters have the following  
*          attributes . . .  
*  
*          Parameter          Description  
*          -----  
*          LENGTH             denotes the number of characters to  
*                               retrieve from the data stream  
*  
*          INDEX              denotes the starting position in the  
*                               80 character data stream of the field  
*                               to retrieve data from. (0 based index).  
*  
*          BUFF               The character array to place the data  
*                               field retrieved from the telemetry stream.  
*/
```

```
int fetch(length,index,buff)  
int length;  
int index;  
int *buff;  
{  
  
    int ccount= 0;  
    char charin= 0;  
    int ones;  
    int number;  
    int I;  
    int sync(void);  
    int getch(char *);
```

```
/*          SYNCronize on the data stream by looking for CR,LF sequence
*/
```

```
    if (!sync())
        return(NO);
```

```
/* Primary data stream FETCH routine
 * skip over to beginning of desired data field
 */
```

```
    for (i=0;i<index;i++)
        if (!getchr(&charin))
            return(NO);
```

```
/* Fill the buffer with proper data field
 */
```

```
    for (i=0;i<length;i++)
        if (!getchr(&buff[i]))
            return(NO);
```

```
    return(YES);
```

```
}
/* Routine to look for CR,LF sequence in data stream
 * lchar is Last character read.
 */
```

```
int sync(void)
{
```

```
    char lchar= 0;
    char charin;
    int i;
    int getchr(char *);
```

```
/* get character – test for LF, if true test last character for
 * CR character.
```

```
*/

    for (i=0;i<110;i++) {
        if (!getchr(&charin))
            return(NO)
        if (charin == LF) {
            if (lchar == CR)
                return(YES);
        } else
            lchar= charin;
    }
    return(NO);
}

/* A subroutine callable from DOS that will attempt to
 * read the serial input port, determine if a character is
 * available - retrieve it if there is - or try up to 32000 times if
 * there isn't. If a failure on the read is encountered the error
 * flag is set and control returns to the caller. If a character is
 * available it is read from the data port, the error flag is reset
 * and control is returned to the caller.
 *
 */

int getchr (value)
char *value;          /* value of received character */
{
    int timeout;

    *value = 0;        /* reset character value */

    for (timeout = 0; timeout < MAXTIMEIN; timeout++)
        if (inp (LSR) & BFMASK) {          /* character available? */
            *value = inp (RBR);             /* - yes! Get it */
            return(YES);
        }
    return(NO);          /* return to caller */
}
```

D.3 Remote Command Checksumming

The following subroutine can be used to modify a preformatted PC-TCS remote serial command to include the checksum necessary for command verification. (See **Remote Serial Access** in Chapter 4.) This checksumming procedure is necessary only if checksumming is enabled while remote serial access is active.

Including the checksum in a remote serial command sent to PC-TCS while checksumming has been disabled, will not affect command verification or execution.

```
#include <string.h>

/*  A subroutine to modify a preformatted command string
 *  with the appropriate checksum for transmission to
 *  PC-TCS.
 *
 *  The proper checksum value is inserted in the passed
 *  command string before the terminating Carriage Return.
 *
 *  The passed command string cmd must be dimensioned large
 *  enough to accommodate at least one more character than
 *  its length before chksum is called.
 *
 *  Written by: D. A. harvey
 */

void chksum(cmd)
char *cmd;
{
    int sum= 0,
        len= 0,
        i,
        subchar;

    for (i=0; i<= strlen(cmd); i++) {
        len= i;
        subchar= cmd[i];
```



```
        if (subchar != 13)
            sum+= subchar;
        else
            break;
    }
    cmd[len]= (sum % 64)+0x20;
    cmd[len+1]= 13;
    return;
}
```

Appendix E

Hardware Connections

The Paddle and Limits hardware pinouts documented on the following pages are for the COMSOFT DCS Servo Control Card and CT Stepper Motor Control Card.

The RS-232 remote serial connections are documented for both DTE to DCE (PC to Modem) and DTE to DTE (PC to Terminal/ PC) devices.

TCS SERVO BOARD MAIN CONNECTOR

Figure E.1: DCS Servo Control Card Main Connector Pin-out

Figure E.2: DCS Servo Control Card AUX Connectors Pin-out

Figure E.3: CT DEC/ Limits Sense Stepper Motor Control Connector Pin-out

Figure E.4: CT RA/ Paddle Sense Stepper Motor Control Connector Pin-out

Figure E.5: Remote RS-232 interface Pin-out

Appendix F

Catalog File Structure

The following table documents the record structure of PC-TCS formatted Catalog files. Users wishing to create their own custom catalogs may use this as a guide to developing their own software to properly format their data. The CATALOG program supplied with PC-TCS, will allow the user to create, modify and print catalogs with this structure. (See Chapter 4 under **User Catalog Generation**.)

Catalog File Structure		
<i>Columns</i>	<i>Format</i>	<i>Description</i>
1 to 4	nnnn	Record Number.
6 to 9	mmmm	Object visual magnitude*100.
11 to 22	r.rrrrrrrrr	Right Ascension in radians.
24 to 36	±d.ddddddddd	Declination in radians.
38 to 43	±ppppp	Right Ascension Proper Motion over 100 years in seconds multiplied by 100.
45 to 50	±ppppp	Declination Proper Motion over 100 years in arcseconds multiplied by 100.
52 to 115	Text	Label Field.
117 to 123	eeee.ee	Coordinate Epoch (optional) – if undefined 1950.0 is assumed.
127 to 128	characters	Carriage Return and line Feed characters (13 and 10 hex).

Table F.1: Catalog file record structure

Appendix G

Error Messages

The following section describes the error messages that might be displayed during the course of PC-TCS operation. Most errors are non fatal and are of an informative nature. The last part of this section describes the procedure necessary to recover from a hard limit induced drive shutdown.

G.1 PC-TCS Runtime Error Messages

Beam 1 Position beyond limits – no move possible. Attempt was made to move to a beam position that would take the telescope beyond positional limits. Move was aborted. Not Fatal.

Beam 2 Position beyond limits – no move possible. Attempt was made to move to a beam position that would take the telescope beyond positional limits. Move was aborted. Not Fatal.

Can not open file FLEXURE.PCF – request aborted. While running the FDC utility, PC-TCS could not find/ open the FLEXURE catalog file. Copy the file FLEXURE.PCF to the PC-TCS working directory. Not Fatal.

Can not Load CATALOG Program – aborted. Either the CATALOG program could not be found on disk or not enough memory was available to load the Program. Error can be Fatal.

Can not shell to DOS – aborted. Not enough memory available to start a DOS shell while leaving PC-TCS in memory. Error can be Fatal.

Command File Already Active – Terminate Active first. Attempt was made to load another Command File while one was currently running. Non Fatal.

Command limit reached at . . . While attempting to load into memory the a command file, PC-TCS detected more than 1000 lines. The command file load has been truncated to 1000 commands. Not Fatal.

DECODE Error in PCTCSD.FLX . . . Flexure aborted. PC-TCS was unable to decode data read from the Declination Flexure File. Flexure corrections were not made. Non Fatal.

DECODE Error in PCTCSR.FLX . . . Flexure aborted. PC-TCS was unable to decode data read from the Right Ascension Flexure File. Flexure corrections were not made. Non Fatal.

Epoch out of range – no adjustment made. Attempt was made to change the Display Epoch to a value that was not a valid Gregorian Date. Non Fatal.

Epoch out of range – No modification made. Attempt was made to change to Coordinate Epoch to a value that was not a valid Gregorian Date. Non Fatal.

Error(s) too large – data ignored. While running the FDC utility, PC-TCS computed flexure errors that were greater than 500 seconds of time in RA and/ or greater than 1000 seconds of arc in declination. Data was not filed for aanlysis. Not Fatal.

File in Use – Epoch not modifiable. Attempt was made to change the coordinate Epoch while a Catalog file is set the source. Epoch may be altered only if keyboard mode is active. Non Fatal.

FDC aborted by user request. While running the FDC utility, PT-TCS was instructed to abort FDC by use of the ESC key. Not Fatal.

FDC Catalog read Error – FDC abort. While running the FDC utility, PC-TCS could not read from the FLEXURE.PCF file. File has probably been corrupted. Copy the file FLEXURE.PCF file from the PC-TCS distribution diskettes. Not Fatal.

FLEXURE DATA COLLECTION not implemented this Version PC-TCS. Attempt to activate the Flexure Data collection Utility in version 1.4 or earlier of PC-TCS. Non Fatal.

Flexure Data File(s) NOT on disk . . . Request Ignored. PC-TCS could not find/ open the Flexure data file(s) PCTCSD.FLX, PCTCSR.FLX. Non Fatal.

Illegal Baud Rate Selection – Ignored. Attempt was made to set the baud rate to a selection window. Non Fatal.

Illegal Catalog Number – rejected. Attempt was made to set the baud rate from a Catalog file that is beyond the end, before the beginning, or undefined in the Catalog file in use. Non Fatal.

Illegal DEC OFFSET specified – rejected. Declination for OFFSET vector contained data that was not in the form or range of $\pm DD:MM:SS.s$. Non fatal.

Illegal DEC specified – position rejected. Declination for NEXT position contained data that was not in the form or range of $\pm DD:MM:SS.s$. Non Fatal.

Illegal DEC WOBBLE specified – rejected. Declination for WOBBLE vector contained data that was not in the form or range of $\pm DD:MM:SS.s$. Non Fatal.

Illegal RA OFFSET specified – rejected. Right Ascension for OFFSET vector contained data that was not in the form or range of $\pm HH:MM:SS.s$. Non Fatal.

Illegal RA specified – position rejected. Right Ascension for NEXT position contained data that was not in the form or range of $\pm HH:MM:SS.s$. Non Fatal.

Illegal RA WOBBLE specified – rejected. Right Ascension for OFFSET vector contained data that was not in the form or range of $\pm HH:MM:SS.s$. Non Fatal.

Illegal time specified – no adjustment made. UT Time contained data that was not in the form or range of $\pm HH:MM:SS.s$. Non fatal.

LIMIT DEC found active – check I/O circuitry. PC-TCS detected that the Declination/ Limits DCS card DEC limit bit was set at system check-out. Check cabling, external hardware and Limits I/O mask for cause of Warning. Non fatal.

LIMIT HORIZON found active – check I/O circuitry. PC-TCS detected that the Declination/ Limits DCS card HORIZON limit bit was set at system check-out. Check cabling, external hardware and Limits I/O mask for cause of Warning. Non Fatal.

Limit out of range – no adjustment made. Attempt was made to set Hour Angle, Declination, Horizon, or Slew Limit beyond system maximum. Non Fatal.

LIMIT RA found active – check I/O circuitry. PC-TCS detected that the Declination/ Limits DCS card RA limit bit was set at system check-out. Check cabling, external hardware and Limits I/O mask for cause of Warning. Non Fatal.

Next Position beyond limits – no move possible. Attempts was made to move to a NEXT object position that is beyond the positional limits of the telescope. Non fatal.

No Beam declared – Ignored. Attempt was made to move to a beam position when no WOBBLE Vector has been declared. Non Fatal.

No BIAS RATE is set – None Started . . . Attempt was made to active a bias rate when none has been declared. Non Fatal.

No Command File Specified – No Activation Possible. Attempt was made to activate a Command File when none has made Specified.

No first Beam Declared. Beam 2 Undefinable.

No Next position Declared – Can Not Initialize. Attempt was made to Initialize the telescope coordinate system when no NEXT object position has been declared. Non Fatal.

No Next Position Declared – Ignored. Attempt was made to move to a NEXT object when no NEXT position has been declared. Non Fatal.

No Offset Declared – Ignored. Attempt was made to move to an OFFSET position when no OFFSET vector has been declared. Non Fatal.

No Reference Position Declared – Ignored. Attempt was made to move to a REFERENCE Object when none has been declared.

No Such Star in Catalog – FDC abort. While running the FDC utility, PC-TCS could not find the requested record/ star. The FLEXURE.PCF file has probably been corrupted – copy a fresh version of the file to the PC-TCS working directory from the distributing diskettes. Non Fatal.

No Such Star in Catalog – rejected. Attempt was made to retrieve an Object from a Catalog file when no such object number exists. Non Fatal.

Not a legal date – no adjustment made. Attempt was made to specify a new UT Date that contained illegal data or was not a valid Gregorian Date. Non Fatal.

Offset Position beyond limits – no move possible. Attempt was made to move to an OFFSET position that was beyond the telescope positional limits. Non Fatal.

Orbital Computation Error – aborted. Internal Error – notify COMSOFT. Non Fatal.

Out of Memory!!! PC-TCS ran out of memory while attempting to open a new windows for display. Error is fatal.

PADDLE EAST found active – check I/O circuitry. PC-TCS detected that the Right Ascension/ Paddle DCS card EAST bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non fatal.

PADDLE NORTH found active – check I/O circuitry. PC-TCS detected that the Right Ascension / Paddle DCS card NORTH bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non Fatal.

PADDLE PANIC found active – check I/O circuitry. PC-TCS detected that the Right Ascension/ Paddle DCS card PANIC bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non Fatal.

PADDLE PERMIT found active – check I/O circuitry. PC-TCS detected that the Right Ascension/ Paddle DCS card Slew PERMIT bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non Fatal.

PADDLE SLEW found active – check I/O circuitry. PC-TCS detected that the Right Ascension/ Paddle DCS card SLEW bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non Fatal.

PADDLE SOUTH found active – check I/O circuitry. PC-TCS detected that the right Ascension/ Paddle DCS card SOUTH bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non Fatal.

PADDLE WEST found active – check I/O circuitry. PC-TCS detected that the right Ascension/ Paddle DCS card WEST bit was set at system check-out. Check cabling, external hardware and Paddle I/O mask for cause of Warning. Non Fatal.

POINT.DAT Update Error – data lost. While running the FDC utility, PC-TCS was unable to write/ read from the POINT.DAT data logging file. The POINT.DAT file must be reconstructed. See Chapter 4 under **The FDC Utility**. Fatal to FDC processing.

Rate(s) beyond slew limits – no adjustment made. Attempt was made to set a bias, Drift, Guide, or Slew rate that was beyond the maximum specified by the system. Non Fatal.

READ Error on PCTCSD.FLX . . . Flexure aborted. PC-TCS attempt to read from the Declination flexure data file was unsuccessful. Non fatal.

READ Error on PCTCSR.FLX . . . Flexure aborted. PC-TCS attempt to read from the Right Ascension flexure data file was unsuccessful. Non Fatal.

Reference position beyond limits – no move possible. Attempt was made to move to a REFERENCE position that is beyond the positional limits of the telescope. Non Fatal.

System Error. Error Displayed on Runtime Display that indicates PC-TCS did not have time to complete the servo control loop before another interrupt was generated. Servo control and position readout was lost. Error is Non Fatal.

Unable to OPEN Flexure data files . . . PC-TCS can not find the PCTCSD.FLX and/ or the PCTCSR.FLX file(s) on disk. Non Fatal.

Unable to parse date – no adjustment made. An illegal formatted UT Date was entered by the user, or read from the system. Error May be Fatal if encountered at system check-out. Non Fatal.

Unable to parse DEC data – position rejected. An illegal formatted NEXT Declination position was entered. Non Fatal.

Unable to parse DEC OFFSET data – rejected. An illegal formatted OFFSET Declination vector was entered. Non fatal.

Unable to parse DEC WOBBLE data – rejected. An illegal formatted WOBBLE Declination position was entered. Non Fatal.

Unable to parse RA data – position rejected. An illegal formatted NEXT Right Ascension position was entered in the form $\pm$ HH:MM:SS.s. Non Fatal.

Unable to parse RA OFFSET data – position rejected. An illegal formatted OFFSET Right Ascension vector was entered. Non fatal.

Unable to parse RA WOBBLE data – position rejected. An illegal formatted WOBBLE Right Ascension vector was entered. Non Fatal.

Unable to parse system time – no adjustment made. PC-TCS was unable to accurately read the UT time from the system while setting a UT Date. Non Fatal.

Unable to parse time – no adjustment made. An illegal formatted UT Time was entered by the user, or read from the system. Error may be Fatal if encountered at system check-out – otherwise Non fatal.

Unable to read from file (EOF) – rejected. PC-TCS was unable to read from a Catalog data file. Non Fatal.

G.2 Hard Limit Recovery Procedure

In most cases, Pc-TCS will give the user ample warning of an impending limit violation. If however, the user has set the software limits to their absolute maximum (matching the hard, or final limits), the telescope drive system will be disabled from moving in any direction, at any speed. Recovery from this hard Limit condition is *EXTREMELY DANGEROUS* and should be undertaken with extreme caution.

Two methods can be used to recover from limit violations. The first method uses the **Override** option of the **Limits** submenu.

The second method requires “fooling” PC-TCS into believing that the telescope is not in a precarious attitude. This method subsequently invalidates the PC-TCS coordinate system and will require a complete reinitialization of the telescope after recovery.

Method 1: Disable all Bias rates and turn off the sidereal tracking (see Chapter 3 – the **Motion** submenu). Cancel any previously commanded motion using the **Cancel** command of the **Motion** submenu.

Using the **Limits** submenu, activate the **Override** option. Since the drives have been shutdown, a “DISABLED” message should be visible in the tracking field of the display. Reactivate the drive system by using the **Disable** option of the **Motion** submenu.

Using either the paddle or the **Stow** option of the **Move** submenu, command the telescope out of the limit. (The second option is recommended.)

Once safely out of the limit, deactivate the **Override** option of the **Limits** submenu.

Method 2: !!! Warning !!! method 2 requires the defined Stow position be the Zenith.

Disable all Bias rates and turn off the sidereal tracking (see Chapter 3 – the **Motion** submenu). Note the Current Local Sidereal Time and using the value of your latitude as the Declination, make this position the NEXT position by using the **Position** submenu while in the **Keyboard** (see Chapter 3 – under the **Input** subsubmenu). The NEXT line of the Runtime display should now show a coordinate close to the position that is displayed in the Current Position boxes. Using the **Declare** submenu, declare this NEXT position to be the Initial position. (See Chapter 3 – **Declare** submenu under **Initial**.) Once this is done, the Limit warnings should vanish (if the Hardware Limits have been deactivated) from the Runtime display – however, the **DISABLED** message should still be visible between the Current position boxes.

Once again, make sure that the bias and sidereal tracking rates are disarmed. Using the Motion Cancel command (see Chapter 3 – **Motion** submenu under the **Cancel** entry), cancel any previously declared motion command.

Reenable the drives using the **Motion** submenu – Disable/ Enable feature. The telescope is now free to move – but PC-TCS thinks it is pointing near the zenith and doesn't know the telescope is really in a limit.

Using the paddle, *slowly* move the telescope toward the zenith until you actually get there, or until you encounter another hard limit disable of the drive system. Repeat the entire procedure until you get the telescope pointed at the zenith.

Once at the zenith, exit PC-TCS via the **Manual** shutdown option of the **Shutdown** menu – MAKING SURE YOU BYPASS THE MOVE TO STOW OPERATION WITH THE *ESC* KEY.

After PC-TCS shutdown is complete – restart PC-TCS and reinitialize the telescope as described in Chapter 3 under the **Declare** submenu option **Initialize**.

Appendix H

Orbit Parameter File Structure

The following table documents the record structure of PC-TCS formatted Orbital parameters files. Users wishing to create their own orbital parameter files may use this as a guide to properly format the data.

File names must have the “.orb” extension to be recognized by PC-TCS. The file name should be the name of the object (as this is what will be displayed in the label field of the runtime display.)

Each file contains only 8 lines which describes either an elliptic, parabolic or hyperbolic orbit. (The second line determines the orbit type.) Leading spaces on each line are NOT allowed and each line must be delimited by a CR LF sequence. A description of the file contents follows.

Orbital Parameter File Structure	
<i>Line</i>	<i>Description</i>
1	Perihelion distance (q) in A.U. Note: for elliptical orbits, $q = a(1 - e)$
2	Eccentricity of orbit – should be < 1.0 for elliptical orbits, $= 1.0$ for parabolic orbits, and > 1.0 for hyperbolic orbits.
3	Inclination (i) of orbit in degrees (Epoch 2000.0).
4	Argument of perihelion (ω) in degrees (Epoch 2000.0).
5	Longitude of the Ascending Node (Ω) in degrees. (Epoch 2000.0).
6	Date and Time of perihelion in the form of . . . YYYY MM DD.ddddddd (Individual fields should be delimited by spaces only.)
7	Absolute object magnitude (g) as viewed from 1 A.U.
8	Factor “k” for asteroid and comet magnitude computation. k usually varies between 5.0 and 15.0 for cometary objects and around 2.3 for planetesimal (asteroid) type objects. Note that the formulae used for magnitude computation is different depending on the type of orbit. The equation used is of the form . . . $m = g + 5 \log (\Delta) + k \log (r)$ for parabolic and hyperbolic orbits and . . . $m = g + 5 \log (\Delta r) + k\beta$ for elliptical orbits. where: Δ = earth-object distance r = object – sun distance, and β = phase angle (sun-object-earth)

Table H.1: Orbital Parameter File Structure

Appendix I

Software License Agreement

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End User Signature

Date

Addendum

PC-TCS PEC OPERATION

PC-TCS DOME CONTROL

PC-TCS –FOCUS CONTROL

PC-TCS –TPOINT POINTING CORRECTIONS

PC-TCS AND TPOINT INSTALLATION AND OPERATION

PC-TCS AND THESKY

PC-TCS HARDWARE CONNECTIONS

PC-TCS SCRIPT AND REMOTE COMMANDS

TLEMETRY STRUCTURE

COMET.ORB FILE FORMAT

ASTEROID.ORB FILE FORMAT

PC-TCS PEC Operation

PC-TCS version 4.9 and later has the capability of compensating for Periodic Error in RA gear driven telescopes. To make use of this capability the user must supply a TTL level “index hack” signal from the RA worm that produces a pulse once per revolution of the RA worm. This signal must be coupled to the **RA Index Input** of the PC-TCS interface hardware (see updated CTR-10 documentation later in this manuscript).

After the hardware connection has been made use the PFEDIT program to setup critical parameters associated with this option. These steps are documented below.

1. Assign an available PC hardware interrupt via the PFEDIT program (use the *Computer-Interrupts* menu item). Make sure the interrupt selected is not in use by another device. (Typically interrupt number 7 is available – check your PC documentation.).
2. Set the number of steps per revolution of the worm using the PFEDIT program (use the *Telescope-Axis* menu item.) This parameters tells PC-TCS how many steps of the stepper motor it takes to complete one full revolution of the worm.
3. Set the **PEC Hardware** menu item using the PFEDIT program (use the *Telescope-PEC Hardware* menu toggle).

Once the hardware is installed and PC-TCS has been informed of its availability, the user must “train” PC-TCS to correct for the periodic error. PC-TCS does this by “memorizing” RA button presses from the paddle during the course of a single worm revolution. Depending on your telescope’s gear into this time may vary between tens of seconds to tens o minutes. To accomplish this training – start PC-TCS and point your telescope and begin tracking on easily visible star near or on the celestial equator (0° Declination) and near the meridian. Use a high power eyepiece with a reticle to ensure critical guiding during the subsequent training procedure.

Before beginning training set the desired RA guide rate using the PC-TCS *Rates-Guide* menu item. The rate set here will be remembered by PC-TCS and used for future PEC corrections. (i.e. once training is complete the user may modify the guide rate accommodate other serving needs.) Use the PC-TCS *Parameters-Utilities-DefPEC* menu item to invoke the PEC training interval. If you have PEC hardware turned on (PFEDIT – *Telescope-PEC hardware* menu item) the software will wait for the RA index pulse before sampling begins. Once sampling has begun the training window will display a progress bar as the interval elapses. When complete the information collected will be saved to a file for future tracking corrections. From this time forward anytime tracking is enabled, PEC will automatically be invoked unless specifically turned off using the PC-TCS *Rates-PEC* menu item.

When tracking is started – after being stopped or after PC-TCS startup, the PC-TCS software will attempt to acquire the index hack to “phase” the tracking PEC corrections. During this interval the enunciator “PHASING” will appear between the Current RA and DEC position display fields and below the tracking field of the Runtime Display. While PC-TCS is phasing the PEC corrections no corrections will be applied to the default track rate. Once phasing is complete, the enunciator “PEC” will be displayed and Periodic Error corrections will be active whenever the telescope is tracking sidereal.

Note: PEC is also available if no PEC hardware is available (the PFDIT *Telescope-PEC Hardware* menu item is disabled). In this case PC-TCS must be “trained” every time PC-TCS is started before PEC can be used. No phasing is performed in this environment as PC-TCS “phases” at DefPEC time.

PC-TCS Dome Control

Version 6.x of PC-TCS has the capability of reading and automatically controlling dome position given optional hardware.

To perform dome encoding PC-TCS requires that an incremental encoder and a third party ISA Bus PC card manufactured by US Digital Corporation. The recommended hardware is listed below:

Encoder:	H1-100 Optical Shaft Encoder with PC4 RS-422 Driver (\$100.00)
Pc Card:	PC-7166 Quadrature Decoding Interface Board (\$350.00)
Cable:	CA-F10-XXFT – cable for XX feet from computer. (@\$2.00 / foot)

Available from:	US Digital Corporation 3800 N.E. 68 th Street #A3 Vancouver, WA. 98661 (360) 696-2468
-----------------	---

Dome control is also available if your dome motor control hardware will accept simple TTL signals to move the dome. COMSOFT can also provide custom electronics given the motor and wiring specifications. (Contact COMSOFT for information and pricing of this option.)

Parameters for dome control are configured via the *Dome* menu item of the PFEDIT program. The user must first configure this parameters before dome readout and/or control is available.

Once configured properly, Pc-TCS can display the difference between the current dome azimuth and the telescope pointing azimuth as seen on the Runtime Display of PC-TCS following the **Dome** label will show Green whenever the telescope has a clear view out the dome slit, or Red when the dome is obstructing all or of the telescope aperture.

Dome motion (if implemented) is accomplished via the invocation of the Pc-TCS *Motion-auto-Dome* menu item. Enabling this menu item will allow PC-TCS to move the dome to a position that provides a Green **Dome** enunciator (unobstructed view). If the user does not wish PC-TCS to automatically move the dome (e.g. During a critical time exposure when dome motion could vibrate the telescope – or during a long slew when the dome can move faster than the telescope and the dome would “jerk” as the telescope moves) he can temporarily disable this feature by deselecting the *auto-Dome* menu item until the situation warrants dome motion.

PC-TCS – Focus Control

With version 5.0a PC-TCS supports pulse driven focus control. This option requires stepper motor like hardware installed on the focusing mechanism. PC-TCS controls the focus by providing TTL compatible step and direction signals to the stepper motor driver hardware and displays a running count of these signals on the PC-TCS Runtime Display.

User control accomplished via 4 TTL input signals CTR-05 interface board. These signals are documented in this manuscript under **PC-TCS Hardware Connections**.

Focus control has two user definable speeds and can operate either open loop or closed loop (if the optional encoder and interface card is available).

PC-TCS – Tpoint Pointing Corrections

PC-TCS has two methods of dealing with pointing corrections. The “native” PC-TCS pointing corrections are handled using a “flexure model” that empirically models telescope pointing errors by constructing a two dimensional “map” of pointing errors and then applying corrections to current position of the telescope. This mapping method is very robust and does not rely on an understanding of the root cause of the errors. The drawback of this modeling method is that the whole sky can not be modeled. Because of memory limitations of 16 bit DOS programs, only ± 5 hours of Hour Angle from the meridian and 120 degrees of declination from the pole can be “mapped”. Because of the limited area serviced by this method, it has been abandoned of Method 2. The manual for Version 4.5 explains this flexure mapping procedure.

Method 2 (Tpoint) is a true mathematical modeling of the telescope mount accounting for such terms as polar misalignment, tube and fork flexure, nonperpendicularity of the HA and Dec axis, etc. This method is used by most of the modern astronomical observatories today and is far superior to the mapping method described above. The method does not suffer from the range restrictions and can model the telescope over the entire sky. The drawback of this method is in the complexity of the data analysis and the cost of the modeling software (Tpoint is required and cost an additional \$350.00 above and beyond the cost of the PC-TCS control software). This software is copyrighted by Patrick Wallace and must be purchased separately if the user wishes to use this modeling method. If you wish to purchase this option separately you can contact Patrick Wallace at the address given below.

It should be noted that PC-TCS is normally distributed with Tpoint “compatibility” and a special request must be made to receive the “Flexure Map” version of Pc-TCS (no charge).

Data collection for the two methods is identical; processing of the data is entirely different however, Chapter 4.4 of the PC-TCS Users Manual describes the data collection and analysis procedure for “Flexure mapping” – the following pages describe the data analysis procedure for Tpoint based modeling.

TPOINT Software

Web: www.tpsoft.demon.co.uk

Email: ptw@tpsoft.demon.co.uk

Phone: +44-1235-531198

PC-TCS and Tpoint Installation and Operation

This is an introduction to using Tpoint with PC-TCS. Currently COMSOFT has implemented the standard equatorial modeling capabilities into PC-TCS and has not attempted to incorporate some of the more “esoteric” features of Tpoint’s powerful capabilities. It has been found that given a typical equatorial telescope with “well-behaved” and repeatable flexure and alignment problems, that the fitting terms implemented by COMSOFT are sufficient to model the telescope to the 1 to 2 arcsecond RMS level. If you find that other Tpoint coefficients better model your telescope than used by our default implementation, a custom model can be implemented by COMSOFT at our regular consulting rates.

Before modeling can be done, pointing data must first be collected using the PC-TCS Flexure Data Collection (FDC) utility built into PC-TCS. This procedure will not collect data on telescope pointing errors and store this information in a formatted ASCII Text file called POINT.DAT. See the PC-TCS User’s Manual on how to perform a pointing run.

Before this data can be digested by Tpoint it must first be converted to a format Tpoint understands. A utility program called TCS2TPT.EXE is distributed with the PC-TCS software suite for this purpose. To convert the POINT.DAT file to Tpoint format simply execute the TCS2TPT program in the same directory that contains the POINT.DAT file – the program will create a Tpoint compatible data file called TP.DAT that contains the pointing information in POINT.DAT.

To analyze your pointing data – install your licensed copy of the Tpoint software from the distribution diskette into the subdirectory where PC-TCS runtime programs are stored. Simply copy the entire contents of the distribution diskette to the TCS subdirectory. Once this is accomplished, type the following commands at the DOS prompt:

```
C:\TCS>tpoint
```

to start the Tpoint program. Tpoint will respond with a copyright banner followed by an “*” prompt. The following instructions are a “recipe” for beginning data analysis, for the meaning and detailed information on the commands given here, refer the PC-Tpoint manual.

At the asterisk prompt, type the following command to load in the pointing data previously collected with PC-TCS.

```
*indat point.dat
```

Type the following commands to set up tpoint for equatorial telescope analysis:

```
*call equal (or call altaz for ALTAZ PC-TCS)
*adjust s
*paral
```

To graphically view the raw pointing data and the telescope “uncorrected” errors type:

```
• call e9 (or call a9 for ALTAZ PC-TCS)
```

Tpoint will display 9 graphs showing various interpretations of the telescope pointing errors. For explanations of these graphs and what they represent review the Tpoint User’s Manual.

To begin trying to “fit” the data type the following command:

***fit**

You may have to type this command several times until the “change” column displayed after every fit command shows zero or nearly zero for each of the coefficients. Note that Value and Variance of each of the coefficients. If the variance is a large fraction of the value of the coefficient this is an indication that the “significance” of the coefficient is small and it can be “removed” from the fit by typing the following command:

***lose xxx**

where “xxx” is the name for the coefficient displayed after the “fit” command is given. To add other terms documented in the Tpoint User’s Manual – use the following three commands:

***use xxx**

***paral**

***fit**

where “xxx” is the new coefficient term to add to the model.

Important: *Always use the “paral” command after each new “use” specification and before the “fit” command of a new coefficient term.*

You can graphically check the fit and look for systematic errors and their dependencies by graphing the errors using the **call e9** command at any time.

PC-TCS recognizes only the following terms in its equatorial model. Using other terms than the ones documented below will result in those being ignored and poor pointing from your telescope.

IH	ID	NP	CH	ME	MA	FO	DAF	HCES	HCES
DCES	DCEC	DNP	X2HS	X2HC	D2HS	D2HC	D2DS	D2DC	D4HS
D4HC	EHC	PDD	PHH	TF	TX				

For the ALTAZ version of PC-TCS, model terms recognized are as follows.

ACEC	ACES	AN	AW	CA	ECEC	ECES	IA	NPAE
TF	TX	PAA	HACA	HASA				

When a good fit is found the coefficient can be saved to a PC-TCS readable file using the following command:

***outmod pctcs#.tp**

where “#” is the “Class” of the pointing data reported when a **call e9** (or **call a9** for ALTAZ telescopes) command is given as part of the “title” of the lower center graph label or individual graphs Titles. It is important to store the file by the approximate name as this is the file that must be read by PC-TCS to implement pointing corrections.

PC-TCS has 5 possible pointing model “classes” it can use. Classes “A” and “C” are used when PC-TCS is run with the “Tube East”; Classes “B” and “D” are used when PC-TCS is run with the “Tube West”. The “Default” class (which can be selected for fork mounted telescopes) will have the file name **pctcs.tp**.

If you have questions about Tpoint or PC-TCS operation with a Tpoint model, please feel free to contact COMSOFT at the numbers or addresses listed in the letterhead of this document.

For reference, Pc-TCS uses the following code to correct a given Hour Angle (HA) and DEC for flexure using the Tpoint coefficients determined during the analysis.

```

dha= IH
      +PHH*ha
      +NP*tandec
      +CH/cosdec
      +ME*sinha*tandec
      -MA*cosha*tandec
      +HCES*sinha
      +HCES*cosha
      +DNP*sinha*tandec
      +X2HS*sinha/cosdec
      +X2HC*cos2ha/cosdec
      +EHS* (sinha*sinha*tandec)
      +EHC*sinha*cosha*tandec
      -DAF* (coslat*cosha+sinlat*tandec) ;
ddec= ID
      +PDD* newdec
      +ME*cosha
      +MA*sinha
      +FO*cosha
      +DCES*sindec
      +DCEC*cosdec
      +D4HS*sin4ha
      +D4HC*cos4ha
      +D2DS*sin2dec
      +D2DC*cos2dec
      +EHS*sinha*cosha
      +EHC*cosha*cosha
      +D2HS*sin2ha
      +D2HC*cos2ha;

*ra+= dhaASECTORAD
*dec-= ddecASECTORAD

```

For the ALTAZ version of PC-TCS, the following code is used to implement the model.

```
delaz= IA
```

+ACEC*cosa
+ACES*sinaz
+AN*sinaz*tanel
-AW*cosaz*tanel
+CA/cosel
+NPAE*tanel;

delel= IE
+AN*cosaz
+AW*sinaz
+ECEC*cosel
+ECES*sinel
-TF*cosel
-TX/sinel;

*az+= -delaz*ASECTORAD
*el+= delel*ASECTORAD

ACL Remote Communications

As of version 6.x (sky) PC-TCS has the capability of accepting commands from a remote computer running *TheSky* or the ability to talk the ACL serial communications language to control PC-TCS. The communication link is a standard RS-232 serial connection using any one of the standard DOS/Windows supported serial ports (COM1, COM2, COM3, COM4). Currently, PC-TCS “speaks” the ACL (Astronomy Command Language) at a fixed speed of 9600 baud which should be compatible with most of today’s high speed modems).

Unless otherwise specified, communications with *TheSky* is available as the default communications protocol – in lieu of COMSOFT’s more powerful remote communications protocol as documented in the PC-TCS User’s Manual, which has now become a special order item (no charge).

To invoke remote communications with other computer running *ThSky*, use the Parameters-Input-Remote menu to select the appropriate COM port. (Note: PC-TCS does not perform any port initialization for modem control (i.e. setting Auto Answer mode, etc.) – if this is required it should be done by the user with other communications software. Once selected PC-TCS will always attempt to communicate with remote computers via this port – even after a PC-TCS could start.

Note: As of version 5.7 PC-TCS now supports focus control via *TheSky*.

As of PC-TCS version 5.8 several commands are now supported via the “Terminal Interface” of TheSky.

Command	Functionality
dome on	Turns Pc-TCS’s automatic dome control “on” this command is equivalent to the <i>Motion-auto Dome</i> menu command placing the dome in “Auto” mode.
dome off	Turns Pc-TCS’s automatic dome control “off” this command is equivalent to the <i>Motion-auto Dome</i> menu command placing the dome in “Manual” mode.
dome open	Commands PC-TCS to open the dome shutter (if this option is utilized).
dome close	Commands PC-TCS to close dome shutter (if this option is utilized).
dome stow	Turns PC-TCS’s automatic dome control “on” this command is equivalent to the <i>Motion-Stow Dome</i> menu command placing the dome in “Stow” mode.
hard stow	Performs the PC-TCS <i>Move-Hard Stow</i> operation. This command must be followed by a “cr” command (see below) to be accepted by PC-TCS. The “esc” command may be used instead to abort the operation.
cr	Sends a “Carriage-Return” to PC-TCS.
esc	Sends an “Escape” to PC-TCS.

PC-TCS New/Updated Menu Items

Several new features have been added to PC-TCS and a few menu items have been reorganized in an attempt to increase efficiency and utility. The following pages are addendum and updates to the version 4.5 PC-TCS users manual Reference section in Chapter 3.

Motion- auto Dome

Direct key Access: Alt-D

Functionality: Enables/ disables PC-TCS automatic control of the dome position. When enables PC-TCS will attempt to move the dome via the dome control TTL signals until **Dome** enunciator is Green – indicating an unobstructed telescope view. When disabled PC-TCS will NOT send dome control signals if dome position indicates dome is obstructing telescope view.

Uses: Used to disable PC-TCS dome update during critical observing times such as long exposure astrophotography when dome motion may cause vibrations that would comprise the observation. Also useful in delaying dome rotation during slews to avoid dome “jerk” in those installations where dome speed is faster than telescope slew speed.

Indicators: The menu item will be preceded by a sideways chevron (») when active.

Errors: None associated with this menu item.

Motion- Paddle

Direct key Access: Alt-L

Functionality: Enables/ disables the hardware paddle functionality. This item allows the user to activate/ deactivate the reading of the hardware associated with the paddle.

Uses: Useful in disabling the paddle for observatories where novice users or the general public may inadvertently gain access to the paddle. Also useful in disabling the paddle in the event of paddle hardware failure.

Indicators: The menu item will be preceded by a sideways chevron (») when active.

Errors: None associated with this menu item.

Motion- Stow dome

Direct key Access: None

Functionality: Moves dome to initial position as defined at the beginning of PC-TCS execution or when the **Declare-Dome Initial** menu item is executed.

Uses: Useful in stowing the dome prior to shutdown or verifying dome initialization.

Indicators: the “Dome” label on the Runtime screen will show “Stow”.

Errors: None associated with this menu item.

Declare – Init Next

Direct key Access: Alt-O

Functionality: a powerful commands that should be used with great caution since it modifies how PC-TCS interprets current telescope position. The command is used to initialize the coordinate system of PC-TCS by making it assume that whatever coordinates displayed in the **NEXT** object line of the Runtime display are the current coordinates the telescope is pointing.

If the telescope is not accurately positioned at the actual coordinates of the **NEXT** line, PC-TCS will function improperly. The display elevation, azimuth and position of the telescope will be in error and the software limits will function improperly – allowing the telescope to assume positions that may be hazardous to the instrumentation and/or the observer.

Uses: Used to initialize PC-TCS coordinate system and therefore the true position of the telescope on the sky. Used whenever PC-TCS is first started, after the telescope has recovered from a powerful failure, or lost accurate registration (pointing) on the sky. This command differs from the **Init Command** menu item in that it uses the **NEXT** position as the coordinates to sync telescope position – and therefore does not require a **Move to- NEXT** command be issued before it can be used. This command is also used in initializing German Equatorial scopes that are clutched and can be manually “flipped” and pointed to objects as the command quires the user as to which side of the mount the tube is currently located.

Indicators: The **NEXT** coordinates will replace the current position coordinates on the Runtime Display.

Errors: If the **NEXT** position is outside the limits of the telescope as set by PC-TCS the operation will display an error and abort the operation.

Declare – Focus Zero

Direct key Access: None

Functionality: Zeros out the focus readout on the Runtime Display.

Uses: Used to zero focus readout for user calibration. Useful only if the optional PC-TCS focus hardware is installed and functional.

Indicators: Focus readout on the Runtime display will be zeroed.

Errors: None associated with this menu item.

Declare – Dome Initial

Direct key Access: None

Functionality: Resets the internal PC-TCS dome registration to the dome “stow” azimuth. When invoked PC-TCS adjusts its dome position to be equal to the initial dome stow position. PC-TCS automatically calls this function whenever PC-TCS is first executed. If the dome needs to be initialized at a later time this function may be used to reset the dome “Stow” position.

Uses: Useful in recalibrating the dome position readout after a power failure or if the dome encoder has lost registration due to slippage.

Indicators: the dome readout will change to reflect a new dome error if the telescope is not aligned with the slit.

Errors: None associated with this menu item.

Position – Planets

Direct key Access: None

Functionality: Used to compute the current position of the Sun, Moon, Planets and user defined solar system objects. This menu item display a File Window showing the names of the planets and any orbit files with the extension “.QOR” that contain orbital information such as documented in the ORBIT.DOC file on the distribution diskette.

Note: processing a given object with this menu item will AUTOMATICALLY invoke the track rate for that object. To disable the computed rate use the **Rates – Object** menu item

Uses: Useful in computing the positions of the Sun, Moon, Planets and user defined solar system objects for later telescope slews using the **Move – Next** menu item.

Indicators: The **Rates – Object** menu item will be preceded by a sideways chevron (») when an object has been successfully computed. The object name and coordinates will appear in the **Next** line of the Runtime Display. The tracking indicator between the current position coordinates will indicate an object rate is active if tracking is active.

Errors: An error window will appear if the orbit is mathematically erroneous.

Position – Comets

Direct key Access: None.

Functionality: Allows the user to compute the position a comet from a catalog of these objects found in the file COMETS.ORB.

Note: processing a given object with this menu item will AUTOMATICALLY invoke the track rate for that object. To disable the computed rate use the **Rates – Object** menu item.

This file may be edited with any standard ASCII text editor – provided that the fixed length record structure of the file is not molested. All lines in the file are 128 characters in length. The format for the orbital elements can be found at the end of the file.

Uses: Useful in computing the positions of cataloged comets for later telescope slews using the **Move – Next** menu item.

Indicators: The **Rates – Object** menu item will be preceded by a sideways chevron (») when an object has been successfully computed. The object name and coordinates will appear in the **Next** line of the Runtime Display. The tracking indicator between the current position coordinates will indicate an object rate is active if tracking is active.

Errors: An error window will appear if the orbit is mathematically erroneous.

Position – Asteroid

Direct key Access: None.

Functionality: Allows the user to compute the position a comet from a catalog of these objects found in the file ASTEROID.ORB.

Note: processing a given object with this menu item will AUTOMATICALLY invoke the track rate for that object. To disable the computed rate use the **Rates – Object** menu item.

This file may be edited with any standard ASCII text editor – provided that the fixed length record structure of the file is not molested. All lines in the file are 99 characters in length and must.

Uses: Useful in computing the positions of cataloged asteroids for later telescope slews using the **Move – Next** menu item.

Indicators: The **Rates – Object** menu item will be preceded by a sideways chevron (») when an object has been successfully computed. The object name and coordinates will appear in the **Next** line of the Runtime Display. The tracking indicator between the current position coordinates will indicate an object rate is active if tracking is active.

Errors: An error window will appear if the orbit is mathematically erroneous.

Rates – Object

Direct key Access: None.

Functionality: Allows the user to disable an object rate previously invoked by processing a comet, planet or asteroid orbit using the **Position – Comet, planet, or Asteroid** menu items.

Uses: Used to turn OFF object rates.

Indicators: The **Rates – Object** menu item preceded by a sideways chevron (») will revert back to its inactive state. The tracking indicator between the current position coordinates will no longer indicate an object rate is active if tracking is on.

Errors: An error window will appear if the orbit is mathematically erroneous.

Parameters-Globals-Display-Screen Saver

Direct key Access: None.

Functionality: Enables/ disables the built in screen saver function of PC-TCS. The screen saver when active, will automatically blank the TCS Runtime display if no keyboard or mouse activity has occurred within a user defined number of minutes. When selected PC-TCS will query the user for the “time-out” interval in minutes before the screen will be blanked.

Uses: useful in keeping the PC-TCS Runtime display from “burning-in” on the CTR over extended PC-TCS operation.

Indicators: The menu item will be preceded by a sideways chevron (») when the screen saver is active.

Errors: None are associated with this menu item.

Parameters-Utilities-FDC-Limits

Direct key Access: None.

Functionality: Allows the user to limit the Azimuth of FDC star selection so that a limited area of the sky is sample during Flexure Data Collection. The elevation selection criteria are based on the Horizon Limit, Declination Limit and hour Angle Limits set in PC-TCS.

Uses: Useful in limiting the scope (and therefore the number of objects) sampled during flexure data collection. Very useful in avoiding sampling objects that would otherwise be acquired “below the pole” during pointing runs.

Indicators: A query window will appear prompting the user for the minimum and maximum Azimuth for data sampling.

Errors: Ann error windows will appear if the orbit is mathematically erroneous.

Parameters-Utilities-Focus-FocusABS

Direct key Access: None.

Functionality: Allows the user to move the focus motor to an absolute focus value.

Uses: Useful in moving the focus position to a know fiducial focus after an instrument or filter change for which the focus value was previously known.

Indicators: The focus display will change as the motor is moved to the new position.

Errors: None are associated with this function.

Parameters-Utilities-Focus-FocusREL

Direct key Access: None.

Functionality: Allows the user to move the focus motor to from its current value to the current value plus the user specified value.

Uses: Useful in moving the focus position to a know offset focus after an instrument or filter change for which the focus offset is known.

Indicators: The focus display will change as the motor is moved to the new position.

Errors: None are associated with this function.

Parameters-Utilities-Focus-FocusPARAM

Direct key Access: None.

Functionality: Allows the user to modify the parameters associated with focus motion. Values alterable include:

Focus speed setting (fast & slow) used with keyboard, paddle, REL and ABS moves.

The number of steps the focus fast and focus slow move the focus motor when controlled from TheSky.

The limits of travel of the focus motor in steps.

Uses: Useful in modifying the operational characteristics of the focus control portion of PC-TCS.

Indicators: A window will be opened prompting the user for values for each of the above related items.

Errors: None are associated with this function.

Parameters-Utilities-Focus-Paddle focus

Direct key Access: None.

Functionality: Allows the user to enable/disable the keyboard and hand paddle focus function. This menu item does not inhibit focus operation via the external serial communications interface to PC-TCS.

Uses: Useful in inhibiting guests from changing focus of the telescope or preventing erroneous focus changes during critical observations.

Indicators: A chevron (») will appear next to the menu item when the keyboard and hand paddle focus control is active.

Errors: None are associated with this function.

Parameters-Utilities-DefPEC

Direct key Access: None.

Functionality: Allows the user “train” PC-TCS to correct for periodic error in the telescope RA gear train. If PEC hardware is installed, PC-TCS will wait for the worm index pulse to occur before recording paddle button presses during the user guiding operation that must be done for one full revolution of the worm.

To accomplish this training – start PC-TCS and point your telescope and begin tracking an easily visible star near or on the celestial equator (0° Declination) and near the meridian. Use a high power eyepiece with a reticle to ensure critical guiding during the subsequent training procedure.

Before being training the user should set the desired RA and DEC guide rates using the PC-TCS *Rates-Guide* menu item. The rate set here will be remembered by PC-TCS and user for future PEC corrections. (Once training is complete, the user may modify the guide rates to accommodate other observing needs.)

Note: If PEC hardware is not installed, the user may still “train” PC-TCS but the training will only be valid until PC-TCS is terminated.

Uses: Useful in correcting the telescope tracking to account for periodic error in the RA gear train.

Indicators: If you have PEC hardware turned on (PFEDIT – *Telescope-PEC hardware* menu item) the software will wait for the RA index pulse before sampling begins. Once sampling has begun the training window will display a progress bar as the interval elapses. When complete the information collected will be saved to a file for future tracking corrections. From this time forward any time tracking is enabled, PEC will automatically be invoked specifically turned off using the PC-TCS *Rates-PEC* menu item.

Errors: An error window will appear if the orbit is mathematically erroneous.

PC-TCS Script and Remote Commands

ABER OFF/ON	Turns the pointing corrections for Aberration OFF/ON.
ACCESS	sssssss Sets the security level for keyboard access to PC-TCS (password – sssssss for desired level of access.)
ASTEROID	nnnn Processes the orbit for the Asteroid catalog number “nnnn” and puts the coordinates into the NEXT position registers of PC-TCS. In addition, the rate of motion of the asteroid is automatically computed and implemented so that if tracking is active PC-TCS will immediately begin tracking at the asteroid’s rate of motion. This command should be followed by a MOVENEXT command to move the telescope to the asteroid’s position on the sky.
AUTODOME	OFF/ON Turns TCS automatic dome motion control OFF/ON
BIASDEC	±nnnn.nnn Sets the declination axis “bias rate” in nnnn.nn arcseconds per second. The rate will not become active unless a BIASON command is given or a previous non-zero rate is already active.
BIAS OFF/ON	Turns any active bias rates OFF/ON.
BIASRA	±nnnn.nnn Sets the declination axis “bias rate” in nnnn.nnn arcseconds per seconds. The rate will not become active unless a BIASON command is given or a previous non-zero rate is already active.
CAMERA	ON/OFF Triggers Apogee Instruments CCD shutter (Requires Optional focus control interface hardware).
CANCEL	Stops a slew in progress with full ramp down.
CDBIAS	OFF/ON Turns the correction for cosine declination in the bias rate OFF/ON.
CDDRIFT	OFF/ON Turns the correction for cosine declination in the paddle drift rate OFF/ON.
CDGUIDE	OFF/ON

	Turns the correction for cosine declination in the paddle guide rate OFF/ON.	
CDWOB	OFF/ON	
	Turns the correction for cosine declination in the wobble vector OFF/ON.	
CLEARDIFF	Resets the difference fields on the PC-TCS Runtime display to zero.	
CLIRESUME	(remote only)	
	Restarts the command file currently suspended by WAITREM command.	
COMET	nnnnn	
	Processes the orbit for the comet catalog number “nnnn” and puts the coordinates into the NEXT position registers of PC-TCS. In addition, the rate of motion of the comet is automatically computed and implemented so that id tracking is active PC-TCS will immediately begin tracking at the asteroid’s rate of motion. This command should be followed by a MOVENEXT command to move the telescope to the comet’s position on the sky.	
COMFILELOAD	filename	(remote only)
	Loads the script file “filename” into PC-TCS memory for processing. Processing of script commands begins after the COMFILEON command. (filename should not contain the “.TCS” extension.	
COMFILE	OFF/ON	(remote only)
	If “OFF” – suspends processing of the script file previously loaded with the COMFILELOAD command and activated by the COMFILE ON command.	
DECLAREINIT	Synchronizes the telescope by forcing the current position to become the same as the commanded position.	
DECLAREREF	Copies the current position into the Reference registers of PC-TCS.	
DECALRESTOW		
	Changes the Stow position of the Telescope to be the current position of the telescope.	
DISEPOCH	nnnn.nnn	
	Changes the display epoch to be “nnnn.nnn” is the date expressed as a decimal year. A value of 0.0 will force the display epoch to be equal to the current date.	
DOME	OPEN/CLOSE	
	Opens or closes the dome.	
DOMEINIT	Declares the current position of the dome to be the stow position. This position is used by the STOWDOME command.	
DRIFTDEC	±nnnn.nnn	

	Set the paddle declination Drift rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.
DRIFTRA	±nnnn.nnn Sets the paddle Right Ascension Drift rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.
ELAZ	ee.ee ±aaa.aa Moves to the Elevation and Azimuth specified by ww.ww and ±aaa.aa degrees respectively.
END	(script only) Used in Script files to terminate execution of script commands.
EPOCH	nnnn.nnn Changes the input coordinate epoch to be “nnnn.nnn” where “nnnn.nnn” is the date expressed as a decimal year. A value of 0.0 will force the input epoch to be equal to the current date. The input epoch is used to process coordinate to true of date coordinates.
ERRORRESET	(script only) Used to reset the error branch of a script file set by the ONERROR command. Nominally an error encountered in a script file will cause immediate termination of the script. Using the ONERROR command allows the processing to continue at the line of the script file specified in the ONERROR argument. The ERRORRESET command will cause the script file to revert back to termination on errors after an ONERROR command had been executed.
FILEPERM	filename Sets the source input mode to PCF catalog mode using the PCF file specified by “filename”. (“filename” should not contain the “.PCF” extension.)
FILEUSER	filename Sets the source input mode to USER catalog mode using the USER file specified by “filename.” (“filename” should not contain the “.CAT” extension.)
FLEX OFF/ON	Turns the correction for flexure OFF/ON when processing new input coordinates.
FLEXRATE	OFF/ON Turns the correction rate for flexure OFF/ON.
FOCUS	±nnnn (remote only) Move the focus motor “nnnn” steps.
GOSUB	nnnn (script only)

	Branches to a subroutine at line “nnnn”.
GOTO	nnnn (script only)
	Immediately branches to script execution at line number ‘nnnn’.
GUIDEDEC	±nnnn.nnn
	Set the paddle Declination Guide rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.
GUIDERA	±nnnn.nnn
	Set the paddle Right Ascension Guide rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.
IFNEXT	nnnn (script only)
	Conditionally branches to script file line “nnnn” if the position stored in the PC-TCS Next register is currently accessible by the telescope.
IFNOTNEXT	nnnn (script only)
	Conditionally branches to script file line “nnnn” if the position stored in the PC-TCS NEXT register is NOT currently accessible by the telescope.
JUPITER	Computes the current position of the planet Jupiter and stores the coordinates in the NEXT position registers of PC-TCS. The command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move to this computed position.
KILL	Disables the PC-TCS drive system. This command acts like an “Emergency Stop” command and should be used only in most severe of circumstances.
LIMTENABLE	(remote only)
	Enables limit detection and processing in PC-TCS.
LIMITOVERRIDE	(remote only)
	Disables PC-TCS limit detection and processing.
LUNARRATEON	
	Invokes the lunar rate corrections in RA and DEC.
MANUAL	Sets the Source Mode of PC-TCS to Manual – (keyboard like) where RA and DEC’s must be individually specified rather than retrieved from a file. This mode command must be given before a NEXTRA and NEXTDEC command can be executed.
MARS	Computes the current position of the planet Mars and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.

MERCURY	Computes the current position of the planet Mercury and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.
MIRROR	OPEN/CLOSE Opens or closes mirror covers.
MOON	Compute the position of the Moon and stores the coordinates in the NEXT position registers of PC-TCS. The command also immediately invokes the tracking rate corrections for this object. A MOVENEXT command is required to move to this computed position.
MOVB1	Moves to the current position plus the value of the Wobble Vector.
MOVB2	Moves to the current position minus the value of the Wobble Vector.
MOVDOME	aaa.aa Moves Dome to the azimuth aaa.aa.
MOVENEXT	Moves to the position stored in the NEXT Registers of PC-TCS.
MOVOFF	Move to the current position plus the value of the Offset Vector.
MOVREF	Moves to the position specified in the Reference Register of PC-TCS.
MOVSTOW	Moves to the current Stow position of the telescope. This command also disables tracking.
MTIME	n (remote only) Sets the focus speed – larger numbers = slower speed.
NEPTUNE	Computes the current position of the planet Neptune and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.
NEXTDEC	±ddmmss.s Sets the Next declination Register in PC-TCS to the value “ddmmss.s” where dd = degrees, mm = minutes, and ss.s = seconds. The MANUAL command should be executed before this command is given.
NEXTOBJECT	Automatically increments the objects file pointer when using the FILENAME, or FILEUSER modes to the next object in the catalog.
NEXTRA	±hhmmss.s Sets the Next Right Ascension Register in PC-TCS to the value “hhmmss.s” where hh = hours, mm = minutes, and ss.s = seconds. The MANUAL command should be executed before this command is given.
NOP	A “No Operation” Command – does nothing. Used as a place holder in script files and to test communications in Remote operations.

NUTAT OFF/ON

Turns the pointing corrections for nutation OFF./ON.

OBJECT nnnnn

Retrieves and processes the coordinates in record number “nnnnn” in the catalog file selected using the FILEPERM or the FILEUSER modes. The processed coordinates are then stored in the NEXT registers of PC-TCS. A MOVENEXT command will then move the telescope to these coordinates.

OBJECTRATEOFF

Disables (turns off) the object rates automatically invoked when an asteroid, comet, sun, moon, or planet position is computed.

OFFDEC ±ddmmss.s

Sets the Declination component of the Offset Vector to “ddmmss.ss” where dd = degrees, mm = minutes, and ss.ss = seconds.

OFFRA ±ddmmss.s

Sets the Right Ascension component of the Offset Vector to “ddmmss.ss” where dd = degrees, mm = minutes, and ss.ss = seconds.

ONERROR nnn (script only)

Sets the script error branch to line “nnn” of the script file. If an error occurs in a script command after this command is executed the CLI will begin executing script commands at line number “nnn”. Normally, without this command, PC-TCS would unconditionally terminate script file execution.

ORBIT filename

Reads and processes the orbital information in the “Quick Orbit” file in the filename “filename.” The computed position is stored in the NEXT registers of PC-TCS for later MOVENEXT command. The rate corrections for the motion of the object is also immediately invoked. (Filename should NOT contain the “.QOR” extension).

PARA OFF/ON Turns the pointing corrections for parallax OFF/ON.

PLUTO Computes the current position of the planet Pluto and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.

PREC OFF/ON Turns the pointing corrections for precession OFF/ON.

PROMPT message (script only)

Pauses script file execution and display the “message” argument to the user. Any key hit on the keyboard will continue script file execution.

PROPER OFF/ON

	Turns the pointing corrections for proper motion OFF/ON.	
RATER	rr.r dd.d RR DD tt.t key	Invokes the Raster Scan utility Right Ascension step size of rr.r arcseconds, declination step size of dd.d arcseconds, for RR step in Right Ascension, DD step in Declination, with a dwell time of tt.t seconds. The “key” parameter is used to decide the input which the Raster Scan will terminate. Zero (0) will perform the full raster scan with no interruption capability, one (1) will terminate the scan if the TTL S1 input is high, two (2) will terminate the scan if the TTL S2 input is high, three (3) will terminate the scan if the TTL S3 input is high, four (4) will terminate the scan if the TTL S4 input is high, five (5) will terminate the scan if a keyboard character input is detected.
RECALLPOS	nn	Recalls the saved position (see SAVEPOS command) and copies the coordinates to the NEXT registers of PC-TCS. The telescope can be moved to this position using the MOVENEXT command.
REFRAC	OFF/ON	Turns the pointing position corrections for refraction OFF/ON.
REFRATE	OFF/ON	Turns the rate corrections for refraction OFF/ON.
REMOTECHK	OFF/ON (not TheSky)	Turns the serial checksumming OFF/ON.
REMOTE	OFF/ON (script only)	Turns remote serial access OFF/ON.
RETURN	(script only)	Returns from previously called subroutine (see GOSUB command).
SATURN	Computes the current position of the planet Saturn and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.	
SAVEPOS	nnn	Saves the current position to the Save register number “nnn”. Use the RECALLPOS command to retrieve the value.
SETDATE	mm/dd/yyyy (remote only)	Sets the UT date of PC-TCS to “mm/dd/yyyy” where mm = month, dd = day, yyyy = year.
SETTIME	hhmmss.dd (remote only)	

Sets the PC-TCS UT time to “hhmmss.ss” where hh = hours, mm = minutes, sss.ss = seconds.

SLEW OFF/ON (deadman slew systems only)

Turns OFF/ON the slew capability of PC-TCS (speed limited to @ 1/5 degree per second).

SOLARRATEON

Invokes the solar rate corrections in RA and DEC.

SPIRAL ss.s LL tt.t key

Invokes the Spiral Scan utility Right Ascension and Declination step size of ss.s arcseconds, for LL loops, with a dwell time tt.t seconds. The “key” parameter is used to decide the input which the Raster Scan will terminate. Zero (0) will perform the full raster scan with no interruption capability, one (1) will terminate the scan if the TTL S1 input is high, two (2) will terminate the scan if the TTL S2 input is high, three (3) will terminate the scan if the TTL S3 input is high, four (4) will terminate the scan if the TTL S4 input is high, five (5) will terminate the scan if a keyboard character input is detected.

STEPDEC nnnnnn (remote only)

Moves the telescope in Declination “nnnnnn” steps (or encoder counts). This command provides guiding capabilities for smart peripherals.

STEPRA nnnnnn (remote only)

Moves the telescope in Right Ascension “nnnnnn” steps (or encoder counts). This command provides guiding capabilities for smart peripherals.

STOWDOME Moves the dome to the Stow position as specified in the PFEDIT program. Note: the stow position is the position the dome is at when TCS is started or the position the dome is at when the **Declare-Dome Stow** menu item is activated or the DOMEINIT command is given.

SUN Computes the current position of the Sun and stores the coordinates in the NEXT position registers of PC-TCS. The command also immediately invokes the tracking rate corrections for this object. A MOVENEXT command is required to move to this computed position.

TRACK OFF/ON

Turns OFF/ON sidereal tracking and correction rates.

UNABER OFF/ON

Turns OFF/ON the deconvolution for Aberration on the RA and DEC display.

UNFLEX OFF/ON

Turns OFF/ON the deconvolution for Flexure on the RA and DEC display.

UNKILL

	Enables PC-TCS drive capabilities. (Used after a previous KILL command).
UNREFR	OFF/ON Turns OFF/ON the deconvolution of Refraction on the RA and DEC display.
URANUS	Computes the current position of the planet Uranus and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.
VENUS	Computes the current position of the planet Venus and stores the coordinates in the NEXT position registers of PC-TCS. This command also immediately invokes the tracking rate corrections for this planet. A MOVENEXT command is required to move this computed position.
WAITDL	tt.t (script only) Delays script execution for tt.t seconds.
WAITLST	hhmmss.s (script only) Delays script execution until a Local Sidereal Time of hhmmss.s where hh = hours, mm = minutes, ss.s – seconds.
WAITMO	(script only) Delays script execution until telescope motion is stable.
WAITNEXT	(script only) Delays script execution until position in PC-TCS NEXT registers is within telescope motion limits.
WAITNONEXT	(script only) Delays script execution until NEXT position is outside of telescope motion limits.
WAITREM	(script only) Delays script execution until the remote command CLIRESUME is received.
WAITUT	hhmmss.s (script only) Delays script execution until UT of hhmmss.s where hh = hours, mm = minutes, ss.s = seconds.
WOBDEC	±ddmmss.s Sets the Declination component of the Wobble Vector to “ddmmss.ss” where dd = degrees, mm = minutes, and ss.ss = seconds.
REMWOB	OFF/ON Turns OFF/ON the hardware remote wobble capability of PC-TCS.

WOBRA $\pm$ hhmmss.s

Sets the Right Ascension component of the Wobble Vector to “hhmmss.ss” where hh = hours, mm = minutes, and ss.ss = seconds.

TELEMETRY STRUCTURE

The following table documents the structure of the PC-TCS telemetry stream broadcast on all active channels during PC-TCS operation. This telemetry stream is applicable to the COMSOFT “native” communications protocol and is not available with PC-TCS “TheSky” communications protocol.

Column	Format	Description
1	Number	Motion Status of Telescope, 0= stable, 1= Declination motion, 2= Right Ascension Motion, 3= Both axes in motion.
2	Number	Current Wobble Beam, blank= Not on a Beam, 1= at Beam 1, 2= at Beam 2.
4-12	HHMMSS.ss	Current Telescope Right Ascension
14-22	±DDMMSS.s	Current Telescope Declination
25-33	±HH:MM:SS	Current Telescope Local hour Angle
35-42	±HH:MM:SS	Current Local Sidereal Time
44-48	±DD.s	Current Telescope Elevation in degrees.
50-55	±DDD.d	Current Telescope Azimuth in degrees. – implies west + implies east.
57-61	DD.ss	Current Secant of the Zenith distance (Secz).
63	Number	COM1 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
64	Number	COM2 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
65	Number	COM3 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
66	Number	COM4 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
67	Number	COM5 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
68	Number	COM6 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
69	Number	COM7 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
70	Number	COM8 Remote command execution code, E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized command, C= checksum error.
72	Number	RA Limit. R= in Limit, blank= not in limit.
73	Number	DEC limit. D= in Limit, blank= not in Limit.
74	Number	Horizon Limit. H= in Limit, blank= not in Limit.
75	Number	Drive Disable. D= disabled, blank= enabled.
76-83	YYYY.YY	Display Equinox.
85-93	JJJJJJ.JJ	Julian Date.
95	Number	Current com channel number.
97	±nnnnn	Focus Position
104-109	±DDD.d	Dome Error in degrees. – implies west, + implies east.

111-120	HH:MM:SS.s	UT Time.
121-148		Reserved for expansion.
149-150	Terminator	CR, LF

COMET.ORB File Format

PC-TCS now has a user definable number of comet orbits available via the ***Position-Comets*** menu item. The cometary orbital elements are stored in a file named COMETS.ORB. The file is a sample ASCII Text file capable of being edited with a standard DOS Text files. Be aware however that the file has one element set for one comet per line of the file and that the lines are fixed at 127 characters each – including the terminating Carriage Return-Line Feed sequence. Care must therefore be taken when editing this file not to corrupt the fixed record length structure of the file.

The fields within each element set record line must be separated by blanks (spaces) and the record must be blank padded to a total record length of 127 characters.

Field	Description
Year ID	The Standard IAU comet designation usually formatted C/YYYYID where YYYY is the year of discovery and ID is a Letter-number designation for the comet.
q	The perihelion distance in astronomical units.
e	The orbital eccentricity (1.0= parabolic; <1.0 = elliptical; > 1.0 = hyperbolic)
i	the orbital inclination
ó	The argument of Perihelion
Ω	The longitude of the Ascending Node
y	The year of the date of perihelion
m	the month of the date of perihelion
d.ddddd	The day and decimal day of the date of perihelion
g l	The absolute magnitude of the comet as seen from 1 astronomical unit from the sun and the earth
k	A coefficient for computing magnitude. For elliptical orbits the magnitude is computed via the formula $m = g_l + 5 \log_{10}(\Delta \cdot r) + k \cdot PA$ For parabolic orbits the formula $m = g_l + 5 \log_{10}(\Delta) + k \cdot \log_{10} \mathbb{R}$, where Δ = the distance to the comet from the earth; r = the distance to the comet from the sun; PA = the phase angle of the comet.
name	The name of the comet – (must have no embedded white space)

ASTEROID.ORB File Format

PC-TCS now has a user definable number of asteroid orbits available via the ***Position-Asteroids*** menu item. The asteroid orbital elements are stored in a file named ASTEROID.ORB. The file is a simple ASCII Text file capable of being edited with a standard DOS Text files. Be aware however that one file has one element set for one asteroid per line of the file and the lines are fixed at 98 characters each – including the terminating Carriage Return-Line Feed Sequence. Care must therefore be taken when editing this file not to corrupt the fixed record length structure of the file.

The fields within each element set record line must be separated by blanks (spaces) and the record must be blank padded to a total record length of 98 characters.

Field	Description
ID Number	The Standard IAU asteroid number designation
Element Epoch	The Epoch of the elements in the format yyyyymmdd
Ma	Mean Anomaly of the orbit in degrees
ω	The argument of perihelion in degrees
Ω	Longitude of the Ascending Node in degrees.
i	inclination in Degrees
e	Eccentricity
a	Semimajor axis in astronomical units
G	Absolute magnitude computation
K	Phase contact for magnitude computation
Name	The Asteroid's proper name

PC-TCS Software Installation and Setup

PC-TCS comes distributed on CD-ROM for installation under Windows 95/98 and DOS operating systems. Before attempting to install PC-TCS on your computer, it is recommended that you complete the installation of the interface cards and ancillary electronic connections. If you purchase your TCS computer from COMSOFT the software has been pre-installed and the installation procedure documented here is provided only for reference.

If you are installing PC-TCS under DOS please skip the section below for Windows and see the DOS Installation section of this document.

Windows 95/98 installation

The distribution CD-ROM is set-up to auto-install using InstallShield – the standard Windows 95/98 installation software binding application. Follow the steps outlined below to install and configure PC-TCS/CT.

1. Boot your computer.
2. Insert the PC-TCS/CT installation CD-ROM into your CD-ROM drive and wait for the InstallShield banner to appear.
3. Follow the instructions on screen to complete the installation – it is highly recommended that you take the default answers to the prompts given during installation.

Windows Shortcut Setup

A part of the installation process InstallShield will create and place several icons on the Windows Desktop similar to those shown below. Each of the icons is a shortcut to the DOS applications associated with PC-TCS.

PC-TCS and Windows Power Management

Users of Windows 95/98 should also make sure that the power saving features of the operating system are disabled. This is to make sure that PC-TCS is not interrupted by the operating system while in running the telescope. To set this condition – go to the Windows control panel using the *Start – Settings – Control Panel* menu under Windows. From the Control Panel window bring up the “Power Management” window and configure it as follows . .

Make sure the “Turn-off monitor” and “Turn-off hard-disk” are set to “Never”.

PC-TCS and Windows Time

For proper PC-TCS operation, the target Windows 95/98 machine needs to have its clock set to report Universal Time (UT or GMT). This is done by either double clicking on the time displayed in the task bar at the bottom of the screen, or by going to the Control panel and double clicking on the Date/Time icon.

Once this window is displayed first select the "Time Zone" tab and set the zone to (GMT) Casablanca, Monrovia as shown below. Do not select the (GMT) Greenwich Mean Time Dublin, Edinburgh, Lisbon, London option as this item adjusts for Daylight Savings Time.

After setting the time zone , set the time using the Date & Time tab to the current UT time. If you are not sure what the conversion between UT time and your local time refer to Sky & telescope Magazine. Every issue of this magazine has a description on UT Time that can be found in the "Celestial Calendar" section.

At this point it will be necessary to run the PFEDIT program to configure PC-TCS/CT for your telescope and site. Please skip to the "**Configuring PC-TCS for your telescope**" section of this document.

DOS Installation

DOS installation of PC-TCS from the distribution CD-ROM is a manual copy operation. Follow the steps below.

1. Create a subdirectory called TCS on your system disk by typing the following command while logged to your system disk (usually drive letter C:)

md tcs ←

2. Insert the PC-TCS distribution CD-ROM into your CD-ROM drive.
3. Change to the CD-ROM drive. This is accomplished by typing the drive of your CD-ROM drive followed by a colon (e.g. if your CD-ROM drive is letter "E" then type ***E:***← where ← is the "Enter"
4. Change to the DOS subdirectory by typing

CD DOS ←

5. Copy the contents of the DOS subdirectory to your system disk by typing the following command

xcopy *.* /s c:\tcs ←

Where C: is the system drive letter.

Once this copy is complete you should have all files associated with PC-TCS in a subdirectory called tcs on your system drive.

Configuring PC-TCS for your telescope

Before attempting to execute PC-TCS for the first time users should check the following.

1. All hardware is installed and all external connection to the PC-TCS computer have been made. (See **PC-TCS Hardware Installation Guide**.)
2. Make sure the PC-TCS computer has conventional memory to run PC-TCS. PC-TCS requires a minimum of 605 Kilobytes of conventional (low) memory to operate properly. To check this, at

the DOS prompt (under Windows 95/98 you will have to bring up a MS-DOS window) type **mem** ←. After typing this command you should get a screen similar to the one below.

After setting the time zone, set the time using the Date & Time tab to the current UT time. If you are not sure what the conversions between UT time and your local time refer to Sky & telescope Magazine. Every issue of this magazine has a description on UT Time that can be found in the “Celestial Calendar” section.

At this point it will be necessary to run the PFEDIT program to configure PC-TCS/CT for your telescope and site. Please skip to the “*Configure PC-TCS for your telescope*” section of this document.

Note the “Largest executable program size” line of the display. If this reads less than 605 K the user should modify the file CONFIG.SYS in the root directory of the system drive to read as follows.

```
Device=c:\windows\himem.sys
Device= c:\windows\emm386.exe m9 d=256 noems
Dos=high,umb
```

These lines should be put at the beginning of the CONFIG.DTD file ahead of previous lines if any. It may be necessary to remove or load into high memory (via the *devicehigh=* command) any additional device drivers that may be itemized in the an existing CONFIG.SYS file to prevent them from using conventional memory space. For DOS only users the paths for *himem.sys* and *emm386.exe* may be different. Check your system disk for the proper paths for these two files.

Running the PFEDIT Program

When run, PC-TCS first reads a “Parameter File” (TCS.PF file) that contains information unique to each installation. The program PFEDIT or (actually PFDSKY or PFEDCOM depending on which version of PC-TCS you can run) allows the user to modify this file with information related to their telescope and site. Parameters such as the site geodetic coordinates, the number of motor steps per arcseconds on the sky, telescope axial limits, catalog file paths, security levels for individual menu items, etceteras, can be modified using the PFEDIT program.

Under the Windows 95/98 installation an icon is created which allows access to this program. To execute the PFEDIT program simply double click on the icon which looks like the one shown below. DOS users need to change to the TCS subdirectory and type **PFDSKY** ← or **PFEDCOM** ←.

The TCS.PF file distributed with PC-TCS is the file used by COMSOFT on our 14 inch telescope in Tucson Arizona and therefore is configured for our site and hardware. You will have to make modifications using the PFEDIT program before PC-TCS will operate properly.

PFEDIT will display a screen similar to the one shown below.

The menu bar at the top shows the main options while the help bar at the bottom gives a brief explanation or instruction as to what information is being requested. Below is a brief explanation of the main menu bar options.

Security: This menu item will display a menu bar identical to the one displayed on the PC-TCS Runtime screen. The user may select any menu item and set security level for that item. Security is an issue only in those observatory environments with multiple operators or where the general public may have access to the PC-TCS control computer. When PC-TCS is first started it automatically sets its security level to “Novice” and allows access to those menu items that have a security level of zero (0).

If the security level for a particular menu item on the PC-TCS is at the NOVICE level. An explanation of the other security levels and how to obtain access to them while running PC-TCS can be found in section 2.8 of the User's manual.

Limits: The limits menu allows the user to set limits on telescope motion. Limits to how far the telescope can travel in each axis can be set in addition to limiting the distance above the horizon, and how much of the earth's atmosphere is acceptable for the telescope to look through. If you have a German Equatorial mount the limits menu allows you to set how far the telescope can move past the meridian in Hour Angle before a German Flip is required.

Dome: This menu item is used to manage the dome control capabilities of PC-TCS. The number of encoder steps per degree, the stow azimuth of the shutter and the error window within which PC-TCS will not move the dome can be modified using this menu item.

Computer: The computer menu item allows the user to set various parameters associated with the support interface cards used by PC-TCS. Base address, interrupt numbers, software switches to enable external joystick and paddle operation are set using this menu item.

Files: This menu item allows the user to set the paths to various catalog files, script files and orbital element files used by PC-TCS.

Telescope: The telescope menu item is used to configure PC-TCS to the user's particular site and telescope features. Here the name of the site, and the site geodetic coordinates can be set. In addition the important constants associated with the drive system can be configured.

Exit: This menu item allows the user to exit the PFEDIT program.

To begin, bring up the PFEDIT program and move to the **Computer** menu and hit the *Enter* key – you will see a vertical menu bar as follows.

Hitting *Enter* again will bring up the address change menu as shown below.

If your CTR-10 card was supplied by COMSOFT then it has already been set for an address of 300 Hex and the default address should be correct. If you do not have some of the cards displayed in the window leave the addresses as they are unless they conflict with other address in your computer. ***Note: Do NOT set any of the addresses to 0 as this will cause the computer crash when PC-TCS attempts to talk such an address.*** If you have to change the base address of any of the PC-TCS computer interface cards – enter the appropriate hexadecimal version of the address at the appropriate line of the address window. When finished with modifications for any window, an update to the information will not be saved until the window is closed by filling out the last line of the window and hitting *Enter*, or by *Ctrl* and *Enter* keys simultaneously. The later method forces an update of the TCS.PF file without closing the window or having to completely fill out the information in the window itself.

After completing this information window, it should close the Computer menu should appear again. Use the arrow key or move the mouse to select the next menu item "Interrupts".

The interrupt menu item is used to set the interrupt number used by PC-TCS to generate its internal system time and servo control. If you received your CTR-10 interface card from COMSOFT it should already be set for interrupt for the TCS system board to 5. If you have a sound card in your computer, you may need to use interrupt seven for the TCS system board. If this is the case, you will also need to set the "back" interrupt jumper on the CTR-10 to 7 also.

The RA Index interrupt is used only if you have Periodic Error Correction (PEC) hardware hooked into PC-TCS. If you are planning on using PEC, you need to have the RA Index interrupt set indirectly than that set for the System Board Interrupt.

The COM4 interrupt is used by PC-TCS native communications drivers and requires an optional “COM4” I/O board. If you are running TCSSKY and not TCSCOM then this interrupt is not important. If however you plan on running TCSCOM and using the COM4 hardware this interrupt settings needs to be unique as well.

The “Delay constant” prompt in this window is used for some computers that require a slight time delay between the initialization of interrupts and PC-TCS verification that interrupts are functional. Typically P5-120 and higher need this value set to 10. Slower computer may be able to handle a smaller number with generating an error message – “*System board interrupts are not functional*” – when attempting to run PC-TCS.

After completing this modification – move to the Telescope-Location menu item as shown below.

Hitting the *Enter* key will open the following window. Modify the values shown with an accurate Latitude and longitude for your telescope – making sure to note that the longitude is in Hours, Minutes and Seconds.

Note that the longitude information must be entered in HH:MM:SS.ss from Greenwich. To compute this value from DD:MM:SS.ss convert the degrees in decimal degrees, divide the result by 15 and then convert this number back into time representation. Once the site location information is entered – move on to the **Telescope-Axis** menu item as shown below.

Hitting the Enter key will open the following window.

This axis windows are extremely important as it is key to how PC-TCS will command the motors of the telescope mount. As documented before the default values of this window are for COMSOFT’s research telescope in Tucson. Unless your ordered or plan to use encoders for closed loop operation the only items that need to be carefully modified are the Steps per arcseconds values for both the RA and DEC Axes. To compute these values use the formula below.

$$\text{Step per arcsecond} = (\text{GR} * \text{Steps per revolution}) / 1296000$$

Where: **GR** = The gear reduction between the motor and the axis.

Step per revolution – The number of microsteps the motor in one complete revolution. This value set into the microstepping driver dip switch for each axis.

If you are using coarse or closed loop operation encoders you will need to make similar computations for The Encoder counts / arcseconds prompts in these windows and set the “Use Coarse Encoders?” or “Run Closed Loop?” prompt to “I” instead of zero in the Feedback menu item window.

The Encoder Sign prompts should not be changed unless the telescope “runs away” in an uncontrolled slew after a commanded move. This should not happen if you are not running closed loop and have not modified the servo constants. If you require further information about encoders or the setting of the values in this window please contact COMSOFT.

TCSSKY.EXE is used if you plan on having PC-TCS talk to another computer running Software Bisque’s “TheSky”. This version of PC-TCS “speaks” the ACL Telescope protocol will communicate with another computer running TheSky via one of the four available serial ports on the PC. For further information on using TheSky with PC-TCS see the supplemental addendum to the manual provided under COMSOFT native protocol which is much more powerful but as yet unsupported by third party software packages. This version is intended for another programmers who are running other operating systems (Linux for example) and wish to control PC-TCS remotely via a serial link. Further information on the COMSOFT native protocol is documented in the **User’s Manual** in Appendix B,C and D and on the COMSOFT Web page.

Once the PFEDIT information is completed you will be able to run PC-TCS for the first time. There are two of PC-TCS/CT distributed on the CD-ROM. To do this exit the PFEDIT program and start PC-TCS using the TCSSKY icon (if you are running Windows 95/98) or by typing **TCSSKY** ← at the DOS prompt while in the TCS subdirectory. You should briefly see a screen like below.

The “COMBOX” referred to in the banner is the Stepper Driver Electronics box. If the addresses and interrupts entered into the PFEDIT program were correct PC-TCS should begin running after a second or two or after any key on the keyboard is hit (except the *ESC* which will abort and bring you back to the operating system.).

If there is a problem with the addresses or interrupts an error window will appear documenting the problem such as shown below.

If you get this window displayed check the following . . .

Hardware:

1. Make sure the Jumpers and DIP switches on the CTR-10 board in the computer are set properly.
2. Make sure the SR interface module is properly connected to the CTR-10 connectors.
3. Make sure no other cards in your computer use the addresses or interrupts set on the CTR-10 interface card.

Software:

1. Using the PFEDIT program make sure the correct addresses and interrupts are set.
2. Using the PFEDIT program increase the value of the “Delay constant” found in the **Computer-Interrupt** Alteration Window to a value no greater than 99.

If after chacking the above you still have problems please contact COMSOFT for advice.

COMSOFT

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